HOW TO USE THIS MANUAL TRANSAXLE REPAIR MANUAL

010BW–03

GENERAL INFORMATION

1. GENERAL DESCRIPTION

(a) This manual is made in accordance with SAE J2008.

(b) Generally repair operations can be separated in the following 3 main processes:
1. Diagnosis
2. Removing and Installing, Replacing, Disassembling, Installing and Checking, Adjusting
3. Final Inspection

(c) This manual explains ”Removing and Installing, Replacing, Disassembling, Installing and Checking, Adjusting”, but ”Final Inspection” is omitted.

(d) The following essential operations are not written in this manual, however these operations must be done in the practical situation.
(1) Operation with a jack or lift
(2) Cleaning of a removed part when necessary
(3) Visual check

2. INDEX

(a) An alphabetical INDEX is provided as a section on the end of the book to guide you to the item to be repaired.

3. PREPARATION

(a) Use of special service tools (SST) and special service materials (SSM) may be required, depending on the repairing condition. Be sure to use SST and SSM when they are required and follow the working procedure properly. A list of SST and SSM is in the Preparation section of this manual.

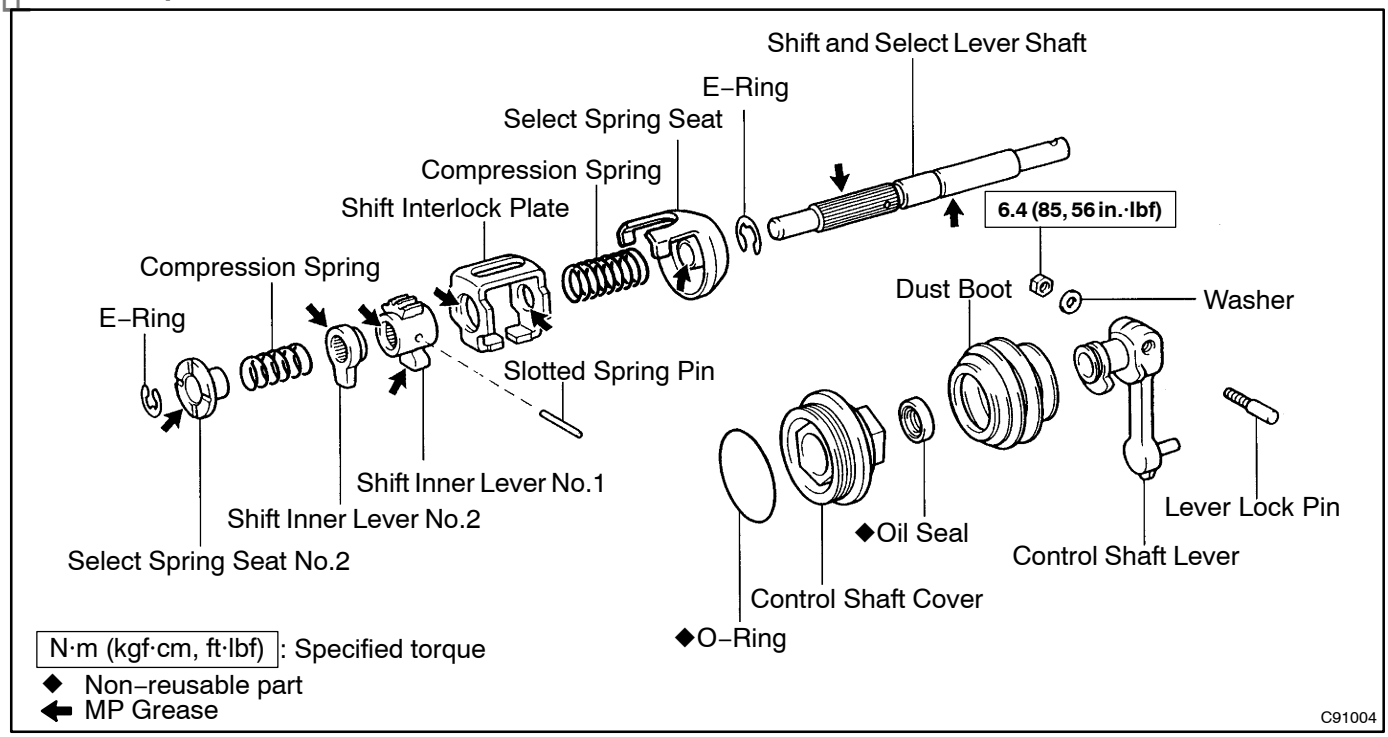
4. REPAIR PROCEDURES

(a) Component drawing is placed as the section or title when necessary.

(b) Illustrations of the parts catalog are placed as the ”disassembled parts drawing” so that it enables you to understand the fitting condition of the components.

(c) Non–reusable parts, grease applied parts, precoated parts and tightening torque are specified in the components drawing.

Example:



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(d) Tightening torque, oil applying position, and non–reusable parts are described as important points in the procedure.

NOTICE:

There are cases where such information can only be indicated by an illustration. In that case, all the information such as torque, oil, etc. are described in the illustration.

(e) Installing procedure of operation items is performed in the reverse order of the removing, and only the important points are described.

(f) Only items with points are described in the procedure, and the operational portion and content are placed using an illustration. In the explanations, details of the operational method, standard value and notice are placed.

(g) There may be a case where the illustrations of similar models are used. In that case the details may be different from the actual vehicle.

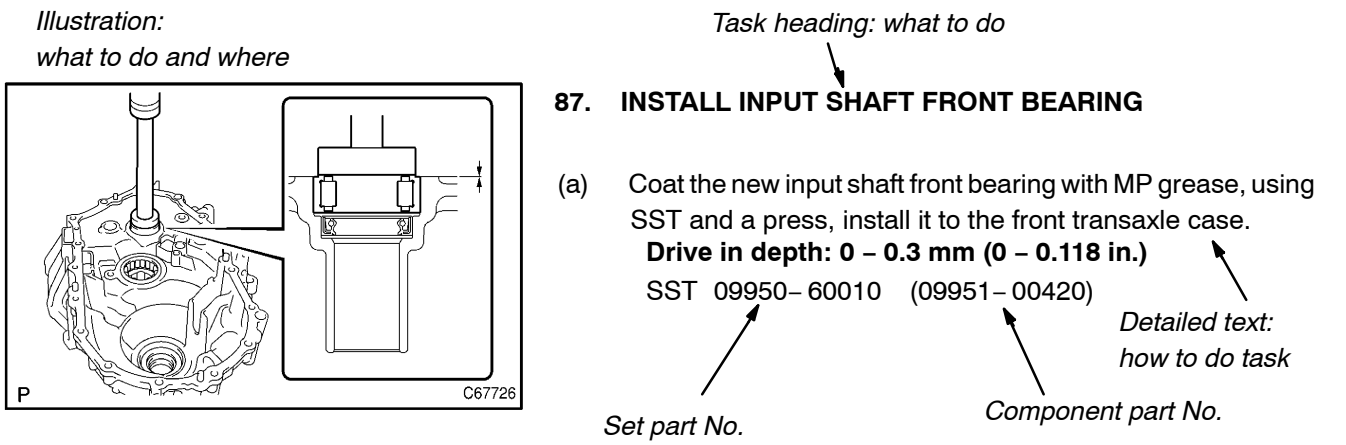
(h) The procedures are presented in a step–by–step format:

(1) The illustration shows what to do and where to do.

(2) The task heading tells what to do.

(3) The detailed text tells how to perform the task and gives other information such as specifications and warnings.

Example:



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HINT:

This format provides an experienced technician with a FAST TRACK to the necessary information. The task heading can be read at a glance when necessary, and the text below provides detailed information. Important specifications and warnings always stand out in bold type.

5. SERVICE SPECIFICATIONS

(a) Specifications are presented in bold type throughout the manual. You never have to leave the procedure to look up your specifications. The specifications are also found in the Service Specifications section for a quick reference.

6. TERMS DEFINITION

CAUTION	Indicate the possibility of injury to you or other people.
NOTICE	Indicate the possibility of damage to the components being repaired.
HINT	Provide additional information to help you perform the repair efficiently.

7. SI UNIT

(a) The UNITS given in this manual are primarily expressed according to the SI UNIT (International System of Unit), and alternately expressed in the metric system and the English System.
Example:
Torque: 30 N·m (310 kgf·cm, 22 ft·lbf)

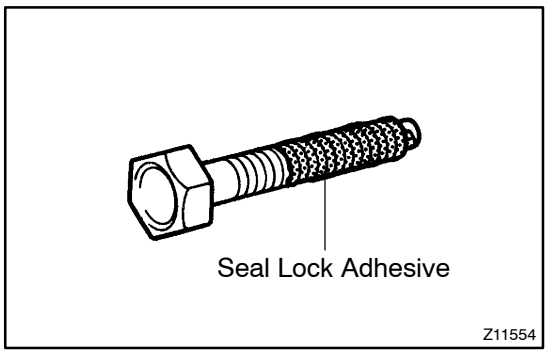
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REPAIR INSTRUCTION FOR MANUAL TRANSAXLE REPAIR MANUAL

010BX–03

PRECAUTION

1. BASIC REPAIR HINT



(a) PRECOATED PARTS

(1) Precoated parts are bolts, nuts, etc. that are coated with a seal lock adhesive at the factory.

(2) If a precoated part is retightened, loosened or caused to move in any way, it must be recoated with the specified adhesive.

(3) When reusing precoated parts, clean off the old adhesive and dry with compressed air. Then apply the specified seal lock adhesive to the bolt, nut or threads.

NOTICE:

Do the torque checking with the lower limit value of the torque tolerance.

(4) Depending on the seal lock agent to apply, there may be a case where it is necessary to leave it for a specified time until it hardens.

(b) GASKETS

When necessary, use a sealer on gaskets to prevent leaks.

(c) BOLTS, NUTS AND SCREWS

Carefully observe all specifications for bolt tightening torques. Always use a torque wrench.

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TERMS FOR MANUAL TRANSAXLE REPAIR MANUAL

ABBREVIATIONS USED IN THIS MANUAL

010BY–03

Abbreviations	Meaning
E/G	Engine
FIPG	Formed In Place Gasket
MAX	Maximum
MIN	Minimum
MP	Multipurpose
M/T	Manual Transmission
N	Neutral
No.	Number
RH	Right–Hand
SPEC	Specification
SSM	Special Service Materials
SST	Special Service Tools
STD	Standard
T/A	Transaxle
T/M	Transmission
TMC	TOYOTA Motor Coporation
w/	With
w/o	Without
1st	First
2nd	Second

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GLOSSARY OF SAE AND TOYOTA TERMS

This glossary lists all SAE–J1930 terms and abbreviations used in this manual in compliance with SAE recommendations, as well as their Toyota equivalents.

SAE ABBREVIATIONS	SAE TERMS	TOYOTA TERMS ( )––ABBREVIATIONS
A/C	Air Conditioning	Air Conditioner
ACL	Air Cleaner	Air Cleaner
AIR	Secondary Air Injection	Air Injection (AI)
AP	Accelerator Pedal	–
B+	Battery Positive Voltage	+B, Battery Voltage
BARO	Barometric Pressure	–
CAC	Charge Air Cooler	Inter cooler
CARB	Carburetor	Carburetor
CFI	Continuous Fuel Injection	–
CKP	Crankshaft Position	Crank Angle
CL	Closed Loop	Closed Loop
CMP	Camshaft position	Cam Angle
CPP	Clutch Pedal Position	–
CTOX	Continuous Trap Oxidizer	–
CTP	Closed Throttle Position	–
DFI	Direct Fuel Injection (Diesel)	Direct Injection (DI)
DI	Distributor Ignition	–
DLC1 DLC2 DLC3	Data Link Connector 1 Data Link Connector 2 Data Link Connector 3	1: Check Connector 2: Total Diagnosis Communication Link (TDCL) 3: OBD II Diagnostic Connector
DTC	Diagnostic Trouble Code	Diagnostic Code
DTM	Diagnostic Test Mode	–
ECL	Engine Control Level	–
ECM	Engine Control Module	Engine ECU (Electronic Control Unit)
ECT	Engine Control Temperature	Coolant Temperature, Water Temperature (THW)
EEPROM	Electrically Erasable Programmable Read Only Memory	Electrically Erasable Programmable Read Only Memory (EEPROM), Erasable Programmable Read Only Memory (EPROM)
EFE	Early Fuel Evaporation	Cold Mixture Heater (CMH), Heat Control Valve (HCV)
EGR	Exhaust Gas Recirculation	Exhaust Gas Recirculation (EGR)
EI	Electronic Ignition	Distributorless Ignition (DI)
EM	Engine Modification	Engine Modification (EM)
EPROM	Erasable Programmable Read Only Memory	Programmable Read Only Memory (PROM)
EVAP	Evaporative Emission	Evaporative Emission Control (EVAP)
FC	Fan Control	–
FEEPROM	Flash Electrically Erasable Programmable Read Only Memory	–
FEPROM	Flash Erasable Programmable Read Only Memory	–
FF	Flexible Fuel	–
FP	Fuel Pump	Fuel Pump
GEN	Generator	Alternator
GND	Ground	Ground (GND)
HO2S	Heated Oxygen Sensor	Heated Oxygen Sensor (HO2S)
IAC	Idol Air Control	Idol Speed Control (ISC)
IAT	Intake Air Temperature	Intake or Inlet Air Temperature
ICM	Ignition Control Module	–
IFI	Indirect Fuel Injection	Indirect Injection
IFS	Inertia Fuel–Shutoff	–

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ISC	Idle Speed Control	–
KS	Knock Sensor	Knock Sensor
MAF	Mass Air Flow	Air Flow Meter
MAP	Manifold Absolute Pressure	Manifold Pressure Intake Vacuum
MC	Mixture Control	Electric Bleed Air Control Valve (EBCV) Mixture Control Valve (MCV) Electric Air Control Valve (EACV)
MDP	Manifold Differential Pressure	–
MFI	Multiport Fuel Injection	Electronic Fuel Injection (EFI)
MIL	Malfunction Indicator Lamp	Check Engine Light
MST	Manifold Surface Temperature	–
MVZ	Manifold Vacuum Zone	–
NVRAM	Non–Volatile Random Access Memory	–
O2S	Oxygen Sensor	Oxygen Sensor, O_2 Sensor (O_{2S})
OBD	On–Board Diagnostic	On–Board Diagnostic (OBD)
OC	Oxidation Catalytic Converter	Oxidation Catalyst Converter (OC), CC_O
OP	Open Loop	Open Loop
PAIR	Pulsed Secondary Air Injection	Air Suction (AS)
PCM	Powertrain Control Module	–
PNP	Park/Neutral Position	–
PROM	Programmable Read Only Memory	–
PSP	Power Steering Pressure	–
PTOX	Periodic Trap Oxidizer	Diesel Particulate Filter (DPF) Diesel Particulate Trap (DPT)
RAM	Random Access Memory	Random Access Memory (RAM)
RM	Relay Module	–
ROM	Read Only Memory	Read Only Memory (ROM)
RPM	Engine Speed	Engine Speed
SC	Supercharger	Supercharger
SCB	Supercharger Bypass	–
SFI	Sequential Multiport Fuel Injection	Electronic Fuel Injection (EFI), Sequential Injection
SPL	Smoke Puff Limiter	–
SRI	Service Reminder Indicator	–
SRT	System Readiness Test	–
ST	Scan Tool	–
TB	Throttle Body	Throttle Body
TBI	Throttle Body Fuel Injection	Single Point Injection Central Fuel Injection (Ci)
TC	Turbocharger	Turbocharger
TCC	Torque Converter Clutch	Torque Converter
TCM	Transmission Control Module	Transmission ECU (Electronic Control Unit)
TP	Throttle Position	Throttle Position
TR	Transmission Range	–
TVV	Thermal Vacuum Valve	Bimetallic Vacuum Switching Valve (BVSV) Thermostatic Vacuum Switching Valve (TVSV)
TWC	Three–Way Catalytic Converter	Three–Way Catalytic (TWC) CC_{RO}
TWC+OC	Three–Way + Oxidation Catalytic Converter	CC_R + CC_O
VAF	Volume Air Flow	Air Flow Meter
VR	Voltage Regulator	Voltage Regulator
VSS	Vehicle Speed Sensor	Vehicle Speed Sensor (Read Switch Type)
WOT	Wide Open Throttle	Full Throttle

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WU–OC	Warm Up Oxidation Catalytic Converter	–
WU–TWC	Warm Up Three–Way Catalytic Converter	Manifold Converter
3GR	Third Gear	–
4GR	Fourth Gear	–

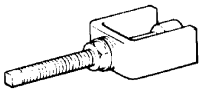
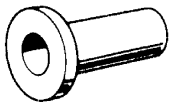
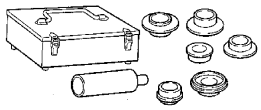
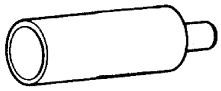
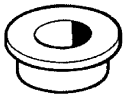
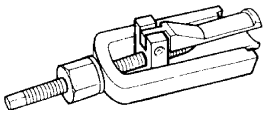
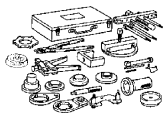
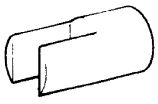
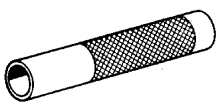
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MANUAL TRANSMISSION/TRANSAXLE

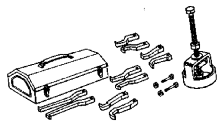
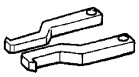
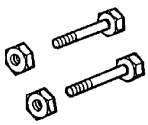
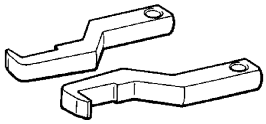
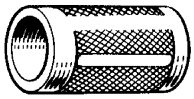
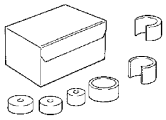
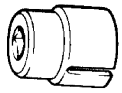
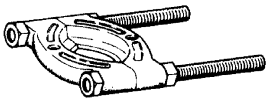
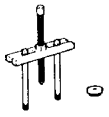
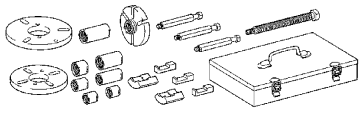
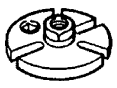
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PREPARATION

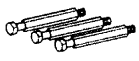
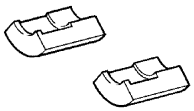
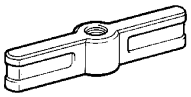
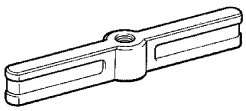
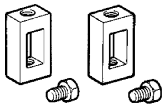
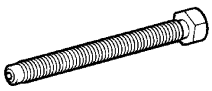
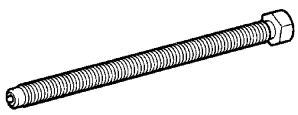
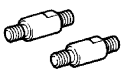
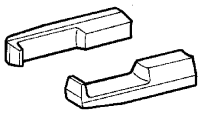
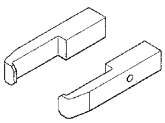
SST

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	09309–12020	5th Driven Gear Replacer	MANUAL TRANSAXLE ASSY(C50)
	09309–36010	Transmission Rear Bearing Replacer	MANUAL TRANSAXLE ASSY(C50)
	09316–60011	Transmission & Transfer Bearing Replacer	MANUAL TRANSAXLE ASSY(C50) INPUT SHAFT ASSY(C50) OUTPUT SHAFT ASSY(C50)
	(09316–00011)	Replacer Pipe	MANUAL TRANSAXLE ASSY(C50)
	(09316–00021)	Replacer "A"	INPUT SHAFT ASSY(C50)
	(09316–00031)	Replacer "B"	OUTPUT SHAFT ASSY(C50)
	09319–60020	Output Shaft Needle Roller Bearing Remover	MANUAL TRANSAXLE ASSY(C50)
	09350–32014	TOYOTA Automatic Transmission Tool Set	MANUAL TRANSAXLE ASSY(C50)
	(09351–32120)	Overdrive Bearing Replacer	MANUAL TRANSAXLE ASSY(C50)
	09564–32011	Differential Preload Adaptor	MANUAL TRANSAXLE ASSY(C50)
	09612–22011	Tilt Handle Bearing Replacer	OUTPUT SHAFT ASSY(C50)

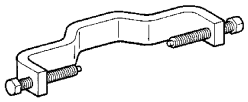
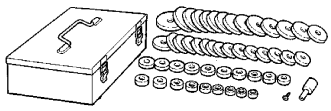
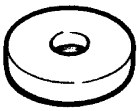
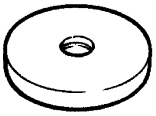
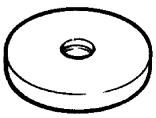
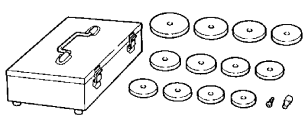
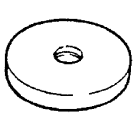
ProCarManuals.com

	09612–65014	Steering Worm Bearing Puller	MANUAL TRANSAXLE ASSY(C50)
	(09612–01040)	Claw ”D”	MANUAL TRANSAXLE ASSY(C50)
	(09612–01050)	Hanger Pin with Nut	MANUAL TRANSAXLE ASSY(C50)
	(09612–01060)	Claw ”E”	MANUAL TRANSAXLE ASSY(C50)
	09636–20010	Upper Ball Joint Dust Cover Replacer	MANUAL TRANSAXLE ASSY(C50) INPUT SHAFT ASSY(C50)
	09710–20011	Front Suspension Bushing Tool Set	MANUAL TRANSAXLE ASSY(C50)
	(09710–06071)	Replacer	MANUAL TRANSAXLE ASSY(C50)
	09817–16011	Back–up Light Switch Tool	MANUAL TRANSAXLE ASSY(C50)
	09950–00020	Bearing Remover	MANUAL TRANSAXLE ASSY(C50) INPUT SHAFT ASSY(C50) OUTPUT SHAFT ASSY(C50)
	09950–00030	Bearing Remover Attachment	MANUAL TRANSAXLE ASSY(C50)
	09950–30012	Puller A Set	MANUAL TRANSAXLE ASSY(C50)
	(09951–03010)	Upper Plate	MANUAL TRANSAXLE ASSY(C50)
	(09953–03010)	Center Bolt	MANUAL TRANSAXLE ASSY(C50)

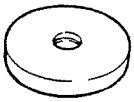
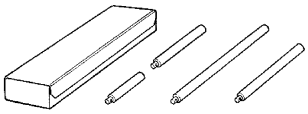
ProCarManuals.com

	(09954–03010)	Arm	MANUAL TRANSAXLE ASSY(C50)
	(09955–03011)	Claw No.1	MANUAL TRANSAXLE ASSY(C50)
	(09957–04010)	Attachment	MANUAL TRANSAXLE ASSY(C50)
	09950–40011	Puller B Set	MANUAL TRANSAXLE ASSY(C50)
	(09951–04010)	Hanger 150	MANUAL TRANSAXLE ASSY(C50)
	(09951–04020)	Hanger 200	MANUAL TRANSAXLE ASSY(C50)
	(09952–04010)	Slide Arm	MANUAL TRANSAXLE ASSY(C50)
	(09953–04020)	Center Bolt 150	MANUAL TRANSAXLE ASSY(C50)
	(09953–04030)	Center Bolt 200	MANUAL TRANSAXLE ASSY(C50)
	(09954–04010)	Arm 25	MANUAL TRANSAXLE ASSY(C50)
	(09955–04061)	Claw No.6	MANUAL TRANSAXLE ASSY(C50)
	(09955–04071)	Claw No.7	MANUAL TRANSAXLE ASSY(C50)
	(09957–04010)	Attachment	MANUAL TRANSAXLE ASSY(C50)

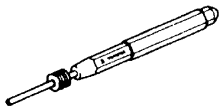
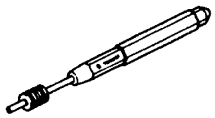
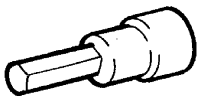
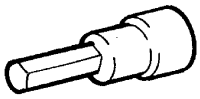
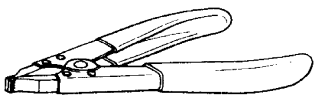
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	(09958–04011)	Holder	MANUAL TRANSAXLE ASSY(C50)
	09950–60010	Replacer Set	MANUAL TRANSAXLE ASSY(C50) SHIFT & SELECT LEVER SHAFT ASSY(C50)
	(09951–00190)	Replacer 19	MANUAL TRANSAXLE ASSY(C50)
	(09951–00200)	Replacer 20	MANUAL TRANSAXLE ASSY(C50)
	(09951–00220)	Replacer 22	MANUAL TRANSAXLE ASSY(C50) SHIFT & SELECT LEVER SHAFT ASSY(C50)
	(09951–00360)	Replacer 36	MANUAL TRANSAXLE ASSY(C50)
	(09951–00370)	Replacer 37	MANUAL TRANSAXLE ASSY(C50)
	(09951–00420)	Replacer 42	MANUAL TRANSAXLE ASSY(C50)
	(09951–00530)	Replacer 53	MANUAL TRANSAXLE ASSY(C50)
	(09951–00550)	Replacer 55	MANUAL TRANSAXLE ASSY(C50)
	(09951–00600)	Replacer 60	MANUAL TRANSAXLE ASSY(C50)
	09950–60020	Replacer Set No.2	MANUAL TRANSAXLE ASSY(C50)
	(09951–00680)	Replacer 68	MANUAL TRANSAXLE ASSY(C50)

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	(09951–00710)	Replacer 71	MANUAL TRANSAXLE ASSY(C50)
	09950–70010	Handle Set	MANUAL TRANSAXLE ASSY(C50) SHIFT & SELECT LEVER SHAFT ASSY(C50)
	(09951–07100)	Handle 100	MANUAL TRANSAXLE ASSY(C50) SHIFT & SELECT LEVER SHAFT ASSY(C50)
	(09951–07150)	Handle 150	MANUAL TRANSAXLE ASSY(C50)

Recomended Tools

	09031–00030	Pin Punch	DIFFERENTIAL CASE ASSY(C50)
	09031–00040	Pin Punch .	MANUAL TRANSAXLE ASSY(C50) SHIFT & SELECT LEVER SHAFT ASSY(C50)
	(09043–20060)	Socket Hexagon Wrench 6	MANUAL TRANSAXLE ASSY(C50)
	(09043–20060)	Socket Hexagon Wrench 6	MANUAL TRANSAXLE ASSY(C50)
	09905–00012	Snap Ring No.1 Expander	MANUAL TRANSAXLE ASSY(C50)

Equipment

V–block	
Cylinder gauge	
Dial indicator or dial indicator with magnetic base	
Feeler gauge	
Micrometer	
Torque wrench	
Vernier calipers	
Chisel	
Plastic hammer	
Slide calipers	
Press	
Heater	

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SSM (Special Service Materials)

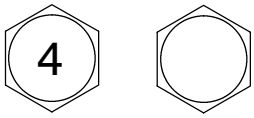
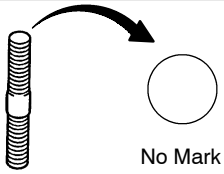
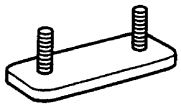
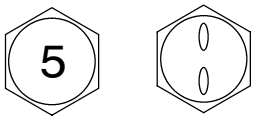
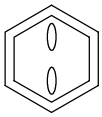
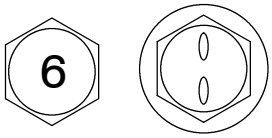
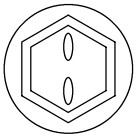
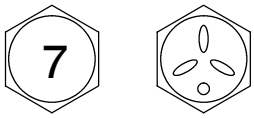
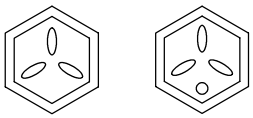
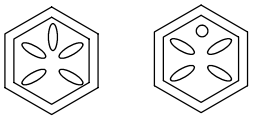
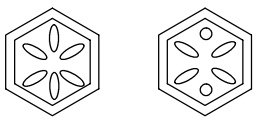
08826–00090	"Seal Packing 1281," THREE BOND 1281 or equivalent (FIPG)	
08833–00080	Adhesive 1344 THREE BOND 1344 LOCTITE 242 or equivalent	

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STANDARD BOLT

HOW TO DETERMINE BOLT STRENGTH

030LK–07

Bolt Type				Class
Hexagon Head Bolt		Stud Bolt	Weld Bolt	
Normal Recess Bolt	Deep Recess Bolt			
 No Mark	 No Mark	 No Mark		4T
				5T
 w/ Washer	 w/ Washer			6T
				7T
				8T
				9T
				10T
				11T

B06431

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030LL–07

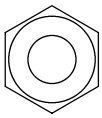
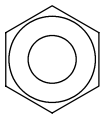
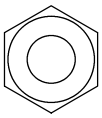
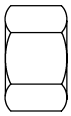
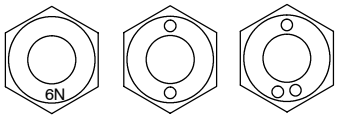
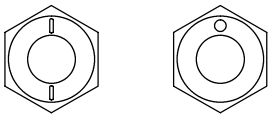
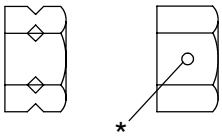
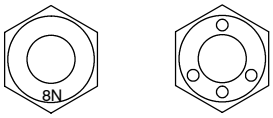
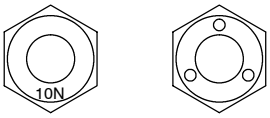
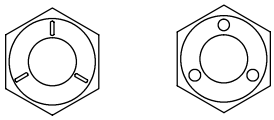
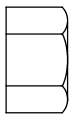
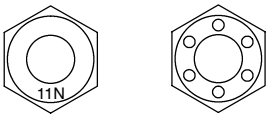
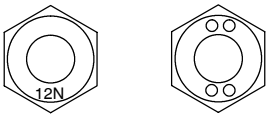
SPECIFIED TORQUE FOR STANDARD BOLTS

Class	Diameter mm	Pitch mm	Specified torque					
			Hexagon head bolt			Hexagon flange bolt		
			N·m	kgf·cm	ft·lbf	N·m	kgf·cm	ft·lbf
4T	6	1	5	55	48 in.·lbf	6	60	52 in.·lbf
	8	1.25	12.5	130	9	14	145	10
	10	1.25	26	260	19	29	290	21
	12	1.25	47	480	35	53	540	39
	14	1.5	74	760	55	84	850	61
	16	1.5	115	1,150	83	–	–	–
5T	6	1	6.5	65	56 in.·lbf	7.5	75	65 in.·lbf
	8	1.25	15.5	160	12	17.5	175	13
	10	1.25	32	330	24	36	360	26
	12	1.25	59	600	43	65	670	48
	14	1.5	91	930	67	100	1,050	76
	16	1.5	140	1,400	101	–	–	–
6T	6	1	8	80	69 in.·lbf	9	90	78 in.·lbf
	8	1.25	19	195	14	21	210	15
	10	1.25	39	400	29	44	440	32
	12	1.25	71	730	53	80	810	59
	14	1.5	110	1,100	80	125	1,250	90
	16	1.5	170	1,750	127	–	–	–
7T	6	1	10.5	110	8	12	120	9
	8	1.25	25	260	19	28	290	21
	10	1.25	52	530	38	58	590	43
	12	1.25	95	970	70	105	1,050	76
	14	1.5	145	1,500	108	165	1,700	123
	16	1.5	230	2,300	166	–	–	–
8T	8	1.25	29	300	22	33	330	24
	10	1.25	61	620	45	68	690	50
	12	1.25	110	1,100	80	120	1,250	90
9T	8	1.25	34	340	25	37	380	27
	10	1.25	70	710	51	78	790	57
	12	1.25	125	1,300	94	140	1,450	105
10T	8	1.25	38	390	28	42	430	31
	10	1.25	78	800	58	88	890	64
	12	1.25	140	1,450	105	155	1,600	116
11T	8	1.25	42	430	31	47	480	35
	10	1.25	87	890	64	97	990	72
	12	1.25	155	1,600	116	175	1,800	130

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030LM–07

HOW TO DETERMINE NUT STRENGTH

Nut Type			Class
Present Standard Hexagon Nut	Old Standard Hexagon Nut		
	Cold Forging Nut	Cutting Processed Nut	
 No Mark			4N
 No Mark (w/ Washer)	 No Mark (w/ Washer)	 No Mark	5N (4T)
 6N			6N
		 *	7N (5T)
 8N			8N
 10N		 No Mark	10N (7T)
 11N			11N
 12N			12N

*: Nut with 1 or more marks on one side surface of the nut.

B06432

HINT:

Use the nut with the same number of the nut strength classification or the greater than the bolt strength classification number when tightening parts with a bolt and nut.

Example: Bolt = 4T

Nut = 4N or more

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MANUAL TRANSMISSION / TRANSAXLE

031EO–02

SERVICE DATA

Manual transaxle assy		
5th gear thrust clearance	STD	0.10 to 0.57 mm (0.0039 to 0.02244 in.)
	MAX	0.57 mm (0.02244 in.)
5th gear radial clearance	STD	0.015 to 0.058 mm (0.0006 to 0.0023 in.)
	MAX	0.058 mm (0.0023 in.)
Synchro nizer ring No.3 clearance	STD	0.75 to 1.65 mm (0.0295 to 0.0649 in.)
	MIN	0.75 mm (0.0295 in.)
Transmission hub sleeve No.3 clearance		0.3 to 0.5 mm (0.0012 to 0.0020 in.)
5th gear clearance		29.915 to 29.931 mm (1.1778 to 1.1783 in.)
Reverse idler gear sub–assy	STD	18.040 to 18.058 mm (0.7102 to 0.7109 in.)
	MAX	18.058 mm (0.7109 in.)
Reverse idler gear shaft outer diameter	STD	17.966 to 17.984 mm (0.7073 to 0.7080 in.)
	MIN	17.966 mm (0.7073 in.)
Front transaxle cace oil seal drive in depth		15.6 to 16.0 mm (0.6141 to 0.6299 in.)
Input shaft front bearing drive in depth		0 to 0.3 mm (0 to 0.0118 in.)
Shift & select lever shaft needle roller bearing drive in depth		0 to 0.5 mm (0 to 0.020 in.)
Differential side bearing preload		0.78 to 1.57 N·m (7.95 to 16.0 kgf·cm, 6.9 to 13.89 in.·lbf)
Front differential case rear plate washer thickness	01	1.10 mm (0.0433 in.)
	02	1.15 mm (0.0453 in.)
	03	1.20 mm (0.0472 in.)
	04	1.25 mm (0.0492 in.)
	05	1.30 mm (0.0512 in.)
	06	1.35 mm (0.0531 in.)
	07	1.40 mm (0.0551 in.)
	08	1.45 mm (0.0571 in.)
	09	1.50 mm (0.0591 in.)
	10	1.55 mm (0.0610 in.)
	11	1.60 mm (0.0630 in.)
	12	1.65 mm (0.0650 in.)
	13	1.70 mm (0.0670 in.)
	14	1.75 mm (0.0689 in.)
	15	1.80 mm (0.0709 in.)
	16	1.85 mm (0.0728 in.)
	17	1.90 mm (0.0748 in.)
	18	1.95 mm (0.0768 in.)
	19	2.00 mm (0.0787 in.)
Transmission case oil seal drive in depth		9.6 to 10.2 mm (0.3779 to 0.4015 in.)
Transaxle case oil seai drive in depth		1.6 to 2.2 mm (0.0629 to 0.0866 in.)
Reverse restrict pin assy drive in depth		15.5 to 16.5 mm (0.6102 to 0.6496 in.)
Transmission clutch hub No.3 shaft snap ring clearance		0.1 mm (0.0039 in.) or less
Transmission clutch hub No.3 shaft snap ring thickness	A	2.25 mm (0.0886 in.)
	B	2.31 mm (0.0909 in.)
	C	2.37 mm (0.0933 in.)
	D	2.43 mm (0.0957 in.)
	E	2.49 mm (0.0980 in.)
	F	2.55 mm (0.1004 in.)
	G	2.61 mm (0.1028 in.)

Input shaft assy		
4th gear thrust clearance	STD	0.1 to 0.55 mm (0.0039 to 0.0217 in.)
	MAX	0.55 mm (0.0217 in.)
3rd gear thrust clearance	STD	0.1 to 0.35 mm (0.0039 to 0.0138 in.)
	MAX	0.35 mm (0.0138 in.)
4th gear radial clearance	STD	0.015 to 0.058 mm (0.0006 to 0.0023 in.)
	MAX	0.058 mm (0.0023 in.)

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3rd gear radial clearance	STD	0.015 to 0.058 mm (0.0006 to 0.0023 in.)
	MAX	0.058 mm (0.0023 in.)
Input shaft runout	MIN	0.015 mm(0.00059 in.)
Input shaft outer diameter	STD: Part A	24.885 to 24.900 mm (0.9797 to 0.9803 in.)
	Part B	28.991 to 29.006 mm (1.1413 to 1.1419 in.)
	Part C	30.985 to 31.000 mm (1.2198 to 1.2204 in.)
	Part D	24.985 to 25.000 mm (0.9836 to 0.9842 in.)
	MIN: Part A	24.885 mm (0.9797 in.)
	Part B	28.991 mm (1.1413 in.)
	Part C	30.985 mm (1.2198 in.)
	Part D	24.985 mm (0.9836 in.)
4th gear clearance	STD	34.015 to 34.031 mm (1.3391 to 1.3398 in.)
	MAX	34.031 mm (1.3398 in.)
3rd gear clearance	STD	36.015 to 36.031 mm (1.4179 to 1.4185 in.)
	MAX	36.031 mm (1.4185 in.)
Synchronizer ring No.2 (For 4th gear) clearance		0.75 to 1.65 mm (0.0295 to 0.0649 in.)
Synchronizer ring No.2 (For 3rd gear) clearance		0.75 to 1.65 mm (0.0295 to 0.0649 in.)
Transmission hub sleeve No.2 clearance		0.15 to 0.35 mm (0.0059 to 0.0137 in.)
Clutch hub No.2 setting shaft snap ring clearance		0.1 mm (0.0039 in.) or less.
Clutch hub No.2 setting shaft snap ring thickness	0	2.30 mm (0.0906 in.)
	1	2.36 mm (0.0929 in.)
	2	2.42 mm (0.0953 in.)
	3	2.48 mm (0.0976 in.)
	4	2.54 mm (0.1000 in.)
	5	2.60 mm (0.1024 in.)
Input shaft rear bearing shaft snap ring clearance		0.1 mm (0.0039 in.) or less
Input shaft rear bearing shaft snap ring thickness	A	2.29 mm (0.0902 in.)
	B	2.35 mm (0.0925 in.)
	C	2.41 mm (0.0949 in.)
	D	2.47 mm (0.0972 in.)
	E	2.53 mm (0.0996 in.)
	F	2.59 mm (0.1020 in.)

Output shaft assy		
1st gear thrust clearance	STD	0.10 to 0.40 mm (0.0039 to 0.0157 in.)
	MAX	0.40 mm (0.0157 in.)
2nd gear thrust clearance	STD	0.1 to 0.45 mm (0.0039 to 0.0177 in.)
	MAX	0.45 mm (0.0177 in.)
1st gear radial clearance	STD	0.015 to 0.058 mm (0.0006 to 0.0023 in.)
	MAX	0.058 mm (0.0023 in.)
2nd gear radial clearance	STD	0.015 to 0.058 mm (0.0006 to 0.0023 in.)
	MAX	0.058 mm (0.0023 in.)
Input shaft runout	MIN	0.015 mm (0.0006 in.)
Input shaft outer diameter	Part A	31.985 to 32.000 mm (1.2592 to 1.2598 in.)
	Part B	37.985 to 38.000 mm (1.4955 to 1.2598 in.)
	Part C	32.985 to 33.000 mm (1.2986 to 1.2992 in.)
2nd gear clearance	STD	38.015 to 38.013 mm (1.4966 to 1.4972 in.)
	MAX	38.031 mm (1.4972 in.)
1st gear clearance	STD	44.015 to 44.031 mm (1.7328 to 1.7335 in.)
	MAX	44.031 mm (1.7335 in.)
1st gear thrust washer thickness	STD	5.975 to 6.025 mm (0.2352 to 0.2372 in.)
	MIN	5.975 mm (0.2352 in.)
Synchronizer ring No.1 (for 2nd gear)clearance	STD	0.70 to 1.30 mm (0.0276 to 0.0512 in.)
	MIN	0.70 mm (0.0276in.)
Synchronizer ring No.1 (for 1st gear)clearance	STD	0.75 to 1.65 mm (0.0295 to 6.8898 in.)
	MIN	0.75 mm (0.0295 in.)
reverse gear and shift fork clearance	MAX	0.35 mm (0.014 in.)

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Clutch hub No.1 shaft snap ring clearance		0.1 mm (0.039 in.) or less
Clutch hub No.1 shaft snap ring thickness	A	2.50 mm (0.0984 in.)
	B	2.56 mm (0.1008 in.)
	C	2.62 mm (0.1031 in.)
	D	2.68 mm (0.1055 in.)
	E	2.74 mm (0.1079 in.)
	F	2.80 mm (0.1102 in.)

Sift & Select lever shaft assy		
Contorol shaft cover oil seal drive in depth		0.2 to 1.2 mm (0.079 to 0.0472 in.)
Sift lever slotted pin (for select inner lever) drivein depth		3.0 to 4.0 mm (0.1181 to 0.1575 in)
Sift lever slotted pin (for sift lever inner No.1) drivein depth		–0.5 to 0.5 mm (–0.0197 to 0.0197 in)
Sift lever slotted pin (for sift lever inner No.2) drivein depth		–0.5 to 0.5 mm (–0.0197 to 0.0197 in)

Differential case assy		
Front differential side gear backlash		0.05 to 0.20 mm (0.0020 to 0.0079 in.)
Front differential pinion thrust washer	MIN	0.94 mm (0.03701 in.)
Front differential pinion shaft No.1 outer diameter	MIN	16.982 mm (0.6685 in.)

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031EP–02

TORQUE SPECIFICATION

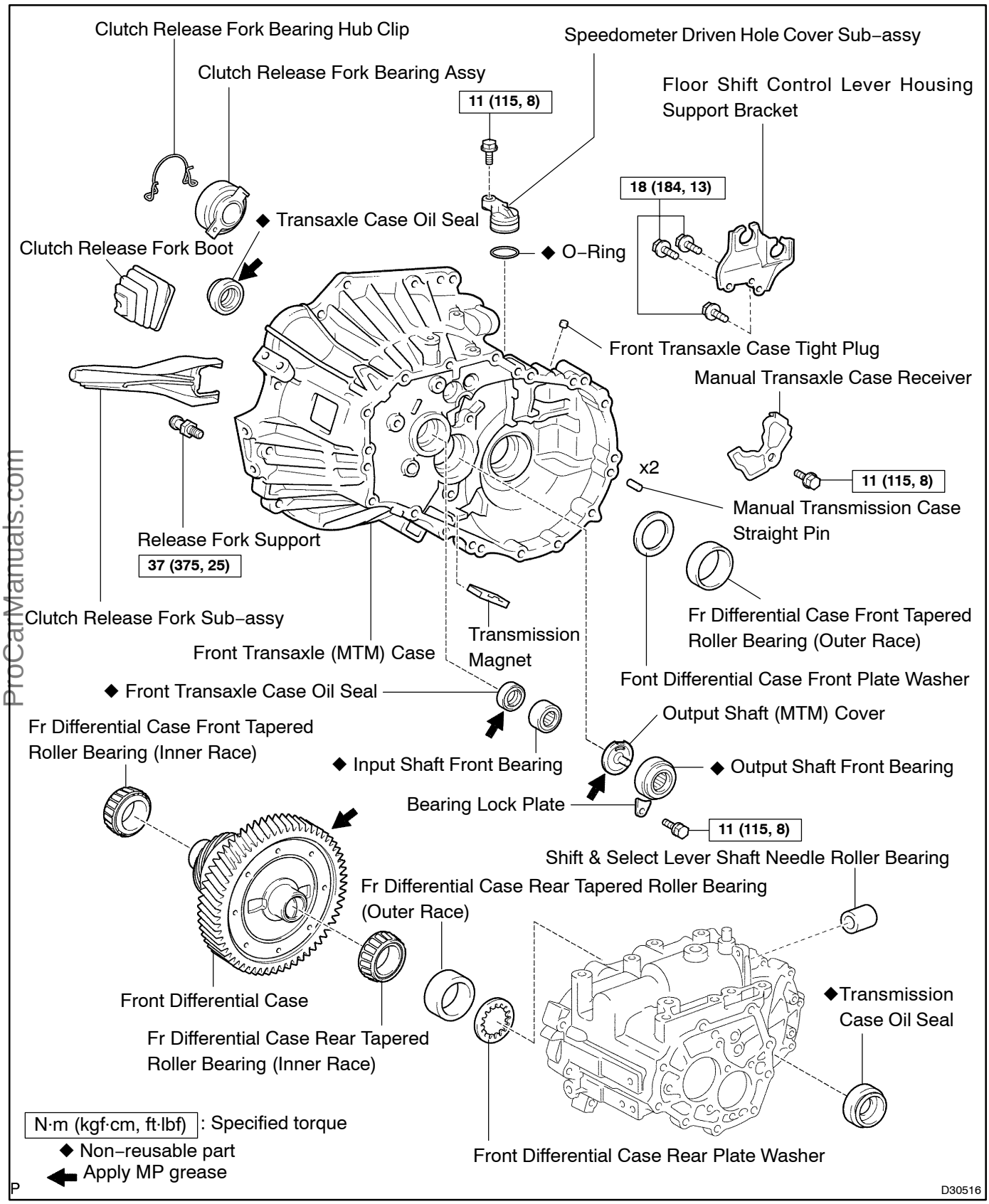
Part Tightened		N·m	kgf·cm	ft·lbf
Bearing lock plate × Front transaxle case		11	115	8
Oil receiver pipe No.1 × Manual transmission case		17	175	13
Oil receiver pipe No.2 × Manual transmission case		17	175	13
Reverse restrict pin plug × Manual transmission case		13	130	9.3
manual transaxle case receiver × Manual transmission case		11	115	8
Gear shift fork No.1 × Gear shift fork shaft No.1		16	160	12
Gear shift fork No.2 × Gear shift fork shaft No.2		16	160	12
Gear shift head No.1 × Gear shift fork shaft No.2		16	160	12
Reverse shift arm bracket assy × Front transaxle case		17	175	13
Manual transmission case × Front transaxle case	bolt	29	300	22
	nut	29	300	22
Reverse idler gear shaft bolt × Manual transmission case		29	300	22
Lock ball assy No.2 × Manual transmission case		39	400	29
Shift detent ball × Manual transmission case		25	250	18
Shift detent ball × Front transaxle case		25	250	18
Bearing retainer rear × Manual transmission case		27	280	20
Gear shift fork No.3 × Gear shift fork shaft		16	160	12
Manual transmission output shaft rear set nut × output shaft		118	1200	87
Manual transmission case cover × Manual transmission case		18	185	14
Control shaft cover × Manual transmission case		20	200	14
Control shift lever lock nut		12	122	9
Lock ball assy No.1 × Manual transmission case		29	300	22
Selecting bell crank assy × Manual transmission case		25	250	18
Floor shift control lever housing support bracket × Front transaxle case		18	184	13
Back–up lamp switch assy × Manual transmission case		40	410	30
Speedometer driven hole cover sub–assy × Manual transaxle		11	115	8
Release fork support × Front transaxle case		37	375	25
Manual transmission filler plug × Manual transaxle		39	400	29
Drain plug sub–assy × Manual transmission case		39	400	29
Selecting bell crank No.2 plate washer × Selecting bell crank		12	120	9
Front defferential ring gear × Front differential case		77	790	57

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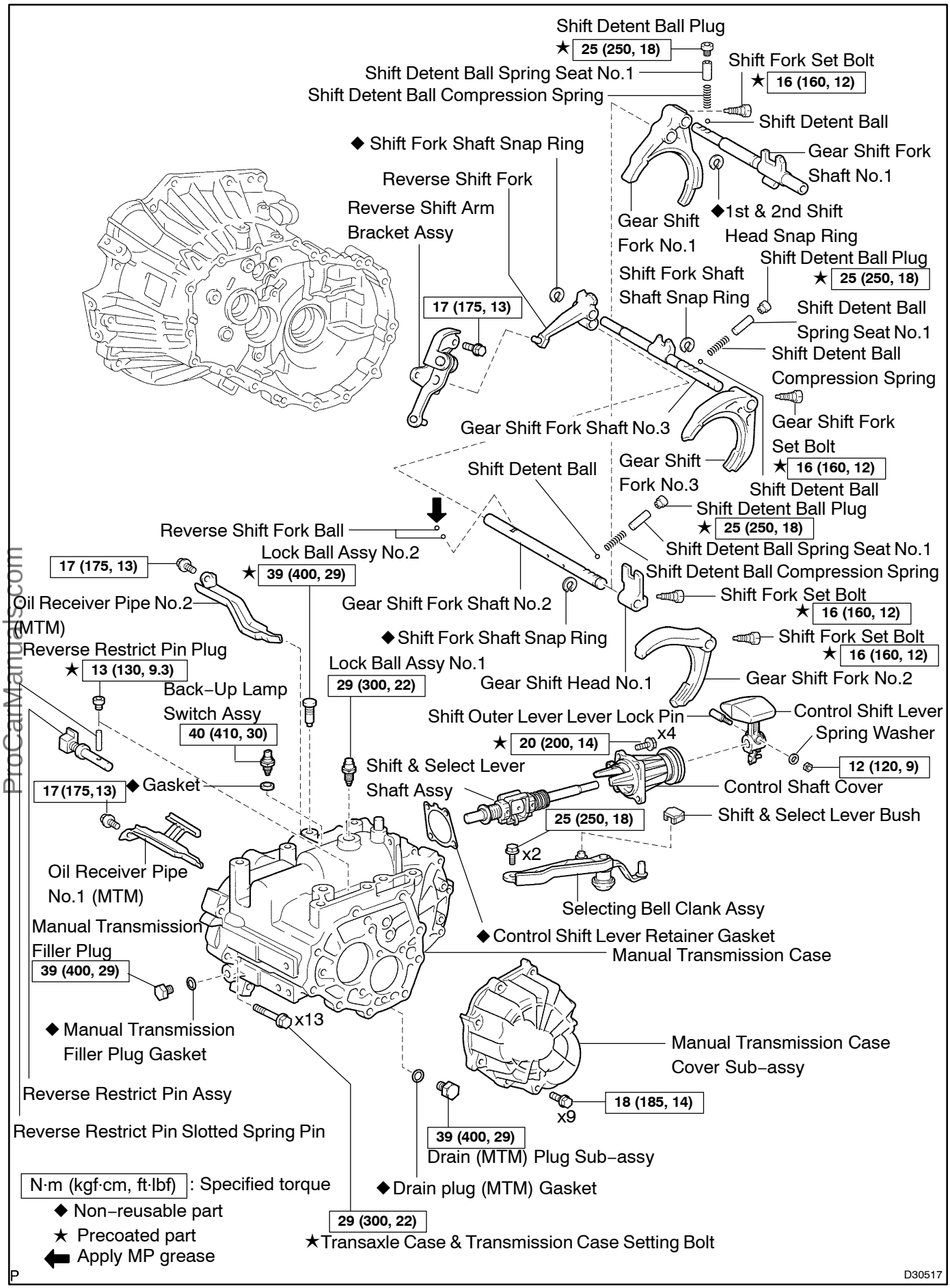
MANUAL TRANSAXLE ASSY (C50)

COMPONENTS

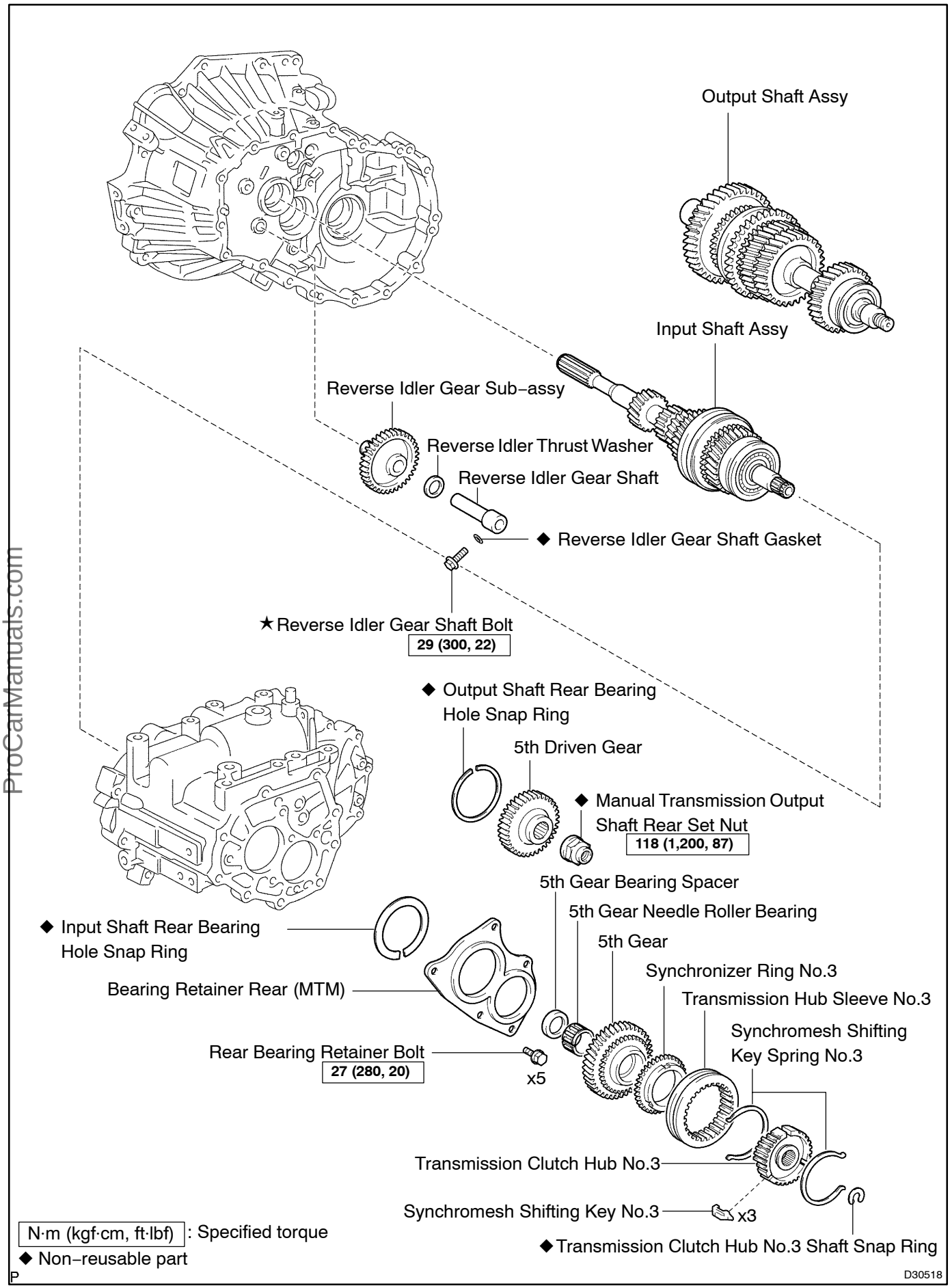
410DC–02



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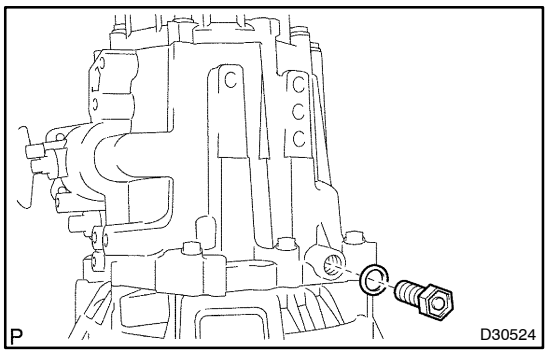
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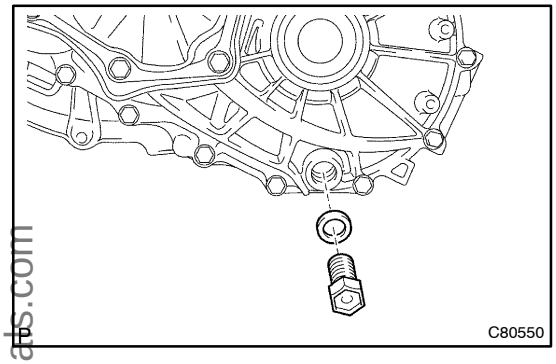
410FF–01

OVERHAUL



1. REMOVE MANUAL TRANSMISSION FILLER PLUG

(a) Remove the manual transmission filler plug and gasket from the manual transmission case.



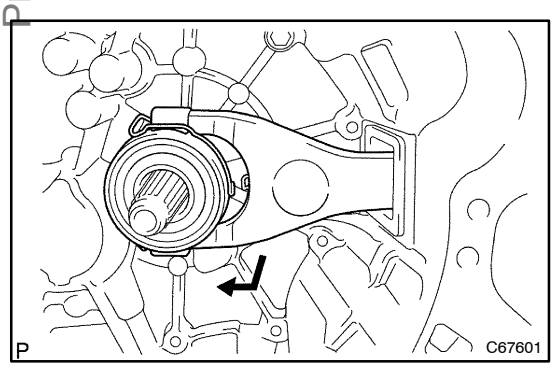
2. REMOVE DRAIN (MTM) PLUG SUB–ASSY

(a) Remove the drain plug and gasket from the manual transmission case.

3. REMOVE SPEEDOMETER DRIVEN HOLE COVER SUB–ASSY

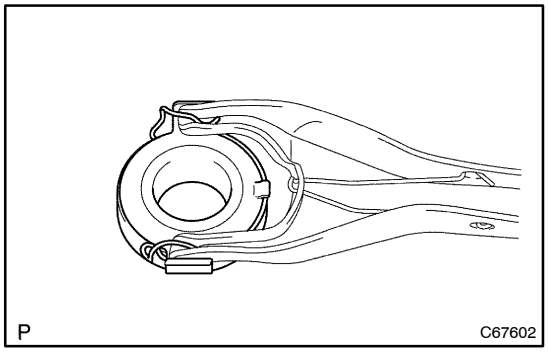
(a) Remove the bolt and speedometer driven hole cover sub–assy.

(b) Remove the O–ring from the speedometer driven hole cover sub–assy.



4. REMOVE CLUTCH RELEASE FORK SUB–ASSY

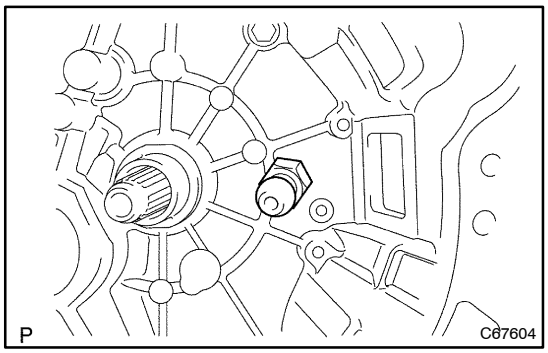
(a) Remove the clutch release fork sub–assy with clutch release bearing assy from the front transaxle case.



5. REMOVE CLUTCH RELEASE BEARING ASSY

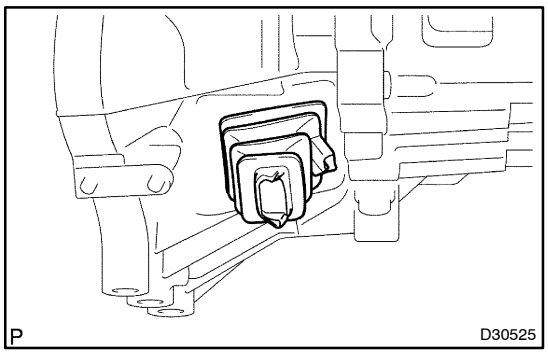
(a) Remove the clutch release bearing hub clip and clutch release bearing assy from the clutch release fork sub–assy.

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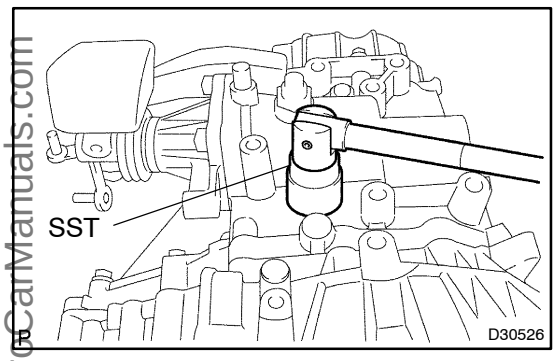
6. REMOVE RELEASE FORK SUPPORT

(a) Remove the clutch release fork support from the front transaxle case.



7. REMOVE CLUTCH RELEASE FORK BOOT

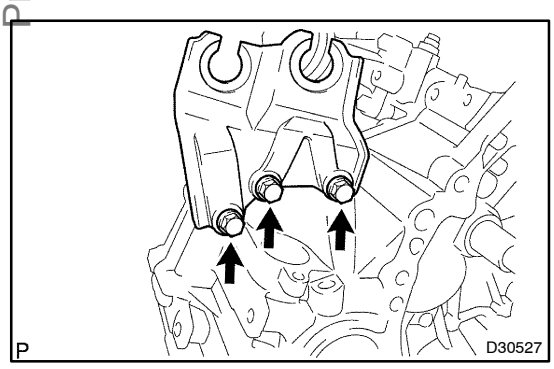
(a) Remove the clutch release fork boot from the front transaxle case.



8. REMOVE BACK UP LAMP SWITCH ASSY

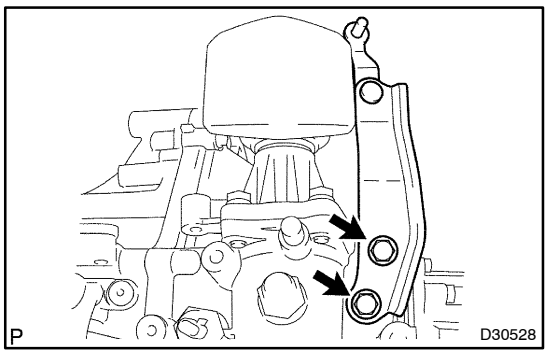
(a) Using SST, remove the back–up lamp switch assy and gasket from the manual transmission case.
SST 09817–16011

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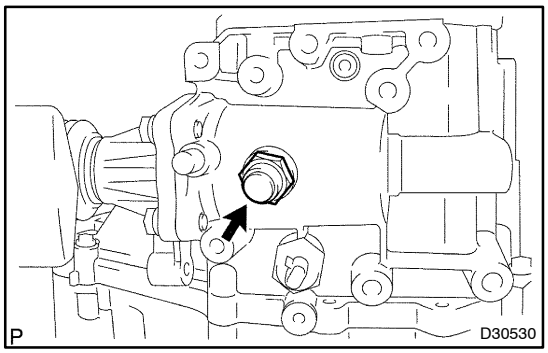
9. REMOVE FLOOR SHIFT CONTROL LEVER HOUSING SUPPORT BRACKET

(a) Remove the 3 bolts and floor shift control lever housing support bracket from the front transaxle case.



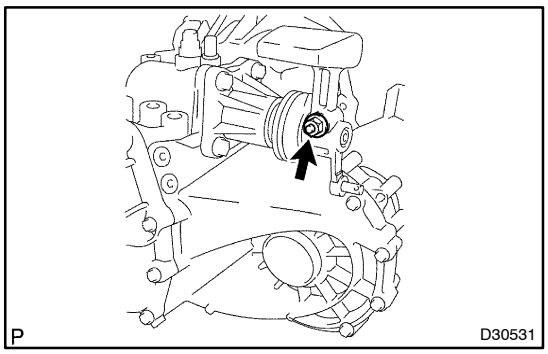
10. REMOVE SELECTING BELL CRANK ASSY

(a) Remove the 2 bolts and selecting bell crank assy from the manual transmission case.



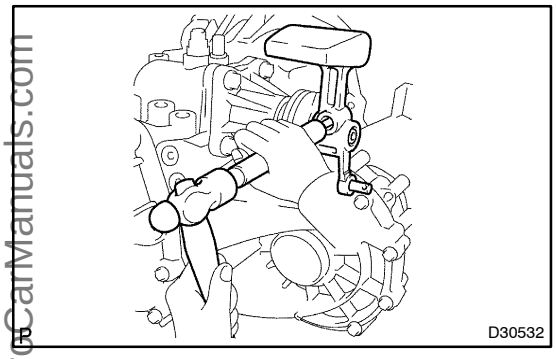
11. REMOVE LOCK BALL ASSY NO.1

(a) Remove the lock ball assy No.1 from the manual transmission case.



12. REMOVE CONTROL SHIFT LEVER

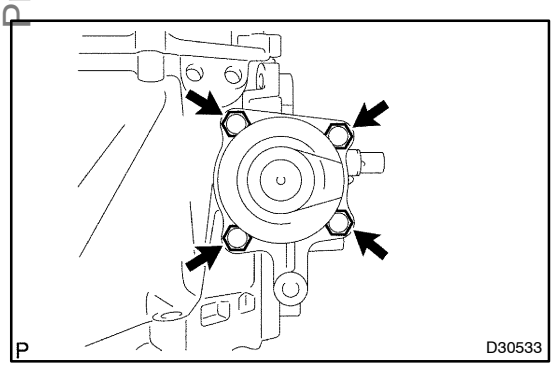
(a) Remove the nut and spring washer.



(b) Using a brass bar and a hammer, remove the shift outer lever lever lock pin.

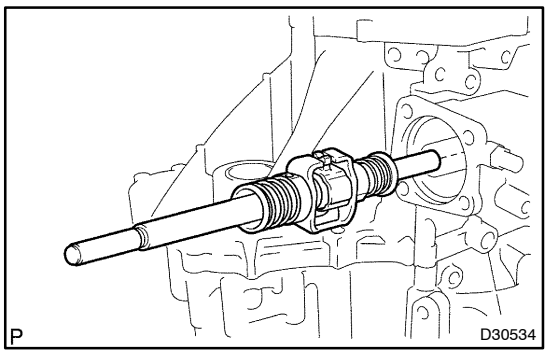
(c) Remove the control shift lever.

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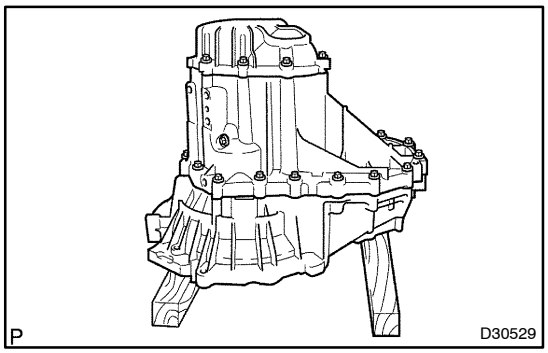
13. REMOVE CONTROL SHAFT COVER

(a) Remove the 4 bolts, control shaft cover and gasket from the manual transmission case.



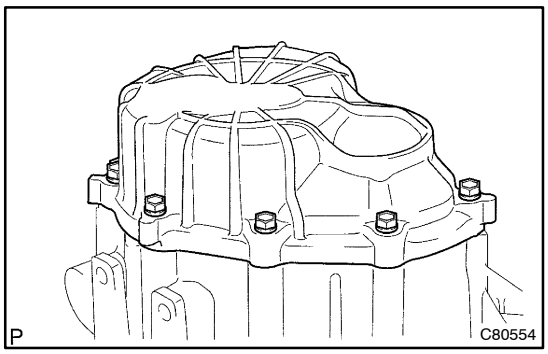
14. REMOVE SHIFT & SELECT LEVER SHAFT ASSY

(a) Remove the shift & select lever shaft assy from the manual transmission case.



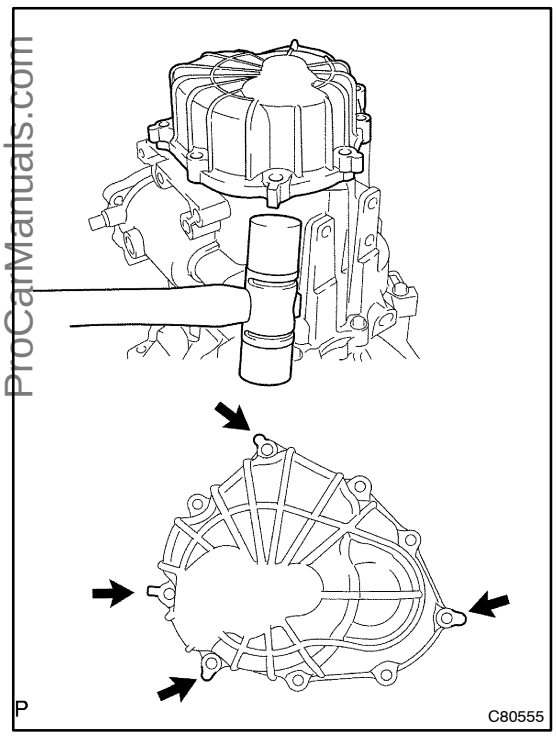
15. FIX MANUAL TRANSAXLE ASSY

(a) Using 2 wooden blocks, fix the manual transaxle assy.



16. REMOVE MANUAL TRANSMISSION CASE COVER SUB–ASSY

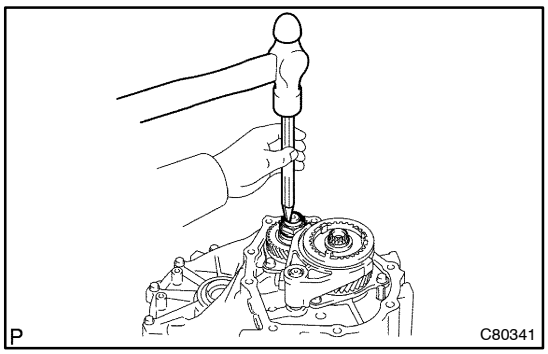
(a) Remove the 9 bolts.



(b) Gently tap the protrusion of the manual transmission case cover sub–assy with a hammer and remove it.

NOTICE:

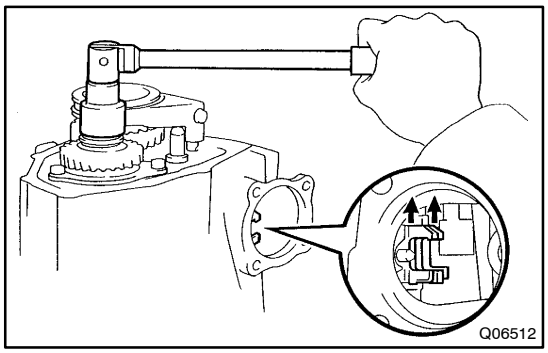
Be careful not to damage the manual transmission case sub–assy.



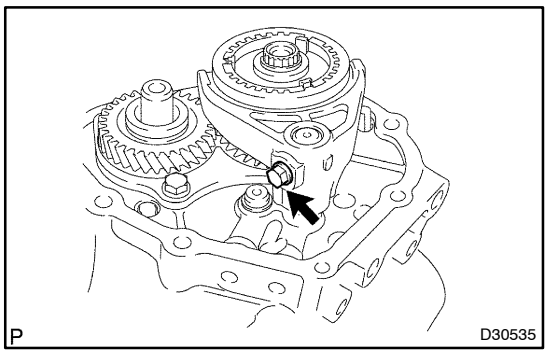
17. REMOVE MANUAL TRANSMISSION OUTPUT SHAFT REAR SET NUT

(a) Using a chisel and a hammer, loosen the staked part of the manual transmission output shaft rear set nut.

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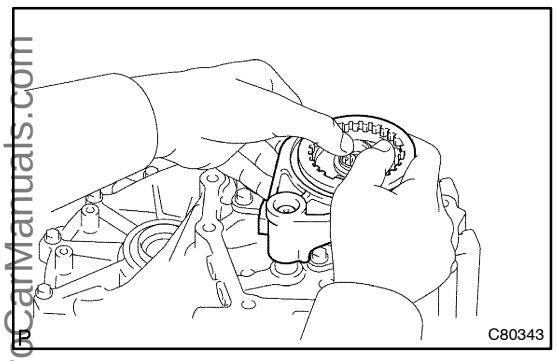


(b) Engage the gears to make them double meshed.
(c) Remove the manual transmission output shaft rear set nut.
(d) Disengage the double meshing of the gear.



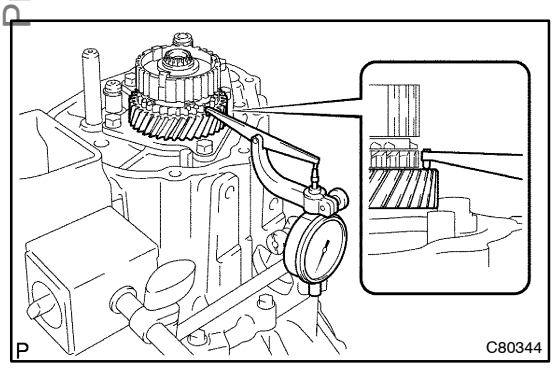
18. REMOVE GEAR SHIFT FORK NO.3

(a) Remove the gear shift fork lock bolt from the gear shift fork No.3.



(b) Remove the hub sleeve No.3 and gear shift fork No.3 from the transmission clutch hub No.3.

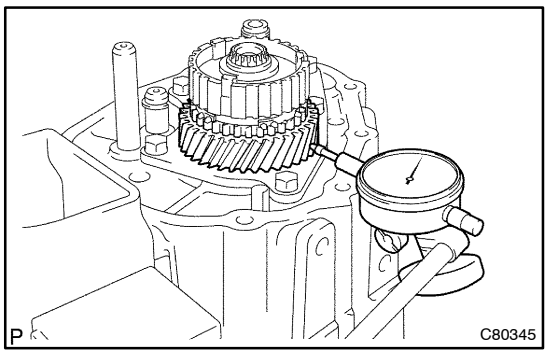
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19. INSPECT 5TH GEAR THRUST CLEARANCE

(a) Using a dial indicator, measure the thrust clearance.
Standard clearance:
0.10 to 0.57 mm (0.0039 to 0.0224 in.)
Maximum clearance:
0.57 mm (0.0224 in.)

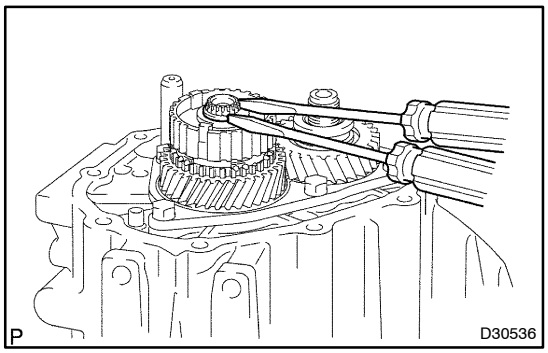
If the clearance exceeds the maximum, replace the transmission clutch hub No.3, 5th gear or input shaft rear radial ball bearing.



20. INSPECT 5TH GEAR RADIAL CLEARANCE

(a) Using a dial indicator, measure the radial clearance.
Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 5the gear, 5the gear needle roller bearing or input shaft.

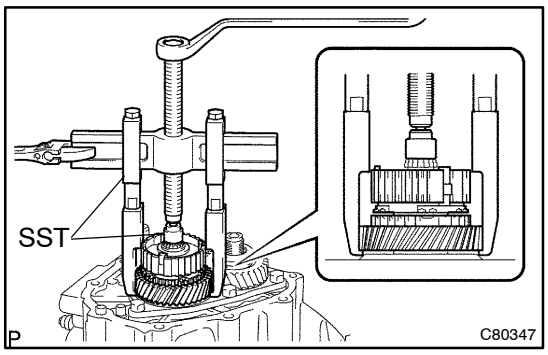


21. REMOVE TRANSMISSION CLUTCH HUB NO.3

(a) Using 2 screwdrivers and a hammer, tap out the snap ring.

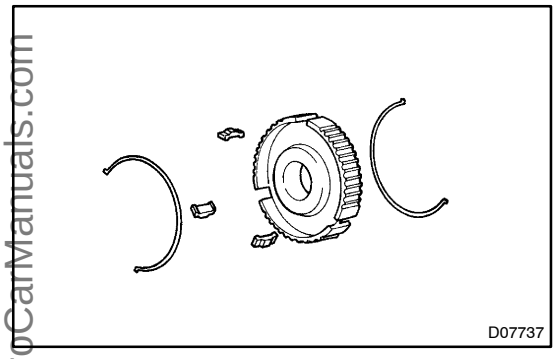
HINT:

Use a shop rag to keep the snap ring from flying.



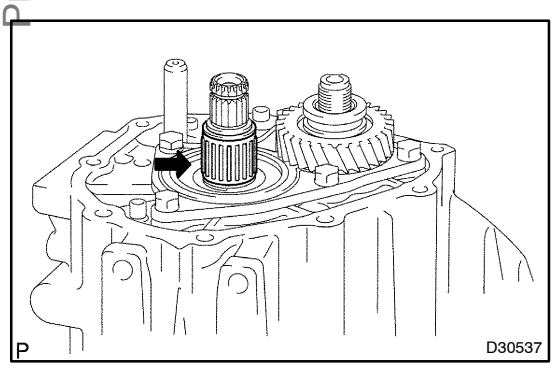
(b) Using SST, remove the transmission clutch hub No.3, 5th gear and synchronizer ring No.3 from the input shaft.

SST 09950–40011 (09951–04020, 09952–04010, 09953–04030, 09954–04010, 09955–04071, 09957–04010), 09950–60010 (09951–00200)



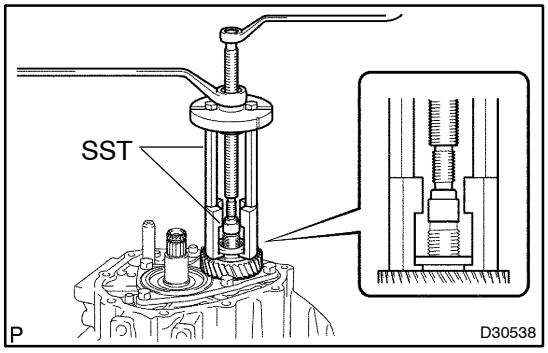
(c) Remove the 3 synchromesh shifting keys and 2 synchromesh shifting key springs from the transmission clutch hub No.3.

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22. REMOVE 5TH GEAR NEEDLE ROLLER BEARING

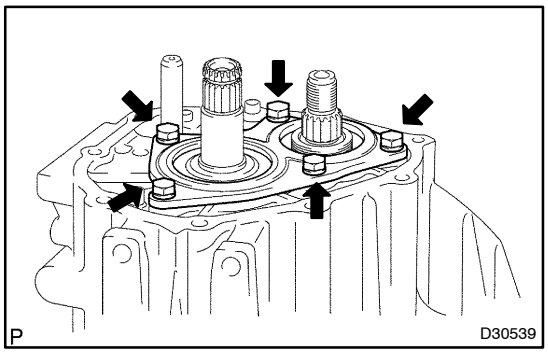
(a) Remove the 5th gear needle roller bearing and 5th gear bearing spacer from the input shaft.



23. REMOVE 5TH DRIVEN GEAR

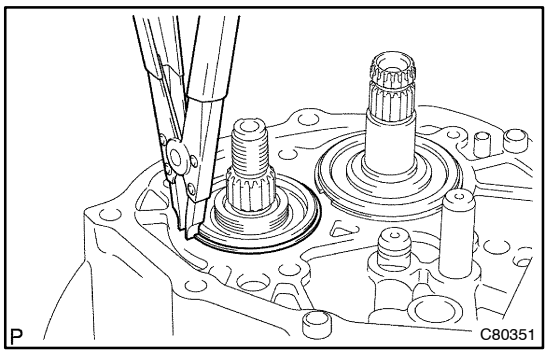
(a) Using SST, remove the 5th driven gear from the output shaft.

SST 09950–30012 (09951–03010, 09953–03010, 09954–03010, 09955–03011, 09957–04010), 09950–60010 (09951–00190)



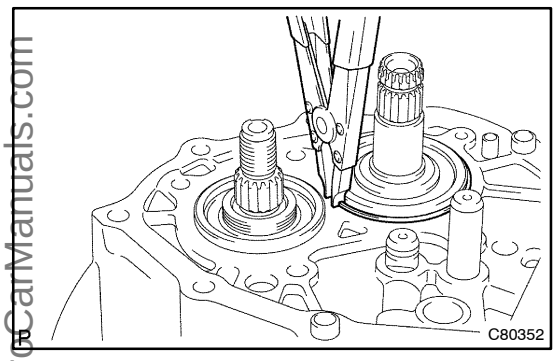
24. REMOVE BEARING RETAINER REAR (MTM)

(a) Remove the 5 bolts and bearing retainer rear from the manual transmission case.



25. REMOVE OUTPUT SHAFT REAR BEARING HOLE SNAP RING

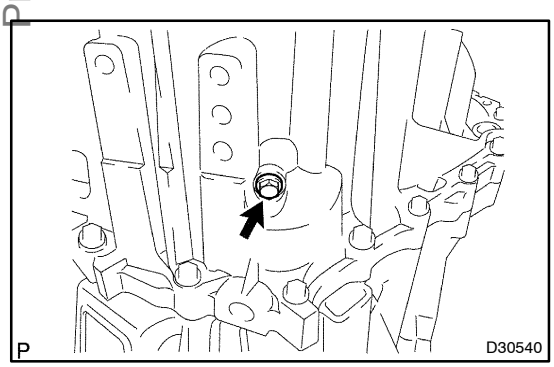
(a) Using a snap ring expander, remove the output shaft rear bearing hole snap ring from the output shaft.



26. REMOVE INPUT SHAFT REAR BEARING HOLE SNAP RING

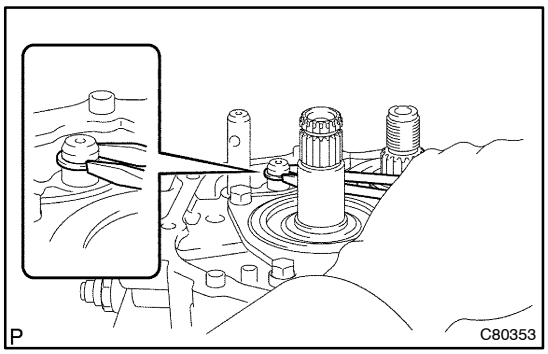
(a) Using a snap ring expander, remove the input shaft rear bearing hole snap ring from the input shaft.

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27. REMOVE REVERSE IDLER GEAR SHAFT BOLT

(a) Remove the reverse idler gear shaft bolt and reverse idler gear shaft gasket from the manual transmission case.

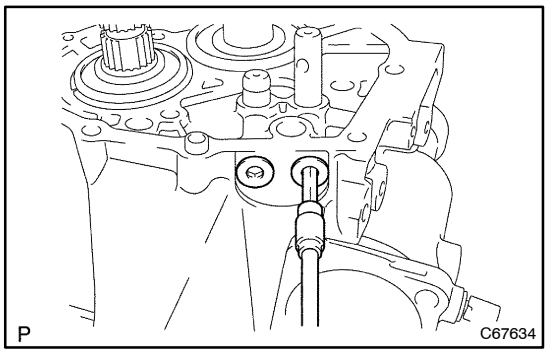


28. REMOVE SHIFT FORK SHAFT SHAFT SNAP RING

(a) Using 2 screwdrivers and a hammer, tap out the snap ring.

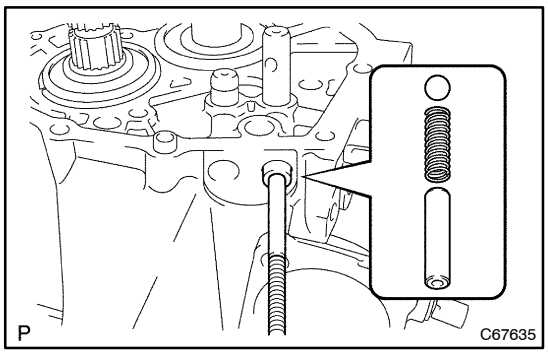
HINT:

Use a shop rag to keep the shifting key spring from flying.

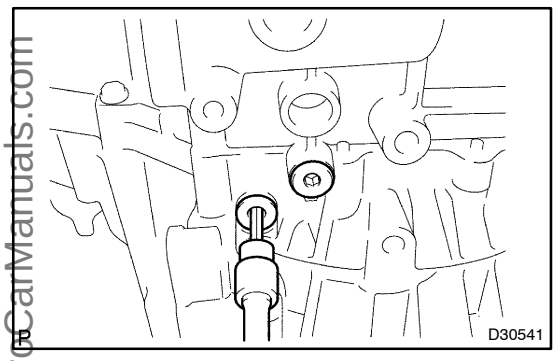


29. REMOVE SHIFT DETENT BALL

(a) Using a hexagon wrench, remove the 2 shift detent ball plugs from the manual transmission case.

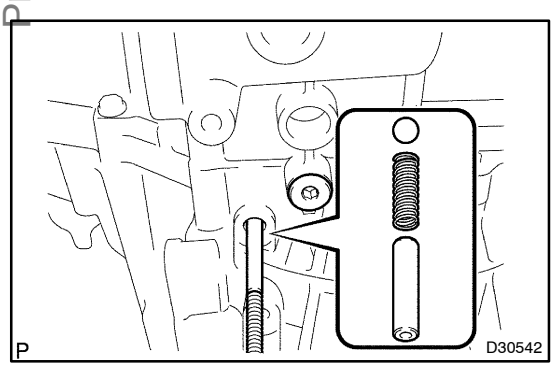


(b) Using a magnetic finger, remove the 2 seats, springs and balls from the manual transmission case.

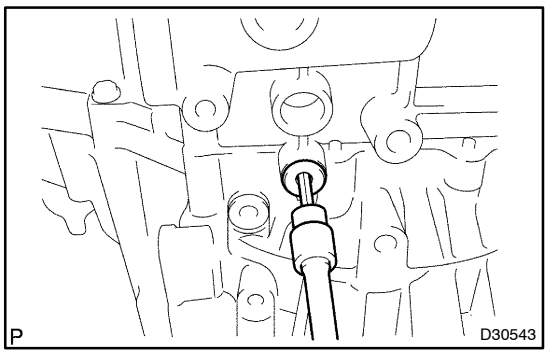


(c) Using a hexagon wrench, remove the shift detent ball plug from the front transaxle case.

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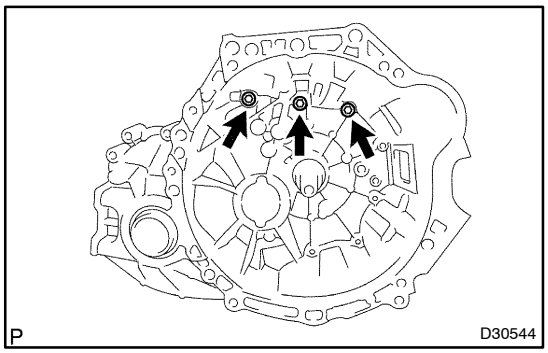


(d) Using a magnetic finger, remove the seat, spring and ball from the front transaxle case.



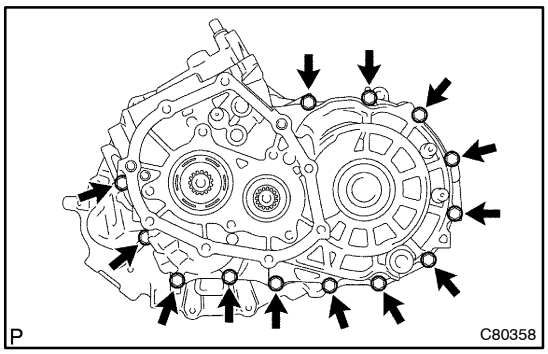
30. REMOVE LOCK BALL ASSY NO.2

(a) Using a hexagon wrench, remove the lock ball assy No.2 from the manual transmission case.

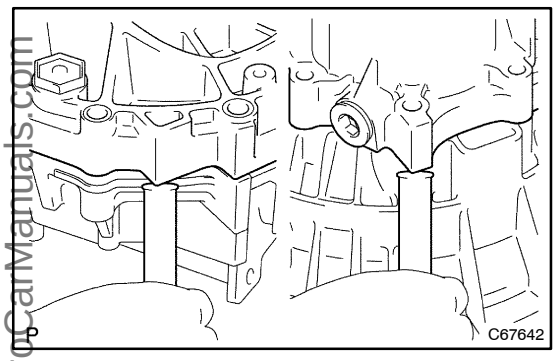


31. REMOVE MANUAL TRANSMISSION CASE

(a) Remove the 3 bolts from the front transaxle case.



(b) Remove the 13 bolts from the manual transmission case.



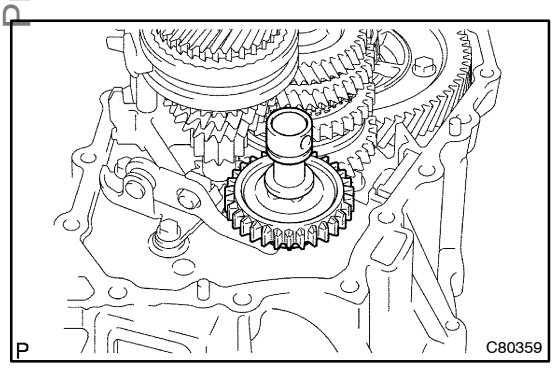
(c) Gently tap the transmission case with a brass bar and hammer and remove it from the transaxle case.

NOTICE:

Do not damage the manual transmission case and transaxle case.

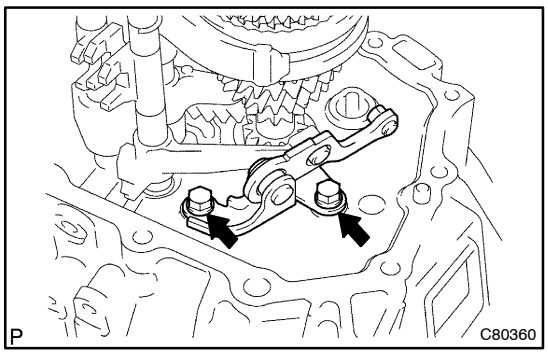
(d) Remove the 2 manual transmission case straight pins from the front transaxle case.

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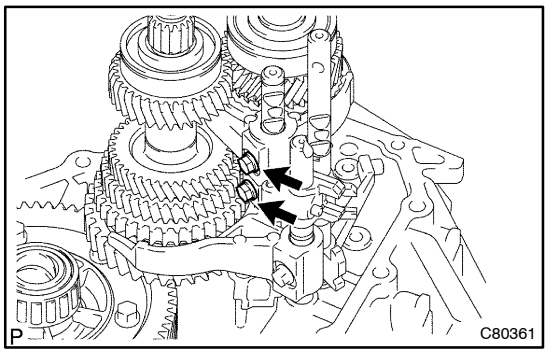
32. REMOVE REVERSE IDLER GEAR SUB–ASSY

(a) Remove the reverse idler gear, thrust washer and reverse idler gear shaft from the front transaxle case.



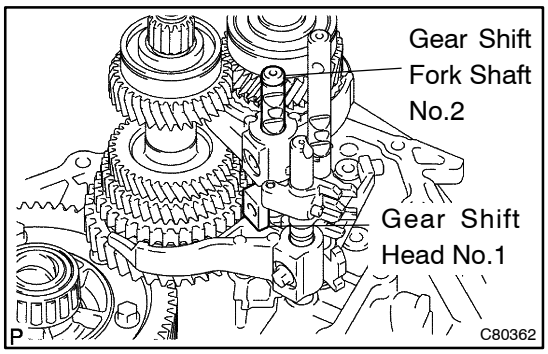
33. REMOVE REVERSE SHIFT ARM BRACKET ASSY

(a) Remove the 2 bolts and reverse shift arm bracket assy from the front transaxle case.

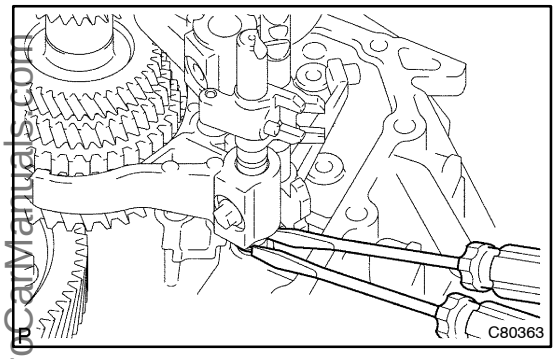


34. REMOVE GEAR SHIFT FORK SHAFT NO.2

(a) Remove the 2 bolts from the gear shift fork No.2 and gear shift head No.1.



(b) Remove the gear shift fork shaft No.2 and gear shift head No.1 from the front transaxle case.

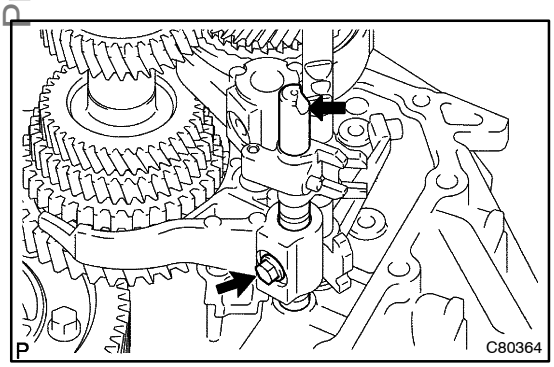


35. REMOVE GEAR SHIFT FORK SHAFT NO.1

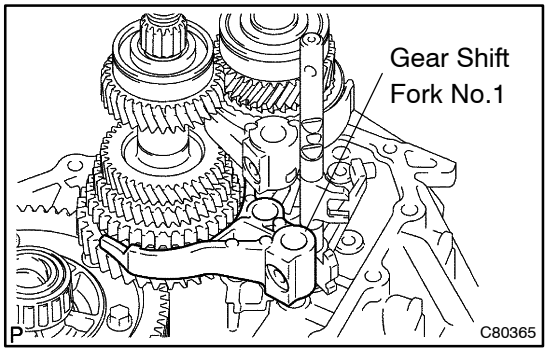
(a) Using 2 screwdrivers and a hammer, tap out the snap rings.

HINT:

Use a shop rag to keep the snap ring from flying.

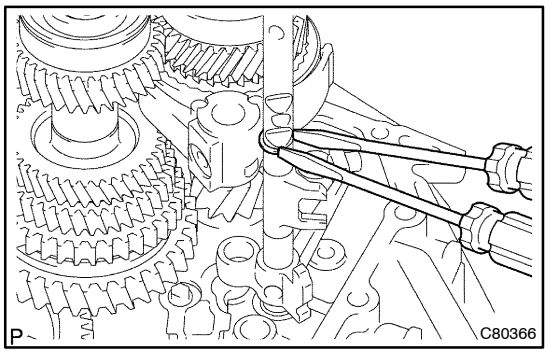


(b) Remove the shift fork set bolt and gear shift fork shaft No.1 from the gear shift fork No.1.



(c) Remove the shift fork No.1.

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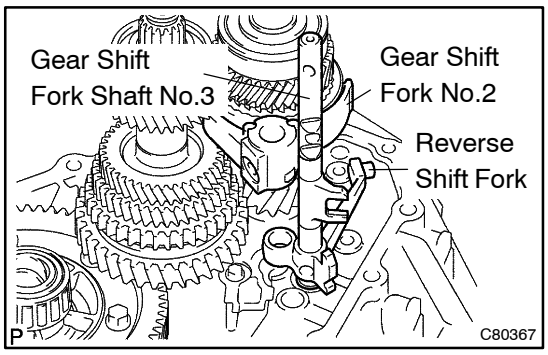


36. REMOVE GEAR SHIFT FORK SHAFT NO.3

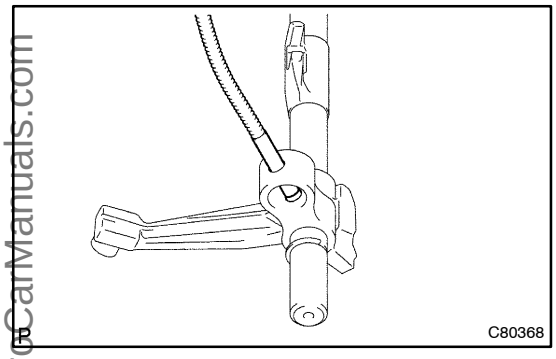
(a) Using 2 screwdrivers and a hammer, tap out the snap ring from the gear shift fork shaft No.3.

HINT:

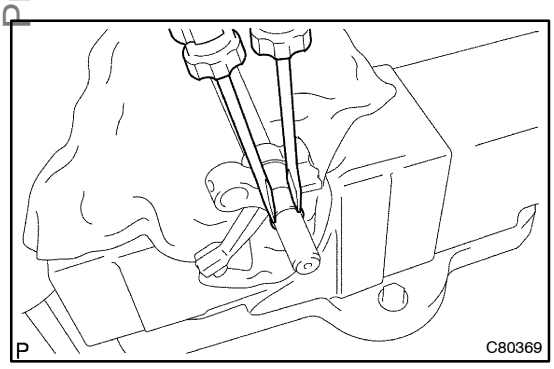
Use a shop rag to keep the snap ring from flying.



(b) Remove the shift fork shaft No.3 with reverse shift fork and gear shift fork No.2 from the manual transaxle case.



(c) Using a magnetic finger, remove the 2 reverse shift fork balls from the reverse shift fork.

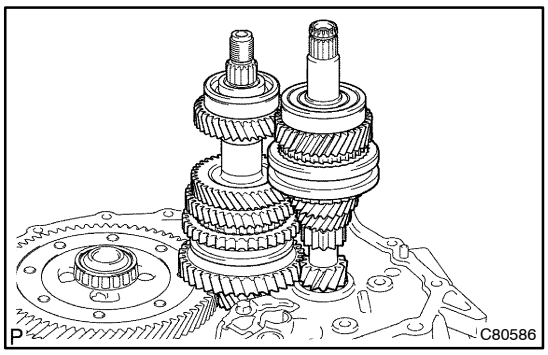


(d) Using 2 screwdrivers and a hammer, tap out the snap ring from the shift fork shaft No.3.

HINT:

Use a shop rag to keep the snap ring from flying.

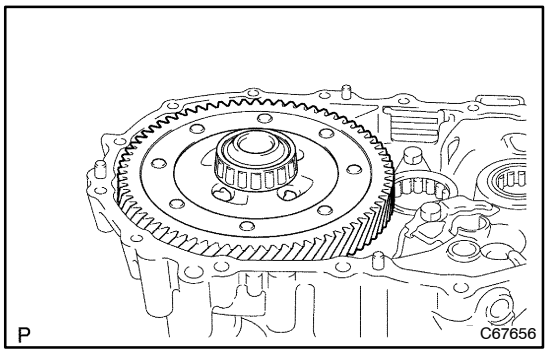
(e) Remove the reverse shift fork from the gear shift fork shaft No.3.



37. REMOVE INPUT SHAFT ASSY

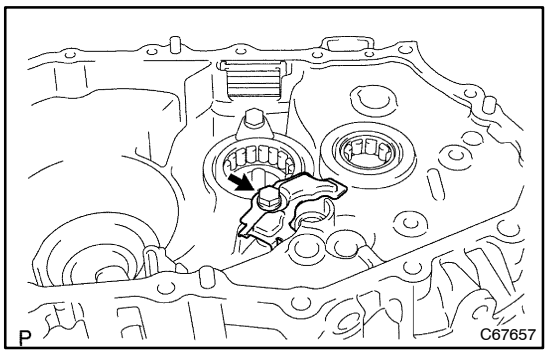
(a) Remove the input shaft assy and output shaft assy from the front transaxle case.

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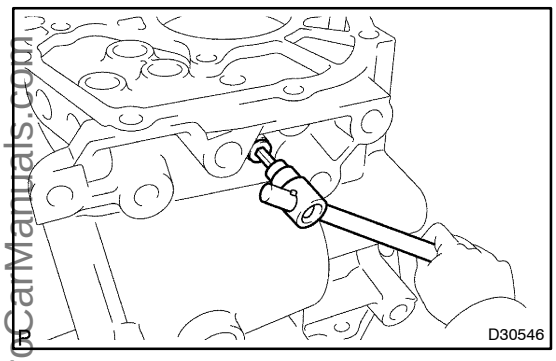
38. REMOVE DIFFERENTIAL CASE ASSY

(a) Remove the differential case assy from the front transaxle case.



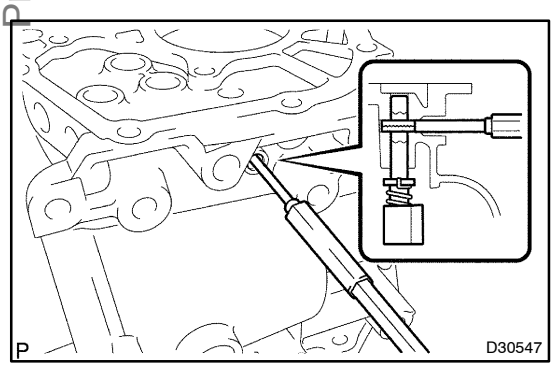
39. REMOVE MANUAL TRANSAXLE CASE RECEIVER

(a) Remove the bolt and manual transaxle case receiver from the front transaxle case.

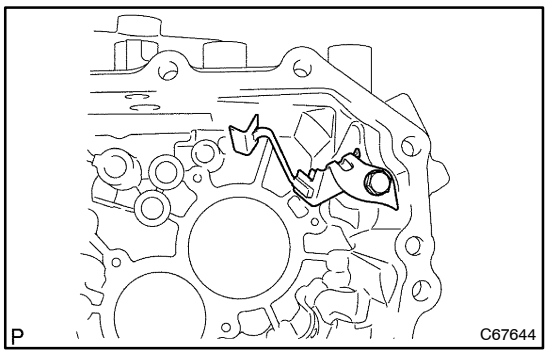


40. REMOVE REVERSE RESTRICT PIN ASSY

(a) Using a hexagon wrench, remove the reverse restrict pin plug from the manual transmission case.



(b) Using a pin punch (ϕ 5 mm) and a hammer, drive out the slotted spring pin and reverse restrict pin assy from the manual transmission case.



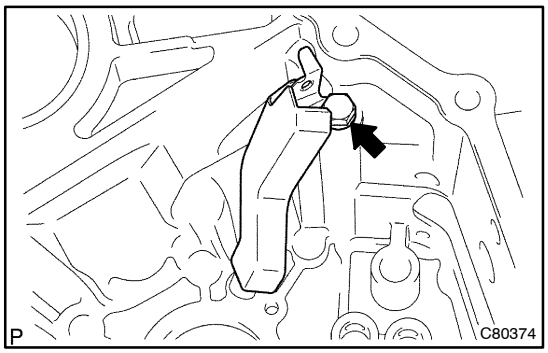
41. REMOVE OIL RECEIVER PIPE NO.1 (MTM)

(a) Remove the bolt and oil receiver pipe No.1 from the manual transmission case.

NOTICE:

Be careful not to damage the oil receiver pipe No.1.

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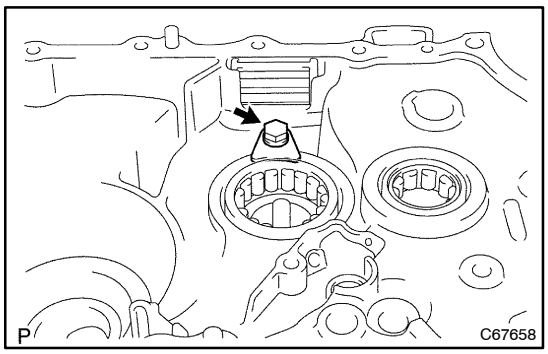


42. REMOVE OIL RECEIVER PIPE NO.2 (MTM)

(a) Remove the bolt and oil receiver pipe No.2 from the manual transmission case.

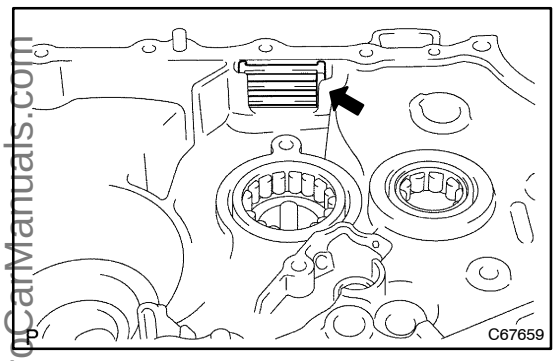
NOTICE:

Be careful not to damage the oil receiver pipe No.2.



43. REMOVE BEARING LOCK PLATE

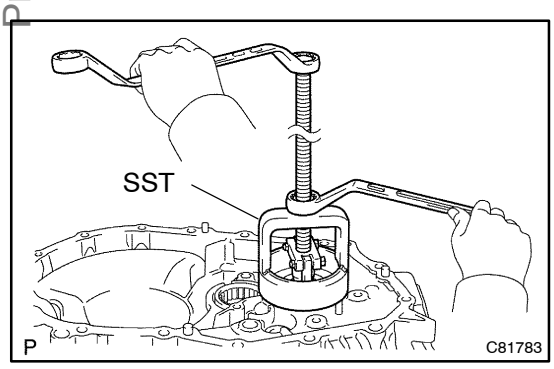
(a) Remove the bolt and bearing lock plate from the front transaxle case.



44. REMOVE TRANSMISSION MAGNET

(a) Remove the transmission magnet from the front transaxle case.

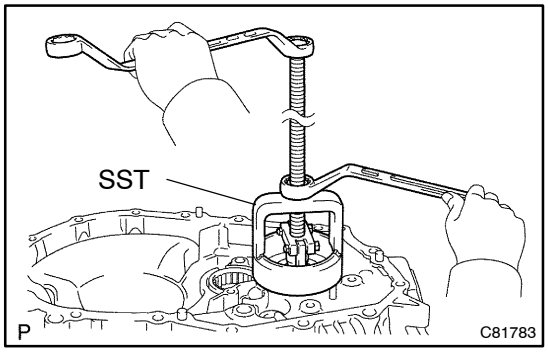
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45. REMOVE INPUT SHAFT FRONT BEARING

(a) Using SST, remove the input shaft front bearing from the front transaxle case.

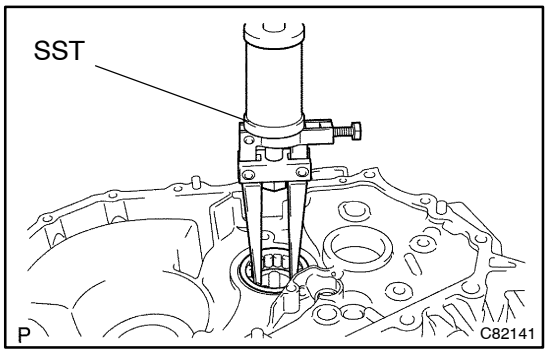
SST 09612–65014 (09612–01050, 09612–01060)



46. REMOVE FRONT TRANSAXLE CASE OIL SEAL

(a) Using SST, remove the front transaxle case oil seal from the front transaxle case.

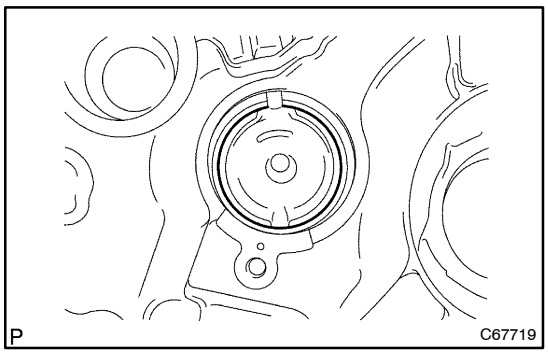
SST 09612–65014 (09612–01040, 09612–01050)



47. REMOVE OUTPUT SHAFT FRONT BEARING

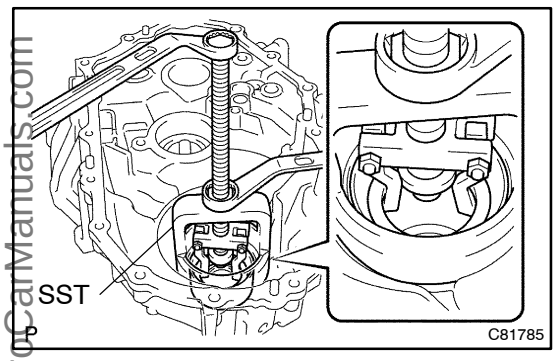
(a) Using SST, remove the output shaft front bearing from the front transaxle case.
SST 09308–00010

NOTICE:
Do not apply excessive force to the front transaxle case when installing it.



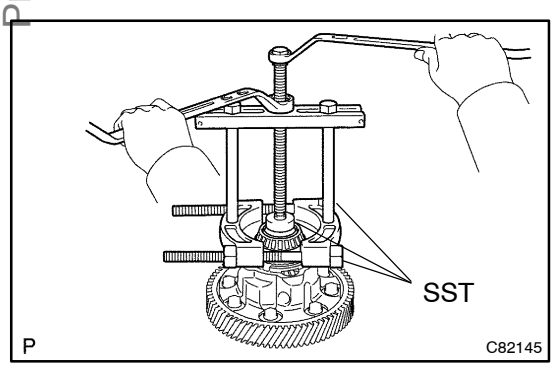
48. REMOVE OUTPUT SHAFT (MTM) COVER

(a) Remove the output shaft cover from the front transaxle case.



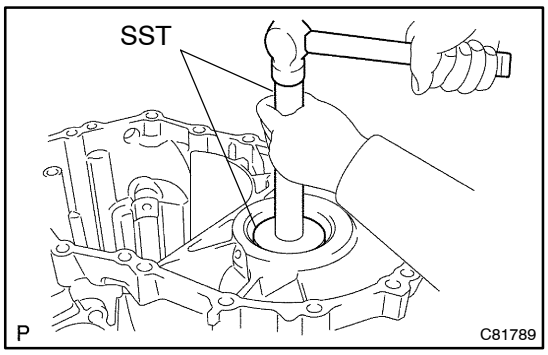
49. REMOVE FR DIFFERENTIAL CASE FRONT TAPERED ROLLER BEARING

(a) Using SST, remove the front differential case front tapered roller bearing (outer race) and plate washer from the front transaxle case.
SST 09612–65014 (09612–01040, 09612–01050),



(b) Using SST, remove the front differential case front tapered roller bearing (inner race) from the front differential case.
SST 09950–00020, 09950–00030, 09950–40011 (09957–04010), 09950–60010 (09951–00360)

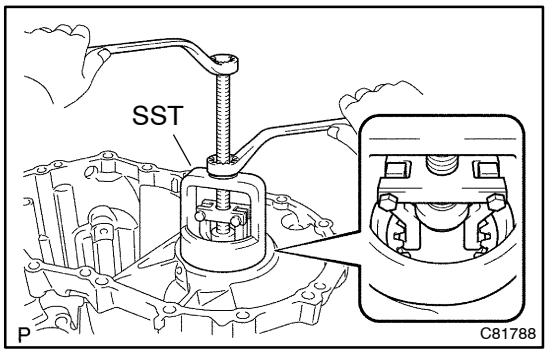
NOTICE:
Be careful not to damage the bearing.



50. REMOVE TRANSAXLE CASE OIL SEAL

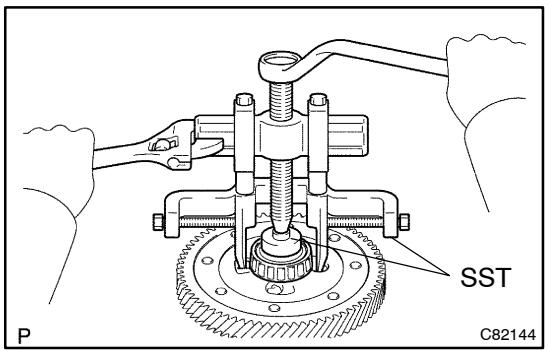
(a) Using SST and a hammer, remove the transaxle case oil seal from the front transaxle case.
SST 09950–60010 (09951–00530), 09950–70010 (09951–07150)

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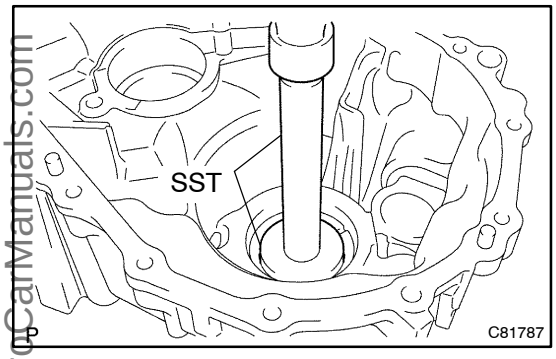
51. REMOVE FR DIFFERENTIAL CASE REAR TAPERED ROLLER BEARING

(a) Using SST, remove the front differential case rear tapered roller bearing (outer race) and plate washer from the transmission case.
SST 09612–65014 (09612–01040, 09612–01050)



(b) Using SST and hammer, remove the front differential case rear tapered roller bearing (inner race) from the front differential case.
SST 09950–40011 (09951–04010, 09952–04010, 09953–04020, 09954–04010, 09955–04061, 09957–04010, 09958–04011), 09950–60010 (09951–00360)

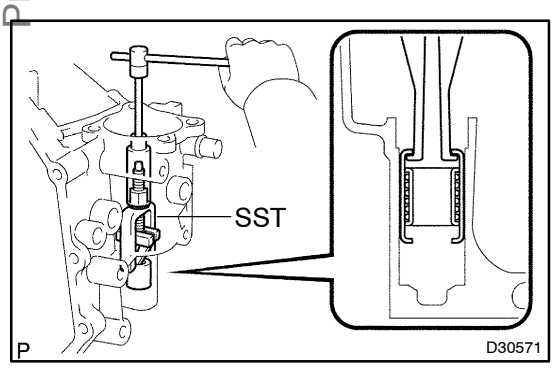
NOTICE:
Be careful not to damage bearing.



52. REMOVE TRANSMISSION CASE OIL SEAL

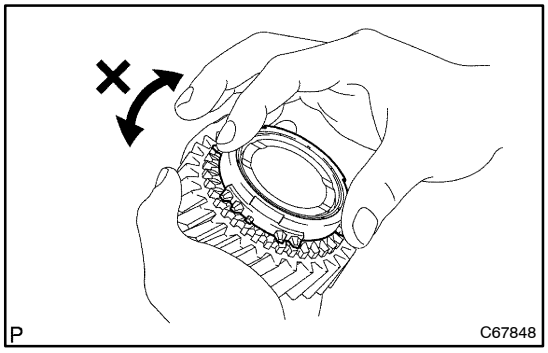
(a) Using SST and a hammer, drive out the transmission case oil seal from the manual transmission case.
SST 09950–60010 (09951–00600), 09950–70010 (09951–07100)

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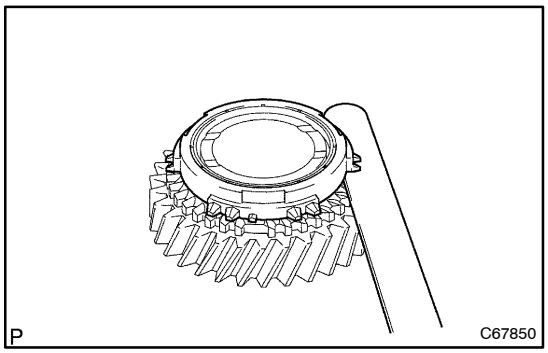
53. REMOVE SHIFT & SELECT LEVER SHAFT NEEDLE ROLLER BEARING

(a) Using SST, remove the shift & select lever shaft needle roller bearing.
SST 09319–60020



54. INSPECT SYNCHRONIZER RING NO.3

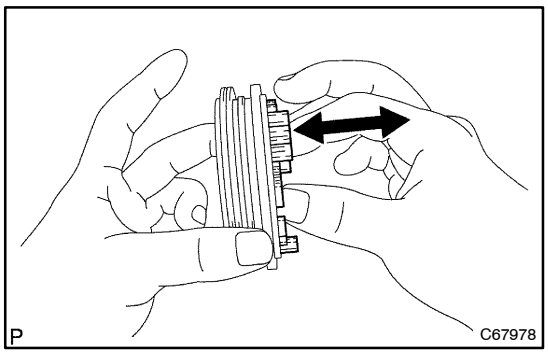
(a) Check for any wear or damage.
(b) Apply gear oil to the cone of the 5th gear and check with the synchronizer ring No.3 contacted by pressure that the gear does not turn in the direction as shown in the illustration.



(c) Using a feeler gauge, check the clearance between the synchronizer ring No.3 and 5th gear with the synchronizer ring No.3 contacted to the cone of the 5th gear.

Standard clearance:
0.75 to 1.65 mm (0.0295 to 0.0649 in.)
Minimum clearance: 0.75 mm (0.0295 in.)

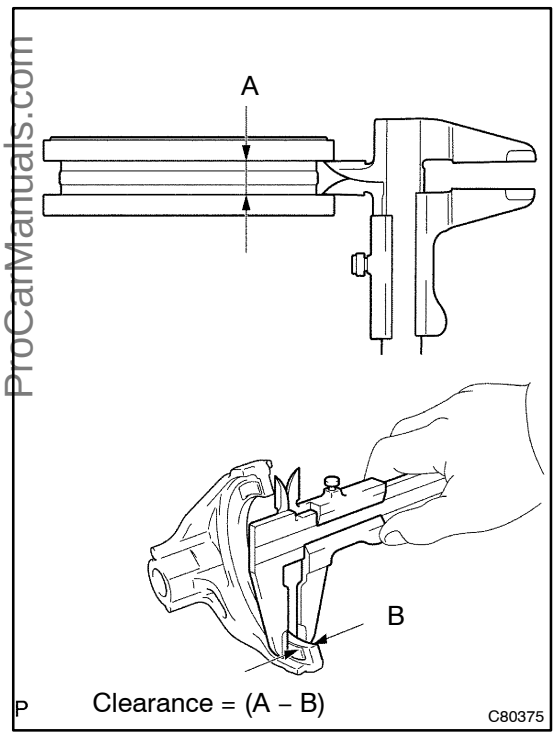
If the clearance is less than the minimum, replace the synchronizer ring No.3.



55. INSPECT TRANSMISSION HUB SLEEVE NO.3

(a) Check the sliding condition between the transmission hub sleeve No.3 and transmission clutch hub No.3.

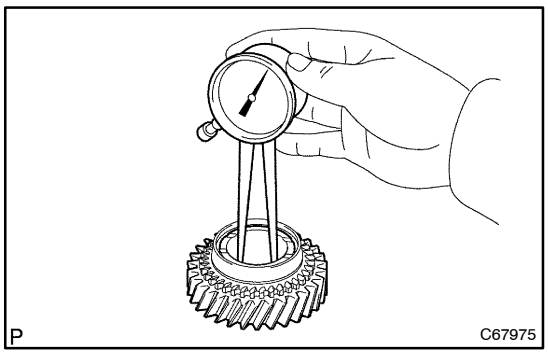
(b) Check that the spline gear's edges of the transmission hub sleeve No.3 are not worn down.



(c) Using vernier calipers, measure the clearance between the transmission hub sleeve No.3 and gear shift fork No.3.

Standard clearance:
0.3 to 0.5 mm (0.0012 to 0.0020 in.)

If the clearance exceeds the standard clearance, replace the transmission hub sleeve No.3 and gear shift fork No.3.



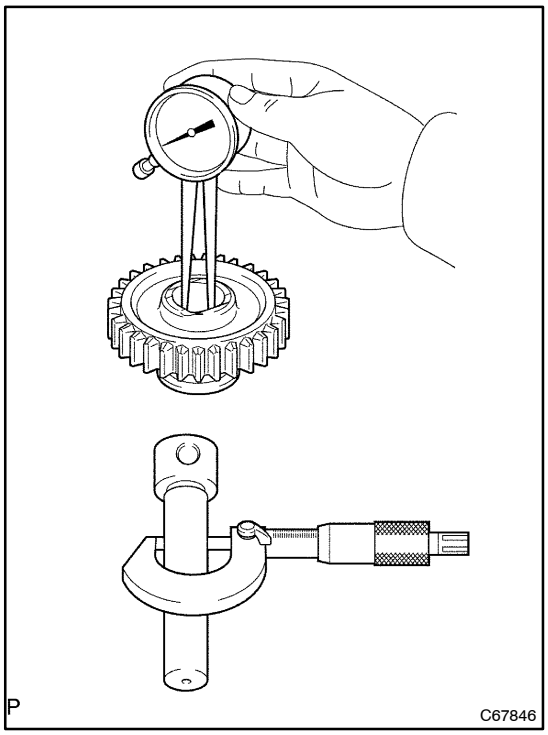
56. INSPECT 5TH GEAR

(a) Using a caliper gauge, inspect 5th gear as shown in the illustration.

Standard diameter:
29.915 to 29.931 mm (1.1778 to 1.1783 in.)

If the diameter exceeds the standard diameter, replace the 5th gear.

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57. INSPECT REVERSE IDLER GEAR SUB–ASSY

(a) Using a caliper gauge, inspect the reverse idler gear sub–assy as shown in the illustration.

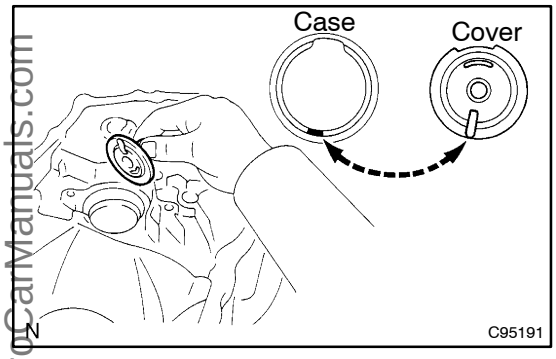
Standard diameter:
18.040 to 18.058 mm (0.7102 to 0.7109 in.)
Maximum diameter:
18.058 mm (0.7109 in.)

If the diameter exceeds the maximum, replace the reverse idler gear sub–assy.

(b) Using a micrometer, inspect the reverse idler gear shaft as shown in the illustration.

Standard outer diameter:
17.966 to 17.984 mm (0.7073 to 0.7080 in.)
Minimum outer diameter:
17.966 mm (0.7073 in.)

If the outer diameter is less than the minimum, replace the reverse idler gear shaft.

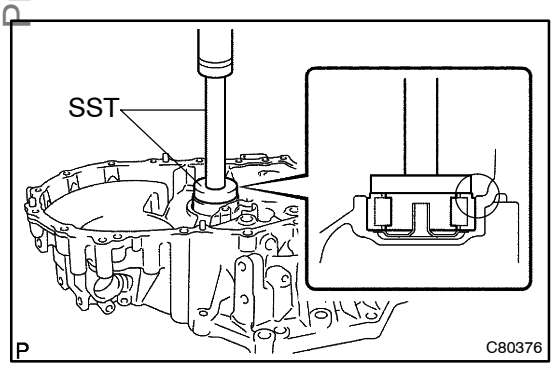


58. INSTALL OUTPUT SHAFT (MTM) COVER

(a) Coat the output shaft cover with MP grease and install it to the front transaxle case.

NOTICE:
Install the output shaft cover projection into the case side hollow.

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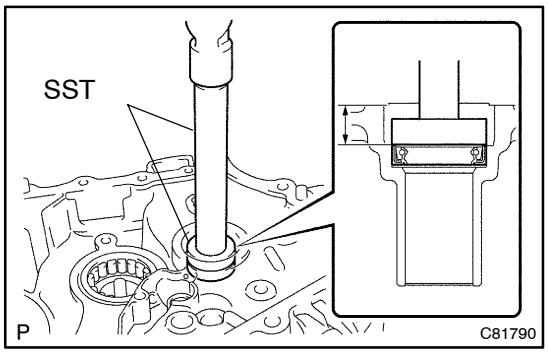
59. INSTALL OUTPUT SHAFT FRONT BEARING

(a) Coat a new output shaft front bearing with gear oil, using SST and a press, install it.

SST 09950–60010 (09951–00550), 09950–70010 (09951–07150)

NOTICE:

- **Be sure to install a new bearing in the correct direction, as shown in the illustration.**
- **When replacing the output shaft front bearing, replace the output shaft front bearing inner race along with it.**



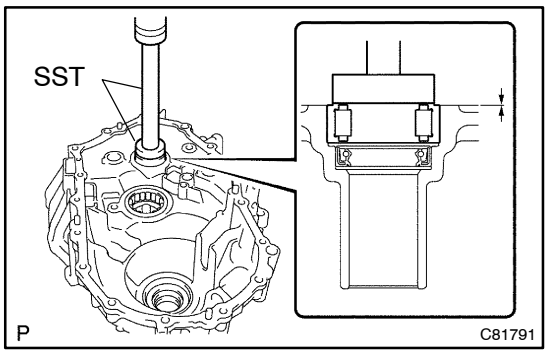
60. INSTALL FRONT TRANSAXLE CASE OIL SEAL

(a) Using SST and a hammer, install a new front transaxle case oil seal to the front transaxle case.

SST 09950–60010 (09951–00370), 09950–70010 (09951–07150)

Drive in depth: 15.6 to 16.0 mm (0.6141 to 0.6299 in.)

(b) Coat the lip of the front transaxle case oil seal with MP grease.

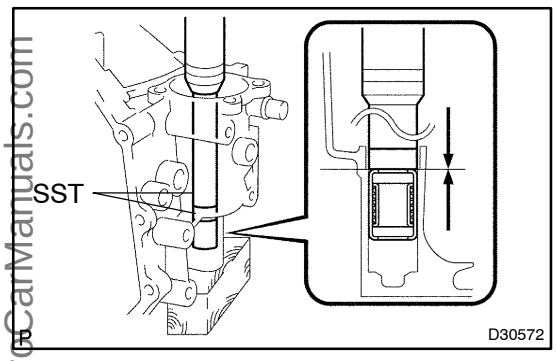


61. INSTALL INPUT SHAFT FRONT BEARING

(a) Coat a new input shaft front bearing with MP grease. Using SST and a press, install it to the front transaxle case.

SST 09950–60010 (09951–00420), 09950–70010 (09951–07150)

Drive in depth: 0 to 0.3 mm (0 to 0.0118 in.)



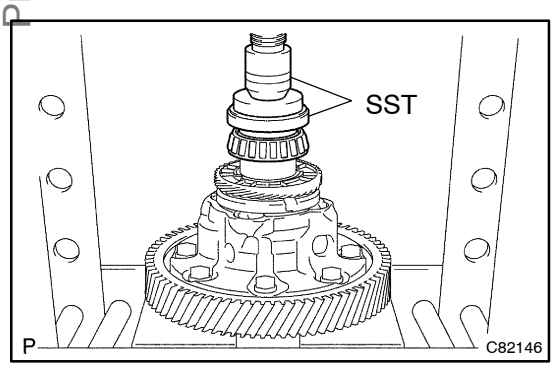
62. INSTALL SHIFT & SELECT LEVER SHAFT NEEDLE ROLLER BEARING

(a) Using SST, install the shift & select lever shaft needle roller bearing.

SST 09950–60010 (09951–00220), 09950–70010 (09951–07100)

Drive in depth: 0 to 0.5 mm (0 to 00020 in.)

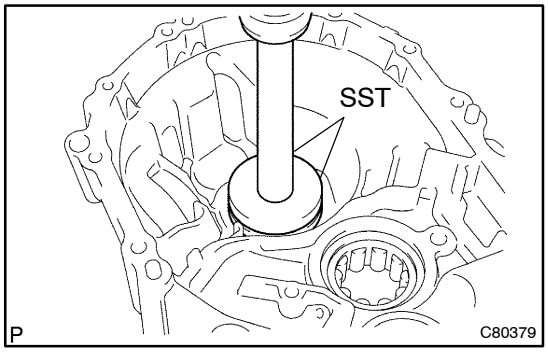
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63. INSTALL FR DIFFERENTIAL CASE FRONT TAPERED ROLLER BEARING

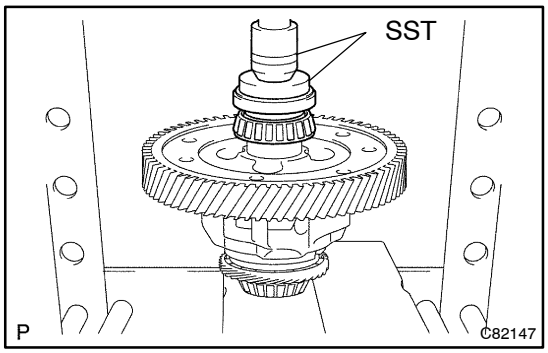
(a) Using SST and a press, install the front differential case front tapered roller bearing (inner race) to the front differential case.

SST 09350–32014 (09351–32120), 09950–60010 (09951–00530)



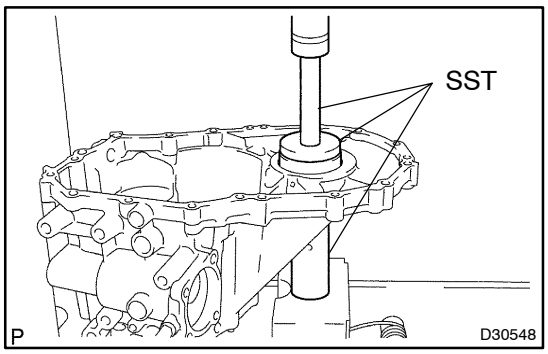
(b) Using SST and a press, install the front differential case front tapered roller bearing (outer race) with plate washer to the front transaxle case.

SST 09950–60020 (09951–00680), 09950–70010 (09951–07150)



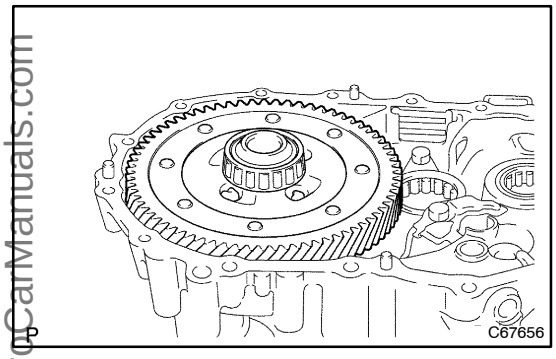
64. INSTALL FR DIFFERENTIAL CASE REAR TAPERED ROLLER BEARING

(a) Using SST and a press, install the front differential case rear tapered roller bearing (inner race) to the front differential case.
SST 09350–32014 (09351–32120), 09950–60010 (09951–00530)



(b) Using SST and a press, install the front differential case rear tapered roller bearing (outer race) with shim to the transmission case.
SST 09309–36010, 09950–60020 (09951–00710), 09950–70010 (09951–07150)

HINT:
Use a front differential case washer rear of the same thickness as the removed one.

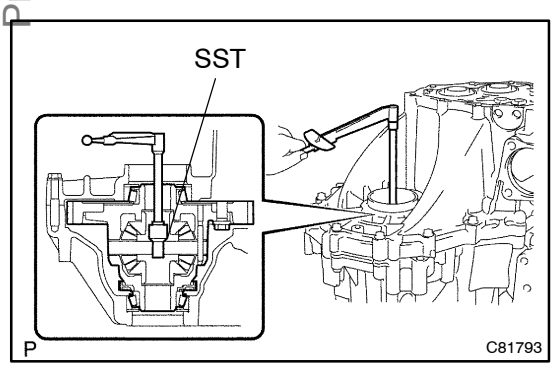


65. ADJUST DIFFERENTIAL SIDE BEARING PRELOAD

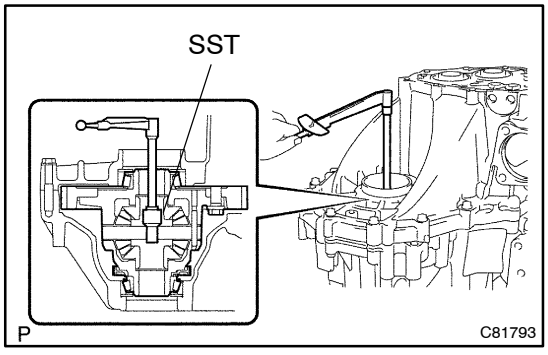
(a) Coat the differential case assy with gear oil, and install it to the front transaxle case.

(b) Install the manual transmission case with 16 bolts.
Torque: 29 N·m (300 kgf·cm, 22 ft·lbf)

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(c) Using SST and a torque wrench, turn the differential case assy to the right and left 2 or 3 times to allow the bearings to settle.
SST 09564–32011



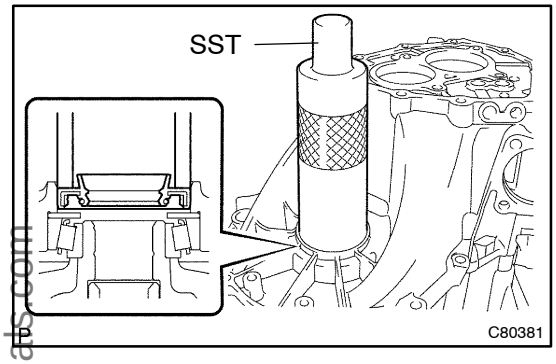
(d) Using SST and a torque wrench, measure the preload.
SST 09564–32011
Preload (at starting):
New bearing:
0.78 to 1.57 N·m (7.95 to 16.0 kgf·cm, 6.9 to 13.89 in.·lbf)
Reuse bearing:
0.49 to 0.98 N·m (5.0 to 10.0 kgf·cm, 4.34 to 8.67 in.·lbf)

(e) Select another plate washer.

Mark	Thickness mm (in.)	Mark	Thickness mm (in.)
01	1.10 (0.0433)	11	1.60 (0.0630)
02	1.15 (0.0453)	12	1.65 (0.0650)
03	1.20 (0.0472)	13	1.70 (0.1060)
04	1.25 (0.0492)	14	1.75 (0.0689)
05	1.30 (0.0512)	15	1.80 (0.0709)
06	1.35 (0.0531)	16	1.85 (0.0728)
07	1.40 (0.0551)	17	1.90 (0.0748)
08	1.45 (0.0571)	18	1.95 (0.0768)
09	1.50 (0.0591)	19	2.00 (0.0787)
10	1.55 (0.0610)	–	–

(f) Remove the 16 bolts and manual transmission case from the transaxle case.

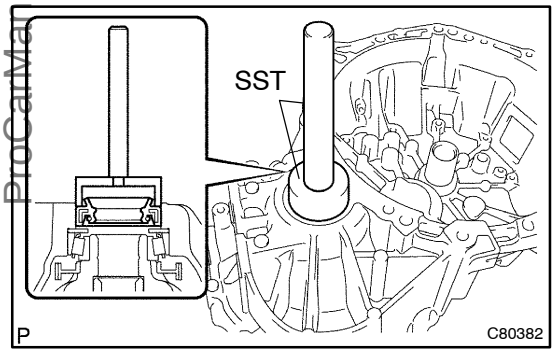
(g) Remove the differential case assy from the transaxle case.



66. INSTALL TRANSMISSION CASE OIL SEAL

(a) Using SST and a hammer, install a new transmission case oil seal to the manual transmission case.
SST 09316–60011 (09316–00011)
Drive in depth: 9.6 to 10.2 mm (0.3779 to 0.4015 in.)

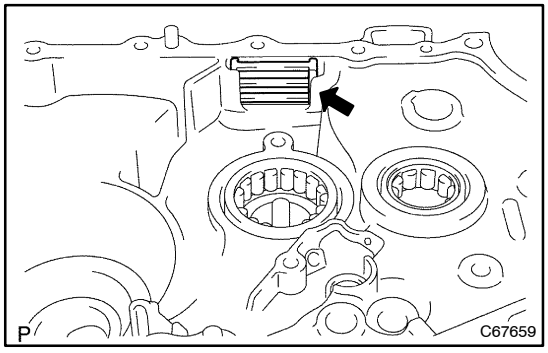
(b) Coat the lip of the front transmission case oil seal with MP grease.



67. INSTALL TRANSAXLE CASE OIL SEAL

(a) Using SST and a hammer, install a new transaxle case oil seal to the front transaxle case.
Drive in depth: 1.6 to 2.2 mm (0.0629 to 0.0866 in.)
SST 09710–20011 (09710–06071), 09950–70010 (09951–07150)

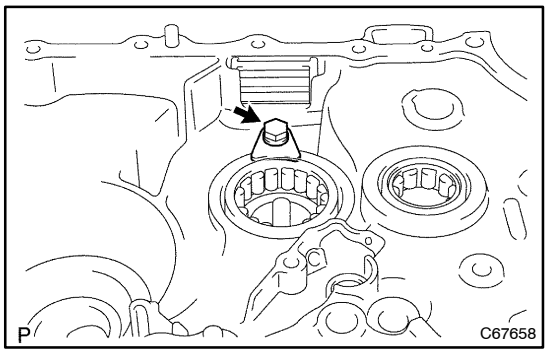
(b) Coat the lip of the transaxle case oil seal with MP grease.



68. INSTALL TRANSMISSION MAGNET

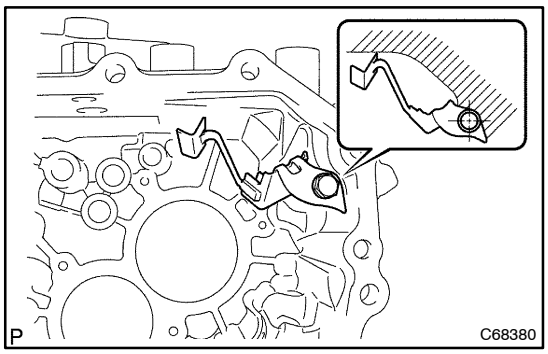
(a) Clean the transmission magnet and install it to the front transaxle case.

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69. INSTALL BEARING LOCK PLATE

(a) Install the bearing lock plate with the bolt.
Torque: 11 N·m (115 kgf·cm, 8 ft·lbf)

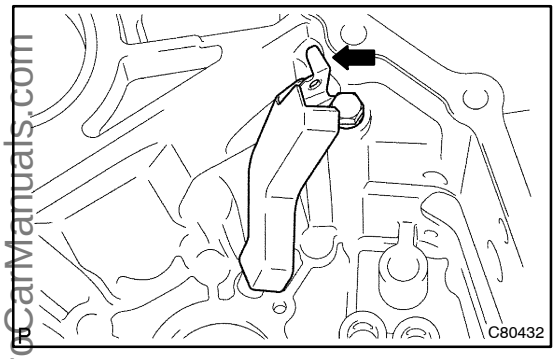


70. INSTALL OIL RECEIVER PIPE NO.1 (MTM)

(a) Install the oil receiver pipe No.1 with the bolt to the manual transmission case.
Torque: 17 N·m (175 kgf·cm, 13 ft·lbf)

NOTICE:

- **Prevent the oil receiver pipe No.1 from being deformed.**
- **Install the oil receiver pipe No.1 while placing it against the manual transmission case, as shown in the illustration.**

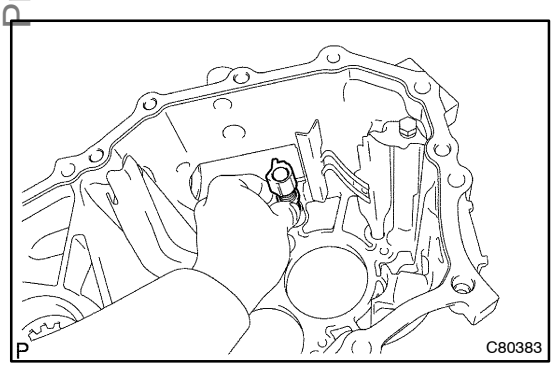


71. INSTALL OIL RECEIVER PIPE NO.2 (MTM)

(a) Install the oil receiver pipe No.2 with the bolt to the manual transmission case.
Torque: 17 N·m (175 kgf·cm, 13 ft·lbf)

NOTICE:

- **Prevent the oil receiver pipe No.2 from being deformed.**
- **Install the oil receiver pipe No.2 while placing it against the manual transmission case, as shown in the illustration.**

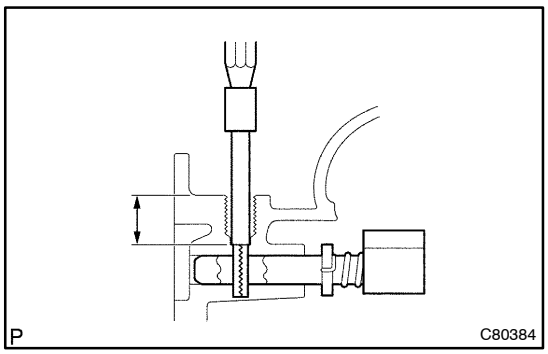


72. INSTALL REVERSE RESTRICT PIN ASSY

(a) Install the reverse restrict pin assy to the manual transmission case.

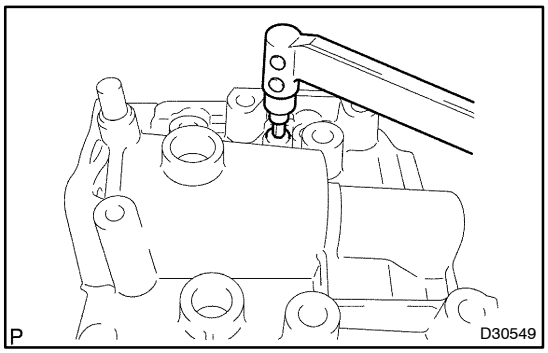
HINT:

Do not set the reverse restrict pin assy in incorrect orientation.



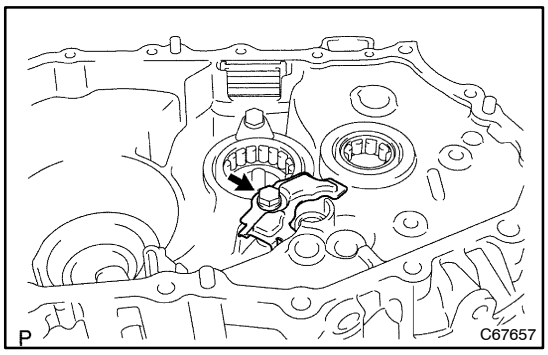
(b) Using a pin punch (ϕ 5 mm) and a hammer, install the slotted pin to the restrict pin assy.
Drive in depth: 15.5 to 16.5 mm (0.6102 to 0.6496 in.)

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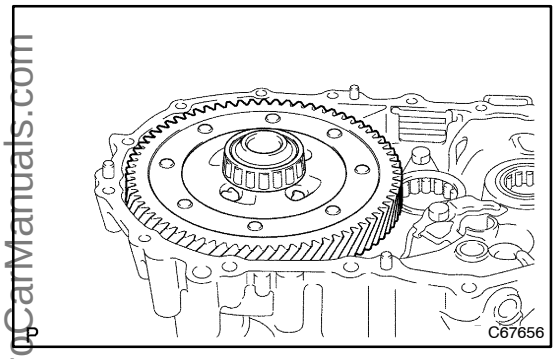
(c) Apply sealant to the reverse restrict pin plug.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent

(d) Using a hexagon wrench and a torque wrench install the reverse restrict pin plug to the manual transmission case.
Torque: 13 N·m (130 kgf·cm, 9.3 ft·lbf)



73. INSTALL MANUAL TRANSAXLE CASE RECEIVER

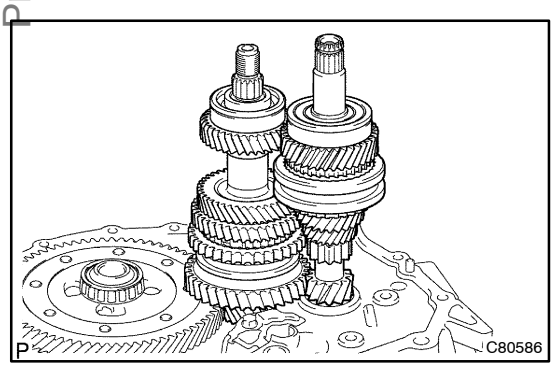
(a) Install the manual transaxle case receiver with the bolt to the front transaxle case.
Torque: 11 N·m (115 kgf·cm, 8 ft·lbf)



74. INSTALL DIFFERENTIAL CASE ASSY

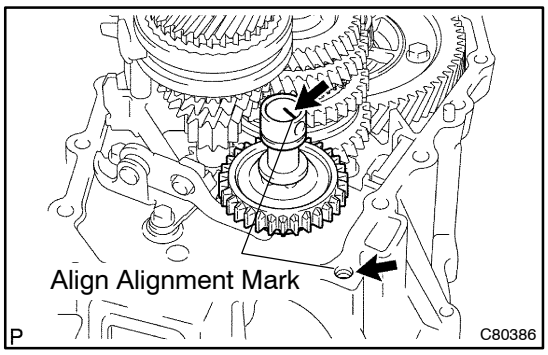
(a) Coat the differential case tapered roller bearings with gear oil, install them to the front transaxle case.

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75. INSTALL INPUT SHAFT ASSY

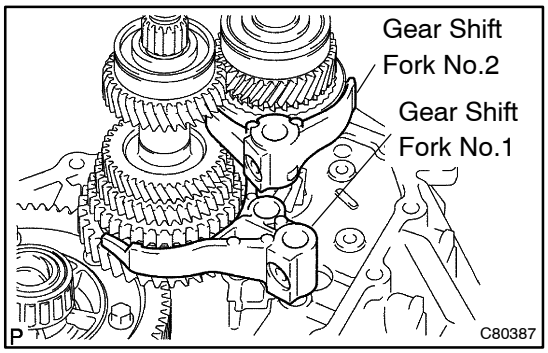
(a) Coat the sliding and rotating surface of the input and output shafts with gear oil, install them to the transaxle case.



76. INSTALL REVERSE IDLER GEAR SUB–ASSY

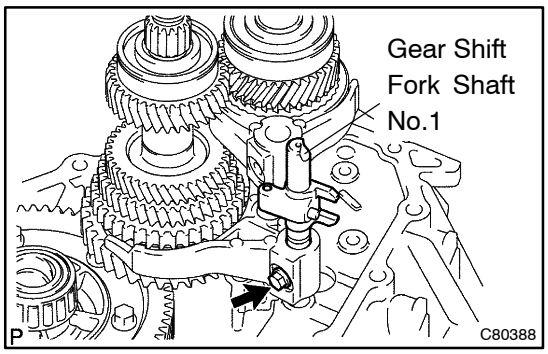
(a) Coat the reverse idler gear, thrust washer and reverse idler gear shaft with gear oil, and install them.

HINT:
Align the mark on the reverse idler gear shaft with the bolt hole as shown in the illustration.



77. INSTALL GEAR SHIFT FORK SHAFT NO.1

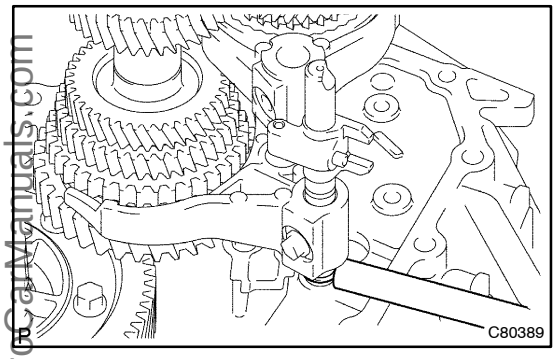
(a) Coat the gear shift fork No.1 and gear shift fork No.2 with gear oil, and install them.



(b) Coat the gear shift fork shaft No.1 with gear oil, and install it.

(c) Apply sealant to the shift fork set bolt.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent

(d) Install the gear shift fork set bolt.
Torque: 16 N·m (160 kgf·cm, 12 ft·lbf)



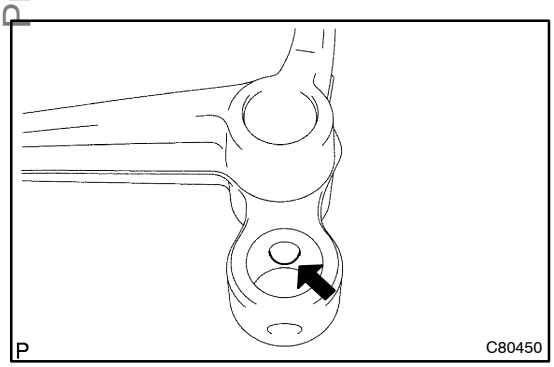
(e) Using a brass bar and a hammer, install a shaft snap ring to the gear shift fork shaft No.1.

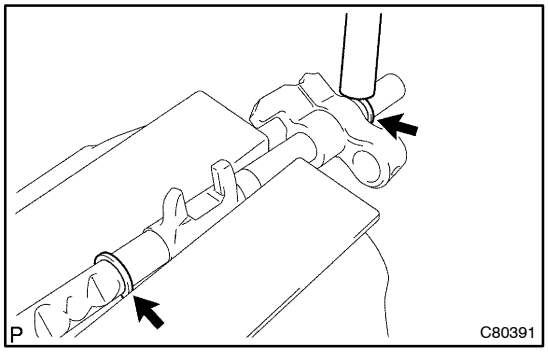
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78. INSTALL GEAR SHIFT FORK SHAFT NO.3

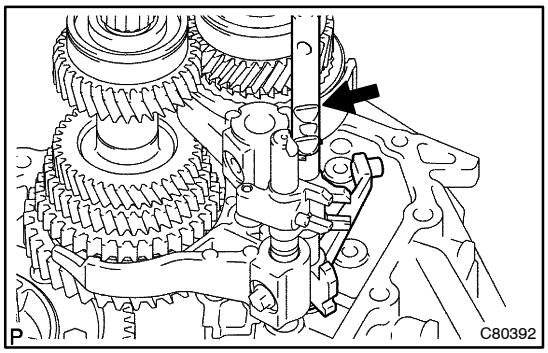
(a) Coat the 2 shift fork balls with MP grease, and install them to the reverse shift fork.

(b) Install the reverse shift fork to the gear shift fork shaft No.3.

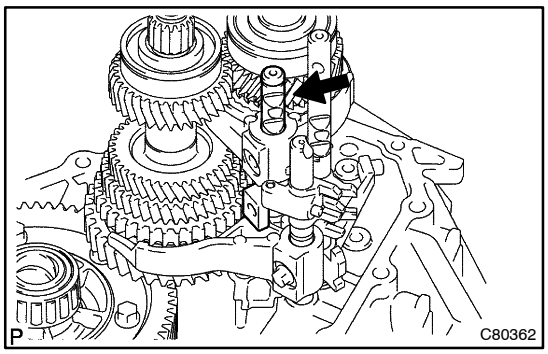




(c) Using a brass bar and a hammer, install 2 new shift fork shaft snap rings to the shift fork shaft No.3.

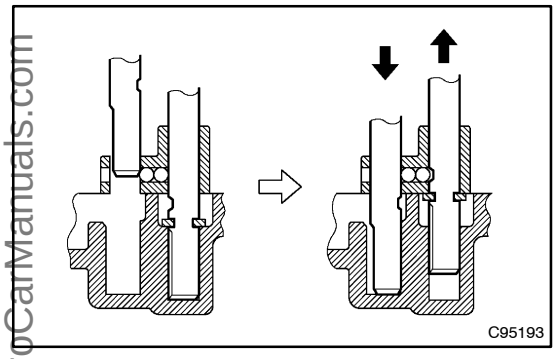


(d) Coat the shift fork shaft No.3 with gear oil, and install it.

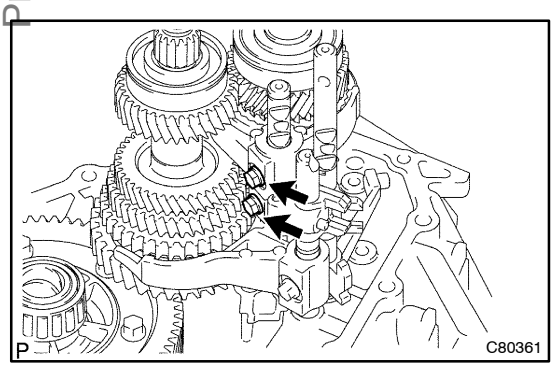


79. INSTALL GEAR SHIFT FORK SHAFT NO.2

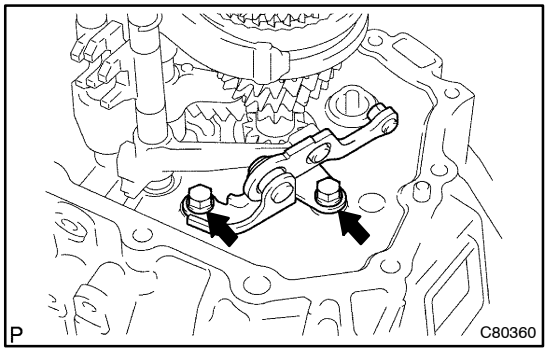
(a) Coat the gear shift head No.1 and gear shift fork shaft No.2 with gear oil, and install them.



NOTICE:
To avoid the interference of the 2 shift fork balls, lift up the gear shift fork shaft NO.3 at the position shown in the illustration.



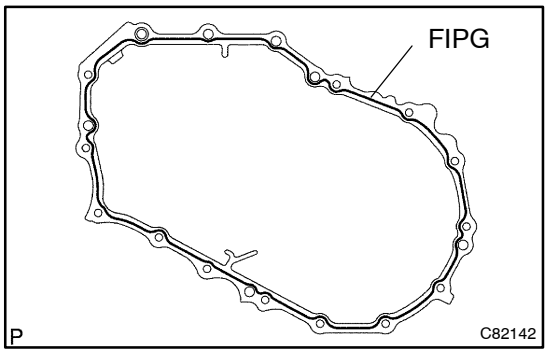
(b) Coat the 2 shift lock bolts with sealant, and install them to the gear shift fork No.2 and shift head No.1.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 16 N·m (160 kgf·cm, 12 ft·lbf)



80. INSTALL REVERSE SHIFT ARM BRACKET ASSY

(a) Install the reverse shift arm bracket assy with the 2 bolts to the front transaxle case.
Torque: 17 N·m (175 kgf·cm, 13 ft·lbf)

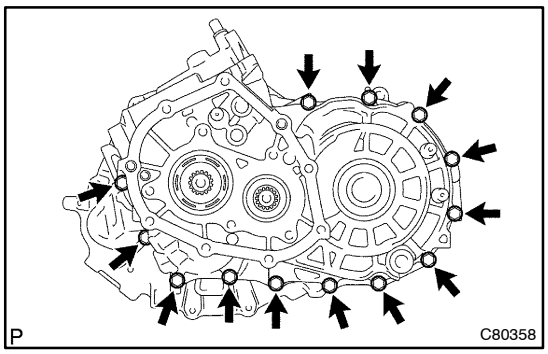
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81. INSTALL MANUAL TRANSMISSION CASE

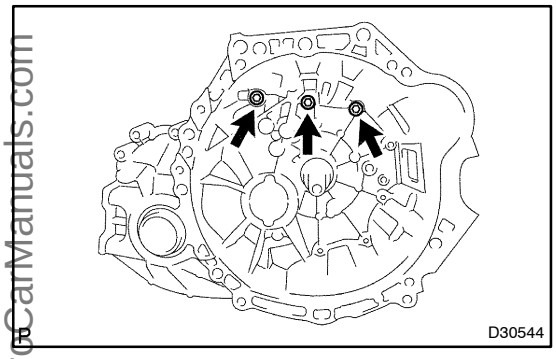
(a) Apply FIPG to the manual transmission case, as shown in the illustration.
FIPG:
Part No. 08826–00090, THREE BOND 1281 or equivalent

NOTICE:
Parts must be assembled within 10 minutes of application. Otherwise, the packing (FIPG) material must be removed and reapplied.



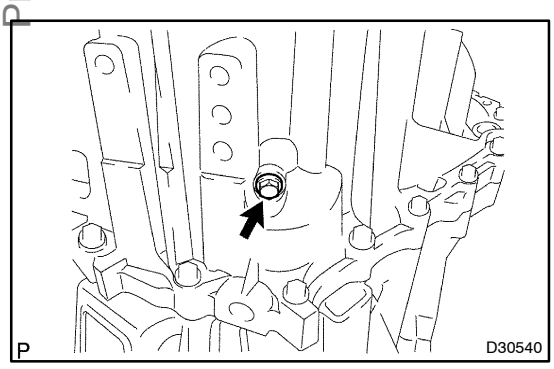
(b) Install the manual transmission case straight pin to the front transaxle case.

(c) Coat the 13 bolts with sealant, and install them to the manual transmission case.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 29 N·m (300 kgf·cm, 22 ft·lbf)



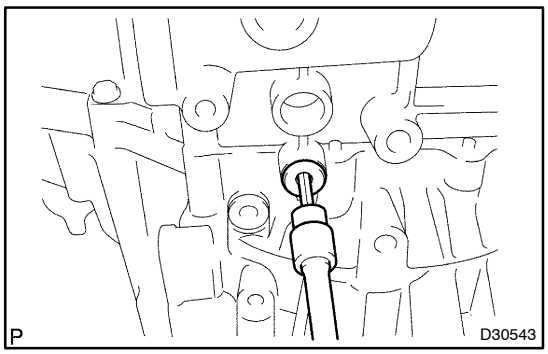
(d) Coat the 3 bolts with sealant, and install them to the front transaxle case.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 29 N·m (300 kgf·cm, 22 ft·lbf)

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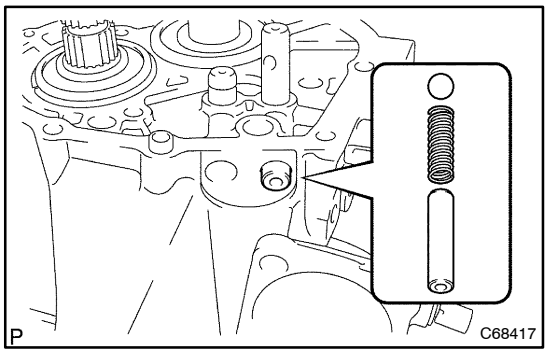
82. INSTALL REVERSE IDLER GEAR SHAFT BOLT

(a) Coat the reverse idler gear shaft bolt with sealant, and install it with a new gasket.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 29 N·m (300 kgf·cm, 22 ft·lbf)



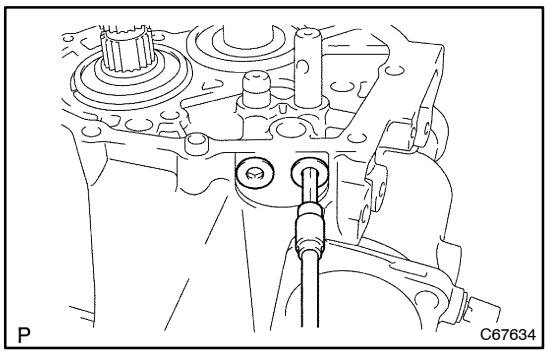
83. INSTALL LOCK BALL ASSY NO.2

(a) Coat the lock ball assy No.2 with sealant, and install it with a hexagon wrench.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 39 N·m (400 kgf·cm, 29 ft·lbf)

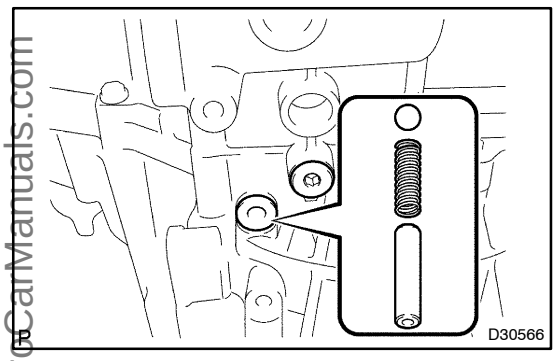


84. INSTALL SHIFT DETENT BALL

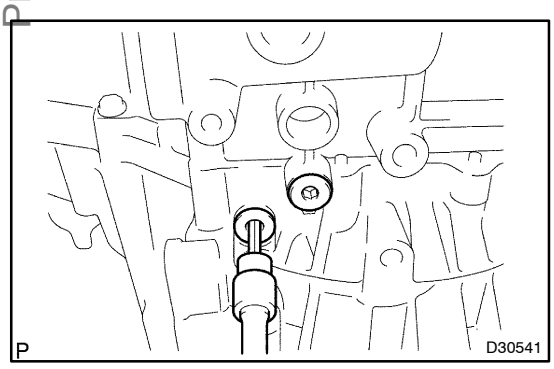
(a) Install the 2 shift detent balls, 2 shift detent ball springs with 2 shift detent ball spring seats No.1 to the manual transmission case.



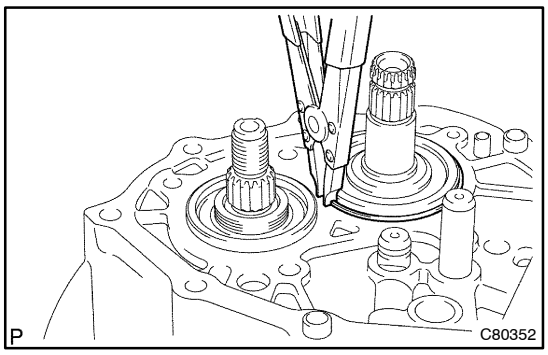
(b) Coat the 2 shift detent ball plugs with sealant, and install them using a hexagon wrench.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 25 N·m (250 kgf·cm, 18 ft·lbf)



(c) Install the shift detent ball, shift detent ball compression spring and shift detent ball spring seat No.1 to the front transaxle case.



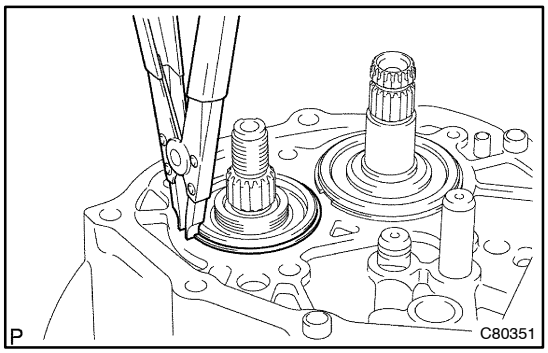
(d) Coat the shift detent ball plug with sealant, and install it using a hexagon wrench.
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 25 N·m (250 kgf·cm, 18 ft·lbf)



85. INSTALL INPUT SHAFT REAR BEARING HOLE SNAP RING

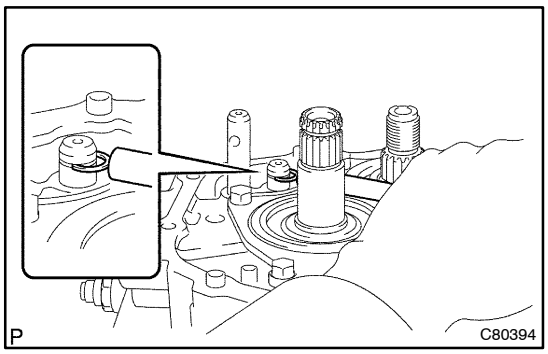
(a) Using a snap ring expander, install the input shaft rear bearing hole snap ring to the input shaft.

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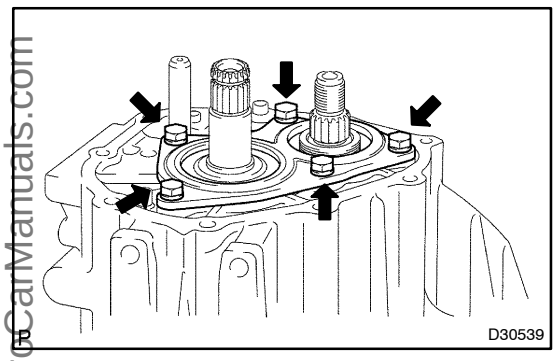
86. INSTALL OUTPUT SHAFT REAR BEARING HOLE SNAP RING

(a) Using a snap ring expander, install the output shaft rear bearing hole snap ring to the output shaft.



87. INSTALL SHIFT FORK SHAFT SHAFT SNAP RING

(a) Using a brass bar and a hammer, install a new shift fork shaft snap ring to the shift fork shaft No.2.

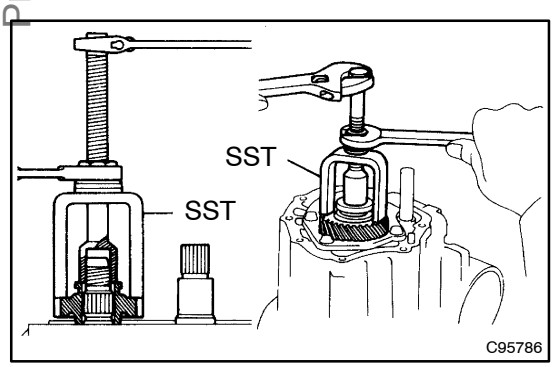


88. INSTALL BEARING RETAINER REAR (MTM)

(a) Coat the 5 bolts with sealant, and install the bearing retainer rear with the 5 bolts to the manual transmission case.

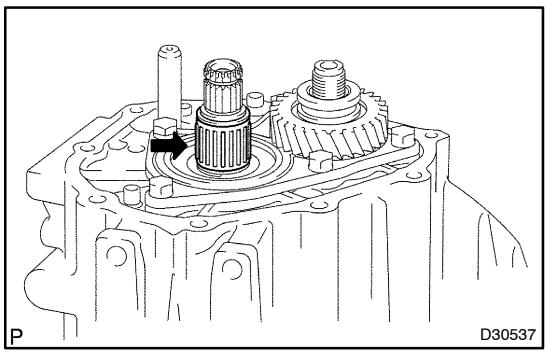
Torque: 27 N·m (280 kgf·cm, 20 ft·lbf)

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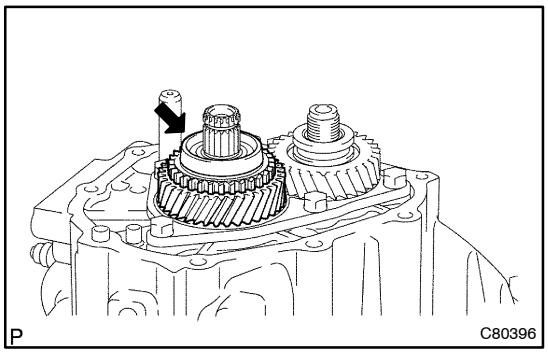
89. INSTALL 5TH DRIVEN GEAR

(a) Using SST, install the 5th driven gear to the output shaft.
SST 09309–12020



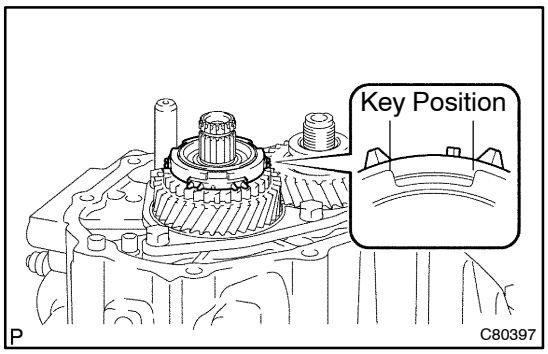
90. INSTALL 5TH GEAR NEEDLE ROLLER BEARING

(a) Coat the 5th gear needle roller bearing and spacer with gear oil, and install them to the input shaft.



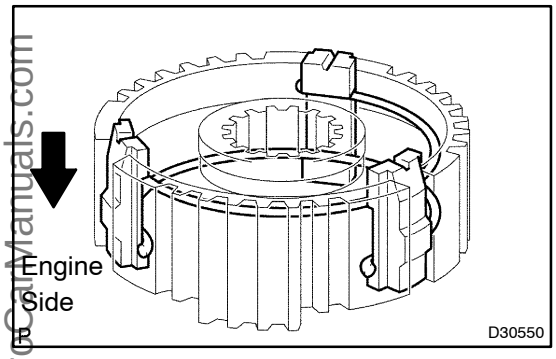
91. INSTALL 5TH GEAR

(a) Coat the 5th gear with gear oil, and install it to the input shaft.



92. INSTALL SYNCHRONIZER RING NO.3

(a) Coat the synchronizer ring No.3 with gear oil, and install it to the 5th gear.

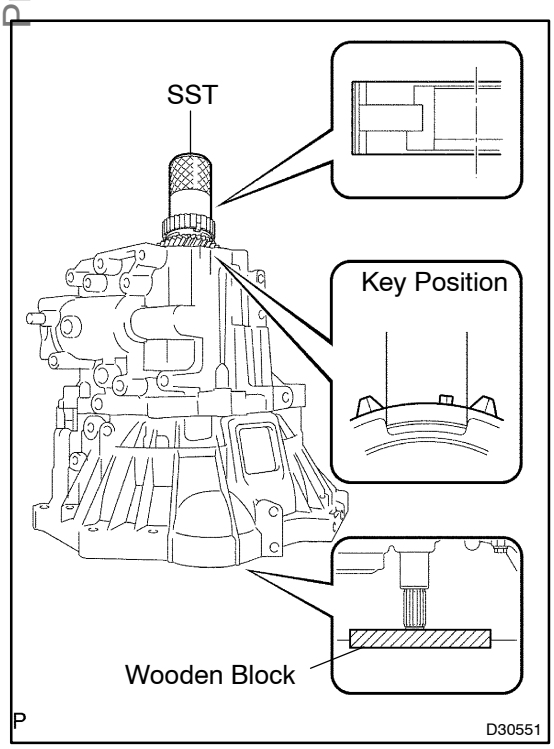


93. INSTALL TRANSMISSION CLUTCH HUB NO.3

(a) Install the 3 synchromesh shifting keys and 2 synchromesh shifting key springs to the transmission clutch hub No.3.

NOTICE:

Do not set both openings of the shifting key springs in the same position.



(b) Using SST and a hammer, install the transmission clutch hub No.3 to the input shaft.

SST 09636–20010

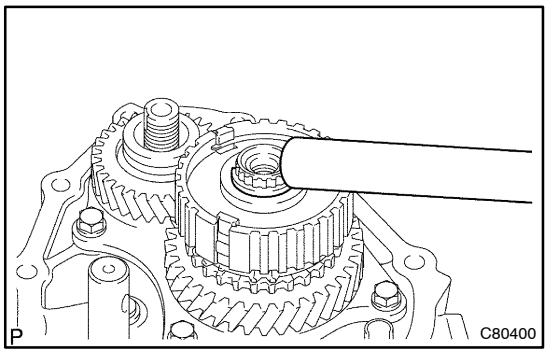
NOTICE:

- **Do not install the transmission clutch hub No.3 with it facing the wrong direction.**
- **Install the transmission clutch hub No.3 checking the positions of the synchronizer ring No.3 key groove and the synchromesh shifting key No.3.**
- **Check that the 5th gear is rotating.**

NOTICE:

Place the suitable sized wooden block to support the input shaft.

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(c) Select a clutch hub No.3 shaft snap ring that will allow minimum axial play.

Clearance: 0.1 mm (0.0039 in.) or less

Mark	Thickness mm (in.)	Mark	Thickness mm (in.)
A	2.25 (0.0886)	E	2.49 (0.0980)
B	2.31 (0.0909)	F	2.55 (0.1004)
C	2.37 (0.0933)	G	2.61 (0.1028)
D	2.43 (0.0957)		

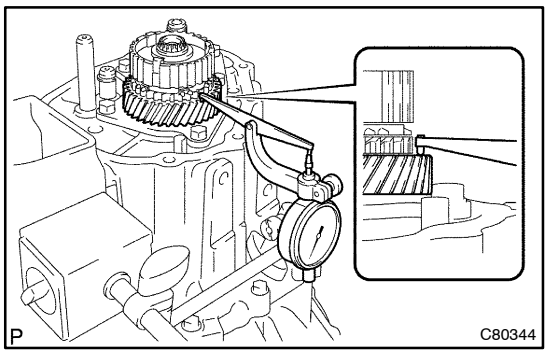
(d) Using a brass bar and a hammer, install the snap ring to the input shaft.

94. INSPECT 5TH GEAR THRUST CLEARANCE

(a) Using a dial indicator, inspect the 5th gear thrust clearance.

Standard clearance:
0.1 to 0.57 mm (0.0039 to 0.02244 in.)
Maximum clearance:
0.57 mm (0.02244 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.3, 5th gear or input shaft rear radial ball bearing.

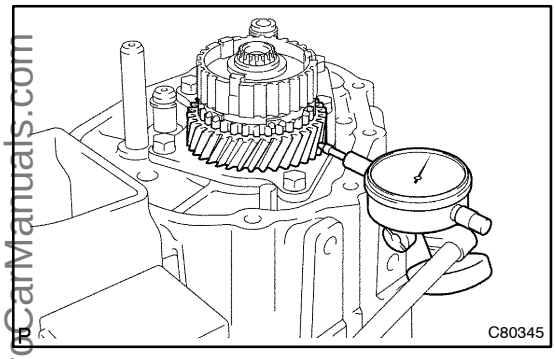


95. INSPECT 5TH GEAR RADIAL CLEARANCE

(a) Using a dial indicator, inspect the 5th gear radial clearance.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 5th gear, 5th gear needle roller bearing or input shaft.

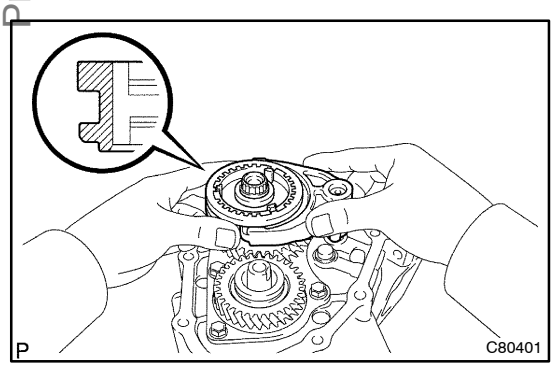


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96. INSTALL GEAR SHIFT FORK NO.3

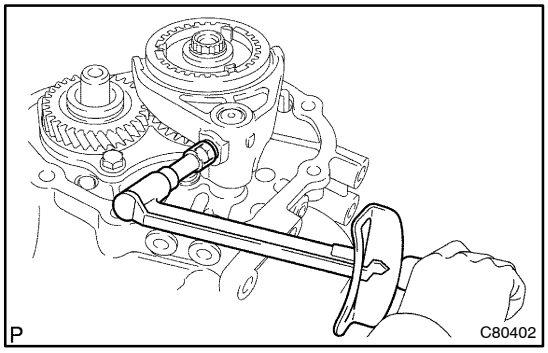
(a) Coat the transmission clutch hub sleeve No.3 with gear oil, and install it and gear shift fork No.3 to the transmission clutch hub No.3.

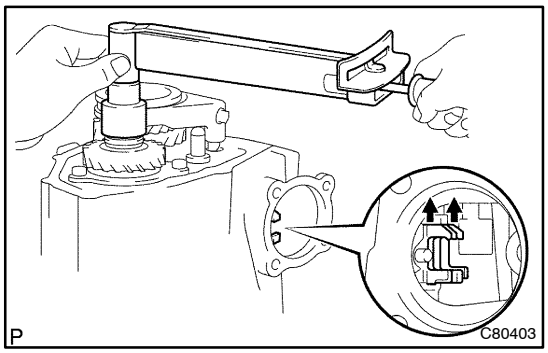
HINT:
Do not set the transmission clutch hub No.3 in incorrect orientation.



(b) Coat the gear shift fork lock bolt with sealant, and install it to the gear shift fork No.3.

Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent
Torque: 16 N·m (160 kgf·cm, 12 ft·lbf)



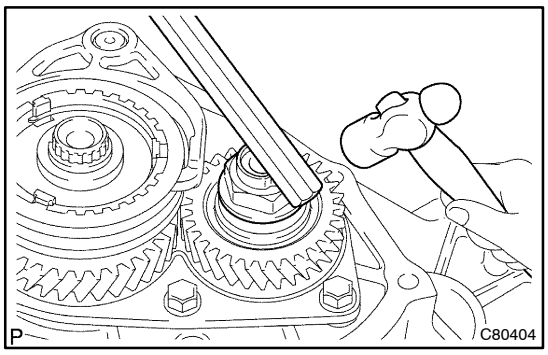


97. INSTALL MANUAL TRANSMISSION OUTPUT SHAFT REAR SET NUT

(a) Engage the gears to make them double meshed.

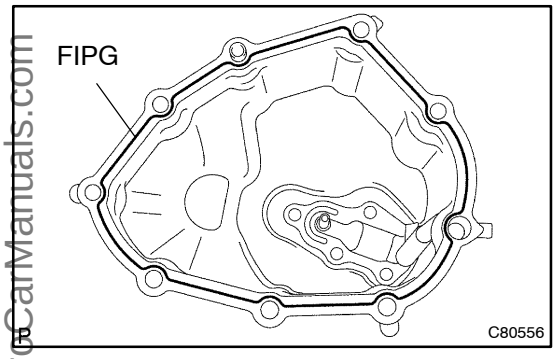
(b) Install a new manual transmission output shaft rear set nut.

Torque: 118 N·m (1200 kgf·cm, 87 ft·lbf)



(c) Using a chisel and a hammer, stake the manual transmission output shaft rear set nut.

(d) Disengage the gears.



98. INSTALL MANUAL TRANSMISSION CASE COVER SUB–ASSY

(a) Apply FIPG to the transaxle case cover as shown in the illustration.

FIPG:

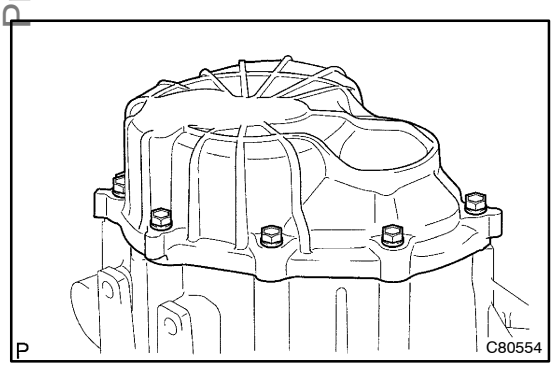
Part No. 08826–00090, THREE BOND 1281

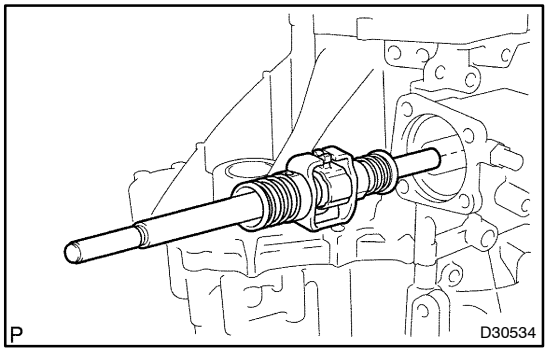
NOTICE:

Parts must be assembled within 10 minutes of application. Otherwise, the packing (FIPG) material must be removed and reapplied.

(b) Install the manual transmission case cover to the manual transmission case with the 9 bolts.

Torque: 18 N·m (185 kgf·cm, 14 ft·lbf)

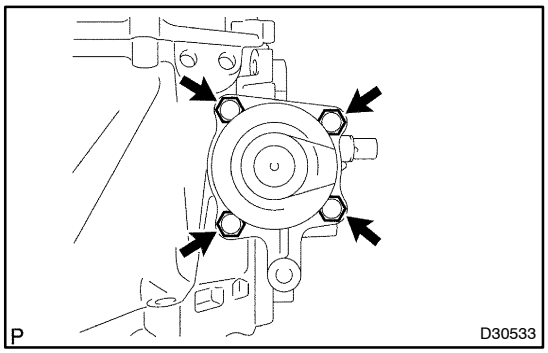




99. INSTALL SHIFT & SELECT LEVER SHAFT ASSY

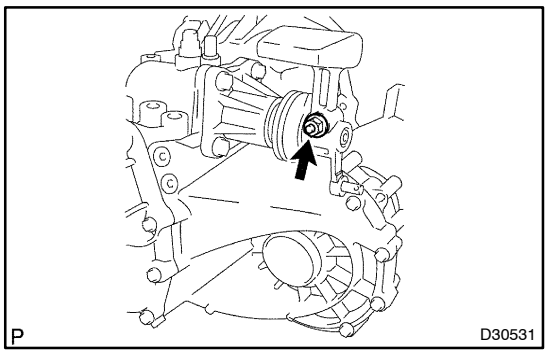
(a) Coat the sift & select lever shaft assy with gear oil, and install it to the manual transmission case.

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100. INSTALL CONTROL SHAFT COVER

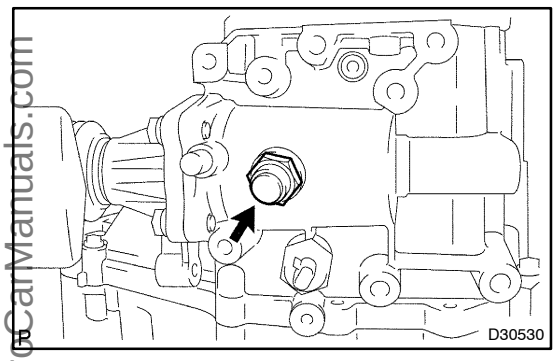
(a) Coat the 4 bolts with sealant. Install a new gasket and control shaft cover with the 4 bolts.
Torque: 20 N·m (200 kgf·cm, 14 ft·lbf)
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent



101. INSTALL CONTROL SHIFT LEVER

(a) Install the control shift lever with the shift outer lever lever lock pin to the shift & select lever shaft assy.

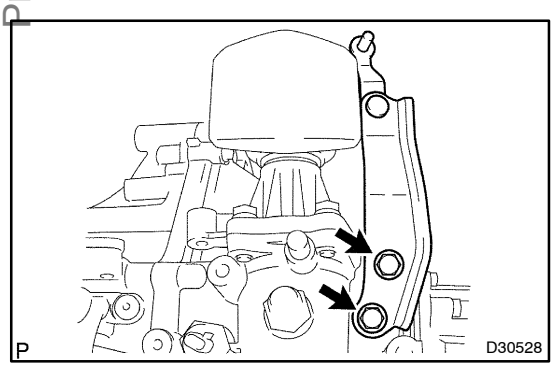
(b) Install the spring washer with the nut.
Torque: 12 N·m (122 kgf·cm, 9 ft·lbf)



102. INSTALL LOCK BALL ASSY NO.1

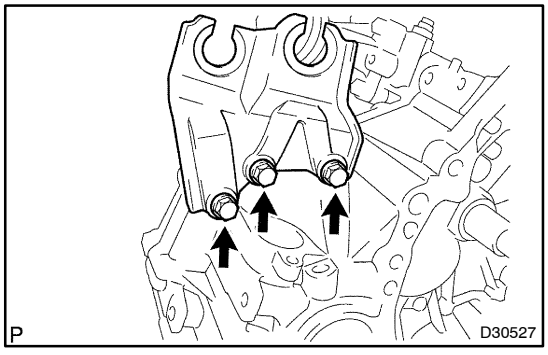
(a) Coat the lock ball assy No.1 with sealant, and install it to the manual transmission case.
Torque: 29 N·m (300 kgf·cm, 22 ft·lbf)
Sealant:
Part No. 08833–00080, THREE BOND 1344, LOCTITE 242 or equivalent

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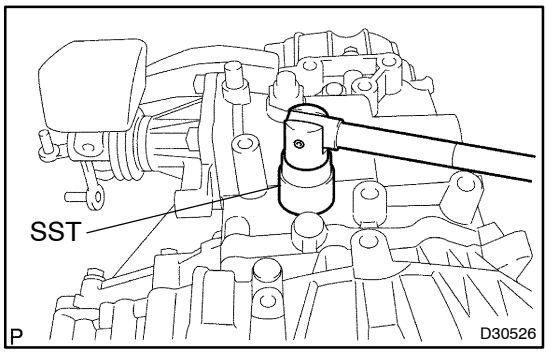
103. INSTALL SELECTING BELL CRANK ASSY

(a) Install the selecting bell crank assy to the manual transmission case with the 2 bolts.
Torque: 25 N·m (250 kgf·cm, 18 ft·lbf)



104. INSTALL FLOOR SHIFT CONTROL LEVER HOUSING SUPPORT BRACKET

(a) Install the floor shift control lever housing support bracket to the front transaxle case with the 3 bolts.
Torque: 18 N·m (184 kgf·cm, 13 ft·lbf)

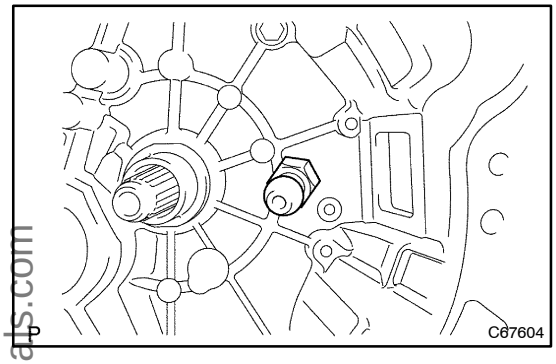


105. INSTALL BACK UP LAMP SWITCH ASSY

(a) Using SST, install the back–up lamp switch assy with a new gasket to the manual transmission case.
SST 09817–16011
Torque: 40 N·m (410 kgf·cm, 30 ft·lbf)

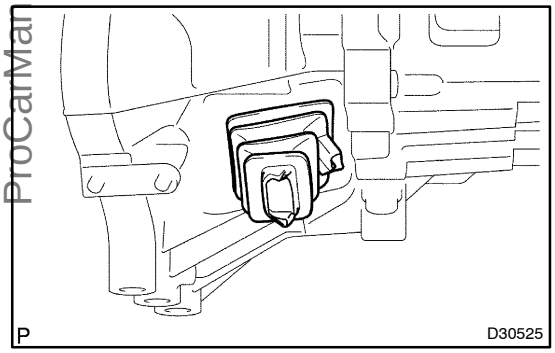
106. INSTALL SPEEDOMETER DRIVEN HOLE COVER SUB–ASSY

(a) Install a new O–ring to the speedometer driven hole cover sub–assy.
(b) Install the speedometer driven hole cover sub–assy to the manual transaxle case with the bolt.
Torque: 11 N·m (115 kgf·cm, 8.3 ft·lbf)



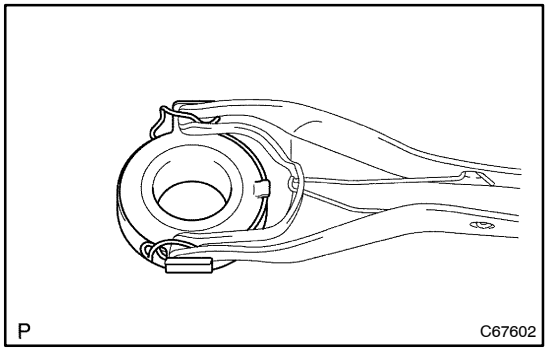
107. INSTALL RELEASE FORK SUPPORT

(a) Install the release fork support to the front transaxle case.
Torque: 37 N·m (375 kgf·cm, 25 ft·lbf)



108. INSTALL CLUTCH RELEASE FORK BOOT

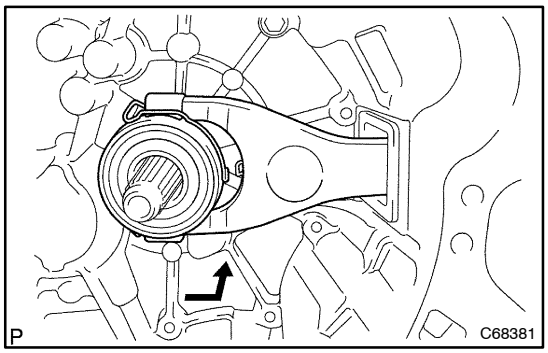
(a) Install the clutch release fork boot to the front transaxle case.



109. INSTALL CLUTCH RELEASE BEARING ASSY

(a) Coat the clutch release bearing assy with release hub grease, and install it to the clutch release fork sub–assy with the clip.
Sealant:
Part No. 08887–01806, RELEASE HUB GREASE or equivalent

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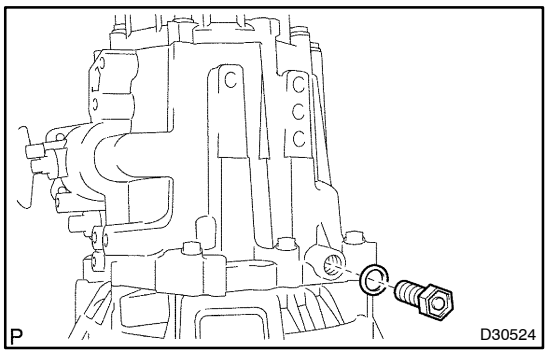
110. INSTALL CLUTCH RELEASE FORK SUB–ASSY

(a) Install the clutch release fork sub–assy to the input shaft.

(b) Apply clutch spline grease to the input shaft spline.

Sealant:

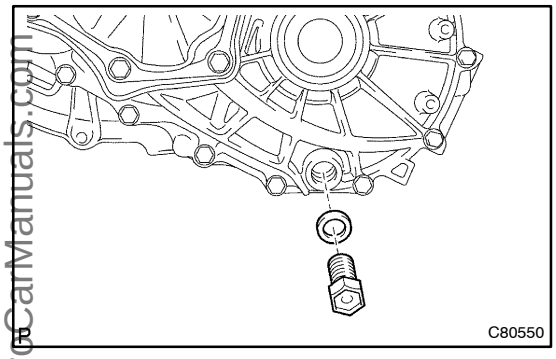
Part No. 08887–01806, CLUTCH SPLINE GREASE or equivalent



111. INSTALL MANUAL TRANSMISSION FILLER PLUG

(a) Install the manual transmission filler plug with a new gasket to the manual transmission case.

Torque: 39 N·m (400 kgf·cm, 29 ft·lbf)



112. INSTALL DRAIN (MTM) PLUG SUB–ASSY

(a) Install the drain plug with a new gasket to the manual transmission case.

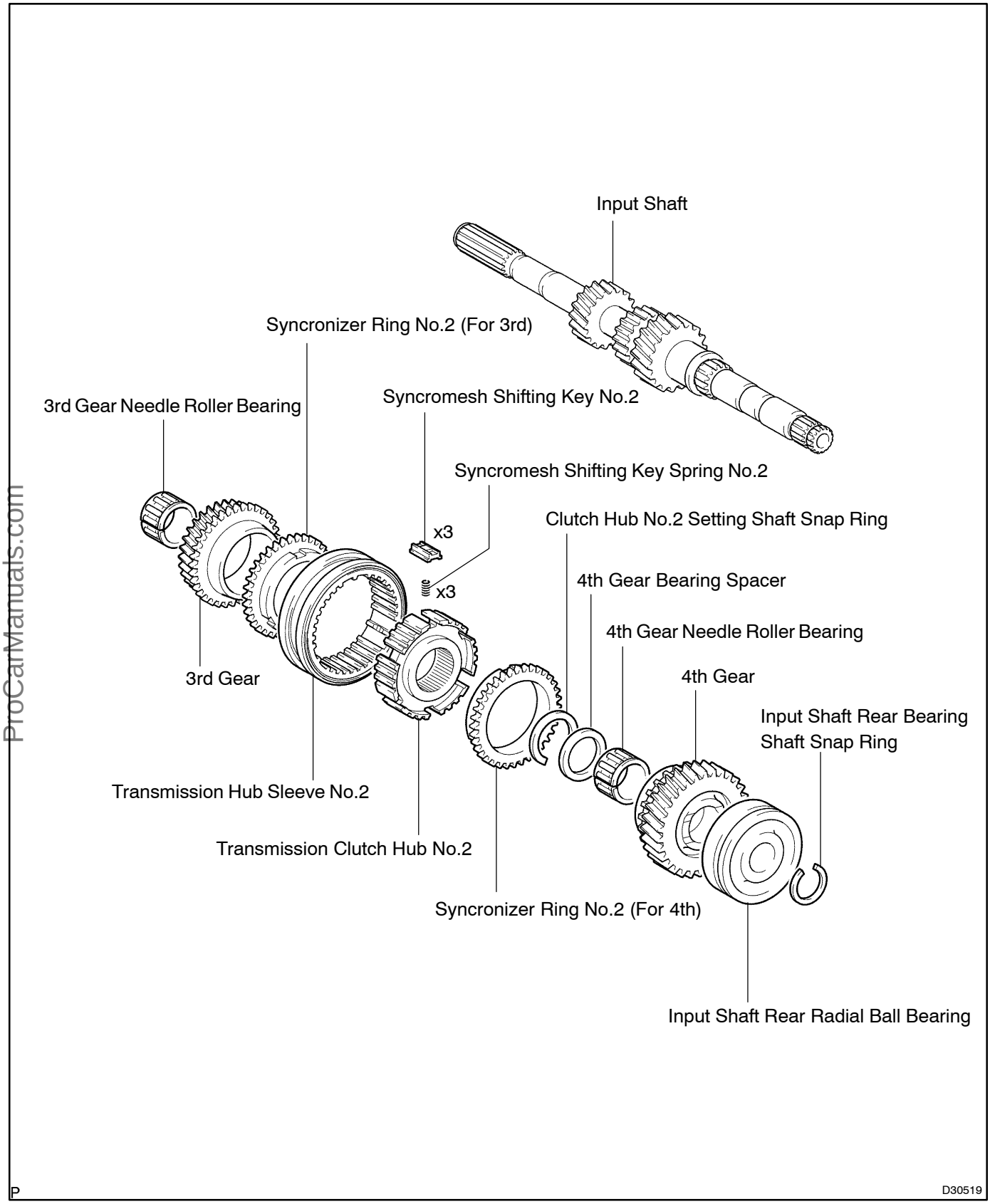
Torque: 39 N·m (400 kgf·cm, 29 ft·lbf)

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INPUT SHAFT ASSY (C50)

COMPONENTS

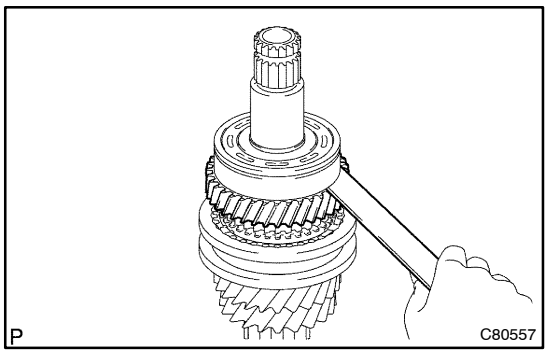
410DE–02



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410DF–02

OVERHAUL

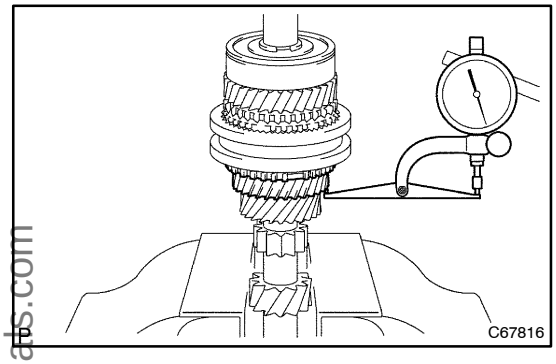


1. INSPECT 4TH GEAR THRUST CLEARANCE

(a) Using a feeler gauge, measure the 4th gear thrust clearance.

Standard clearance:
0.1 to 0.55 mm (0.0039 to 0.0217 in.)
Maximum clearance:
0.55 mm (0.0217 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.2, 4th gear or input shaft rear radial ball bearing.

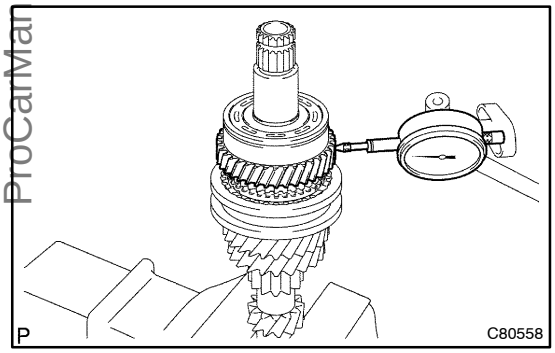


2. INSPECT 3RD GEAR THRUST CLEARANCE

(a) Using a dial indicator, measure the 3rd gear thrust clearance.

Standard clearance:
0.1 to 0.35 mm (0.0039 to 0.0138 in.)
Maximum clearance:
0.35 mm (0.0138 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.2, 3rd gear or input shaft.

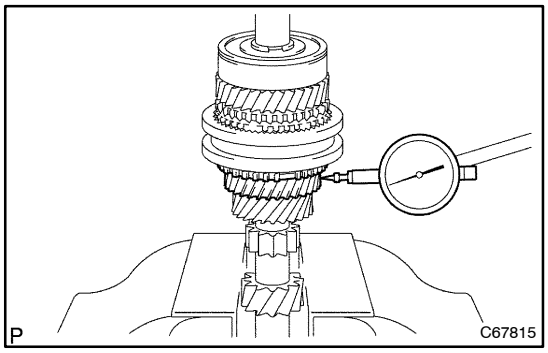


3. INSPECT 4TH GEAR RADIAL CLEARANCE

(a) Using a dial indicator, measure the 4th gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 4th gear, needle roller bearing or input shaft.



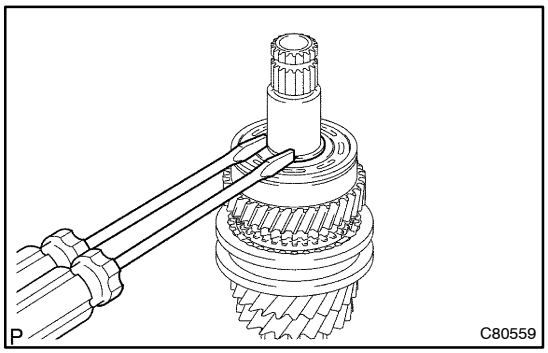
4. INSPECT 3RD GEAR RADIAL CLEARANCE

(a) Using a dial indicator, measure the 3rd gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 3rd gear, needle roller bearing or input shaft.

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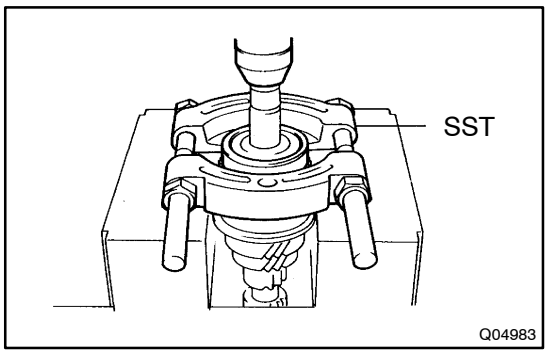


5. REMOVE 4TH GEAR

(a) Using 2 screwdrivers and a hammer, remove the input shaft rear bearing shaft snap ring from the input shaft.

NOTICE:

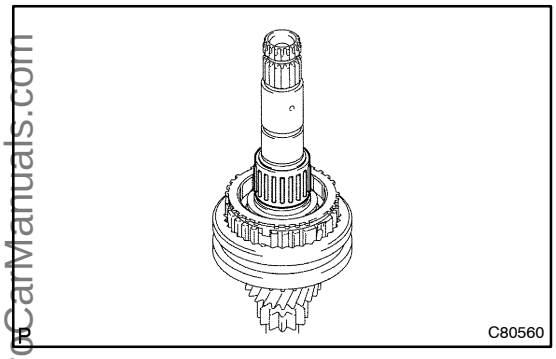
Use a shop rag to keep the snap ring from flying.



(b) Using SST and a press, remove the input shaft rear radial ball bearing rear and 4th gear from input shaft.
SST 09950–00020

NOTICE:

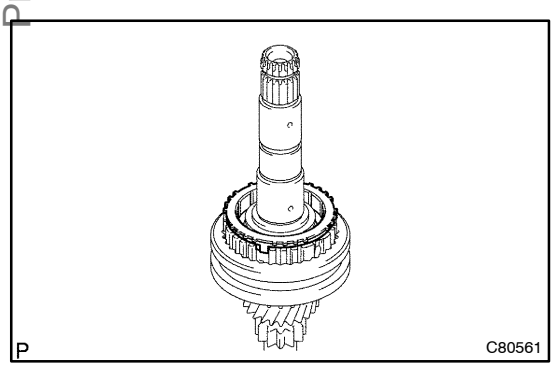
- **Do not tighten SST excessively.**
- **Support the input shaft assy by hand so that it will not be dropped off.**



6. REMOVE 4TH GEAR NEEDLE ROLLER BEARING

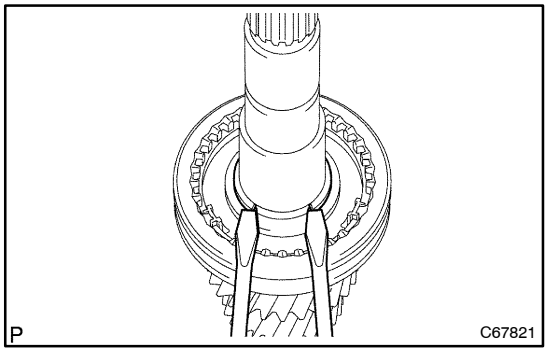
(a) Remove the 4th gear needle roller bearing and 4th gear bearing spacer from input shaft.

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7. REMOVE SYNCHRONIZER RING NO.2 (FOR FOURTH SYNCHRONIZER RING)

(a) Remove the synchronizer ring No.2 from the transmission clutch hub No.2.

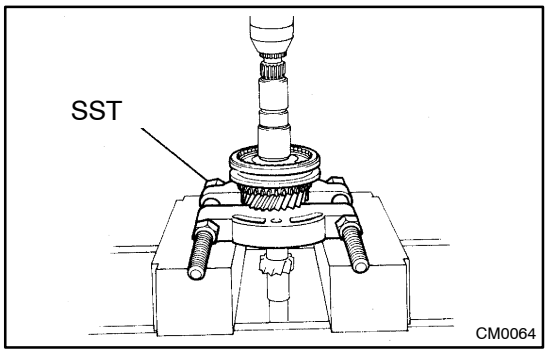


8. REMOVE 3RD GEAR

(a) Using 2 screwdrivers and a hammer, remove the clutch hub No.2 setting shaft snap ring from the input shaft.

NOTICE:

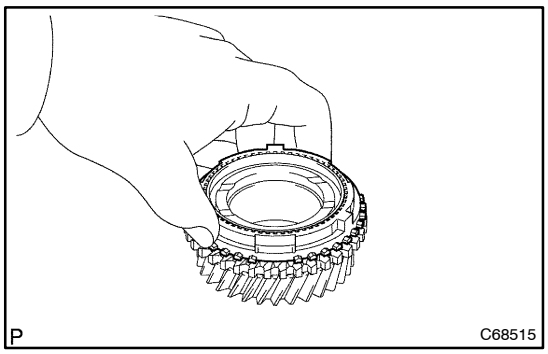
Use a shop rag to keep the snap ring from flying.



(b) Using SST and a press, remove the transmission clutch hub No.2 and 3rd gear from the input shaft.
SST 09950–00020

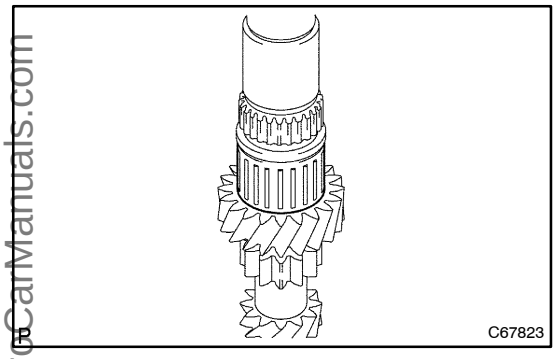
NOTICE:

- **Do not tighten SST excessively.**
- **Support the input shaft by hand so that it will not be dropped off.**



9. REMOVE SYNCHRONIZER RING NO.2 (FOR THIRD SYNCHRONIZER RING)

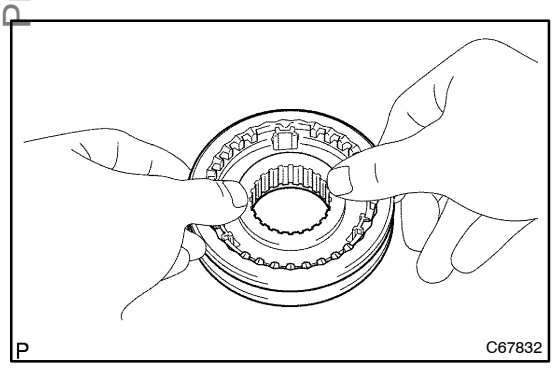
(a) Remove the synchronizer ring No.2 from the 3rd gear.



10. REMOVE 3RD GEAR NEEDLE ROLLER BEARING

(a) Remove the 3rd gear needle roller bearing from the input shaft.

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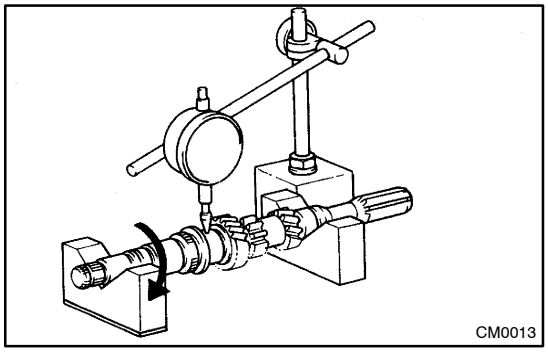


11. REMOVE TRANSMISSION HUB SLEEVE NO.2

(a) Remove the transmission hub sleeve No.2, 3 synchromesh shifting keys No.2 and 3 synchromesh shifting key springs No.2 from the transmission clutch hub No.2.

NOTICE:

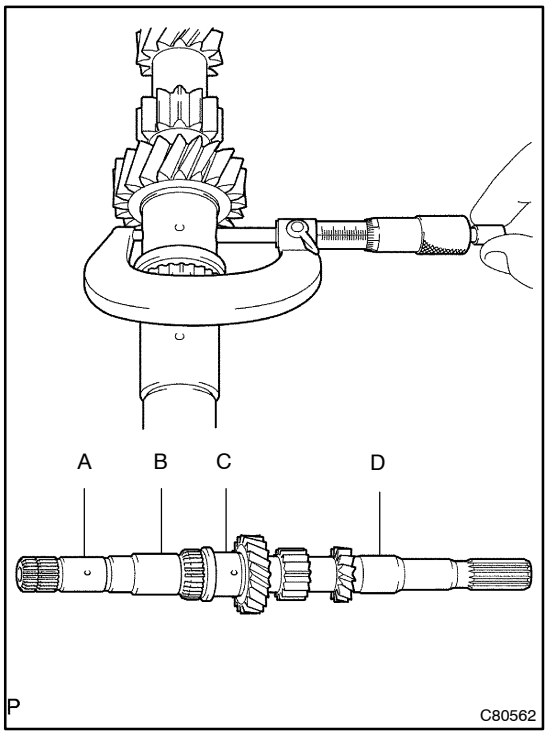
Use a shop rag to keep the shifting key and shifting key spring from flying.



12. INSPECT INPUT SHAFT

(a) Using a dial indicator, check the input shaft runout.
Maximum runout: 0.015 mm (0.00059 in.)

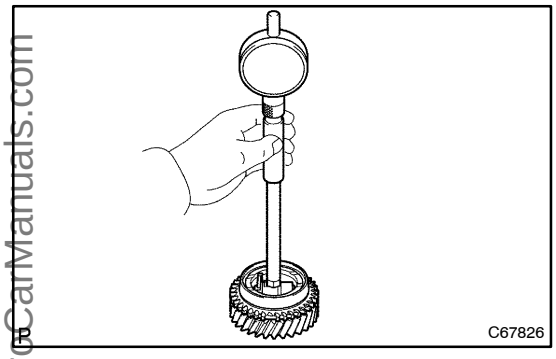
If the runout exceeds the maximum, replace the input shaft.



(b) Using a micrometer, measure the outer diameter of the input shaft journal surface.

Standard outer diameter:
Part A: 24.885 to 24.900 mm (0.9797 to 0.9803 in.)
Part B: 28.991 to 29.006 mm (1.1413 to 1.1419 in.)
Part C: 30.985 to 31.000 mm (1.2198 to 1.2204 in.)
Part D: 24.985 to 25.000 mm (0.9836 to 0.9842 in.)
Minimum outer diameter:
Part A: 24.885 mm (0.9797 in.)
Part B: 28.991 mm (1.1413 in.)
Part C: 30.985 mm (1.2198 in.)
Part D: 24.985 mm (0.9836 in.)

If the outer diameter is less than the minimum, replace the input shaft.

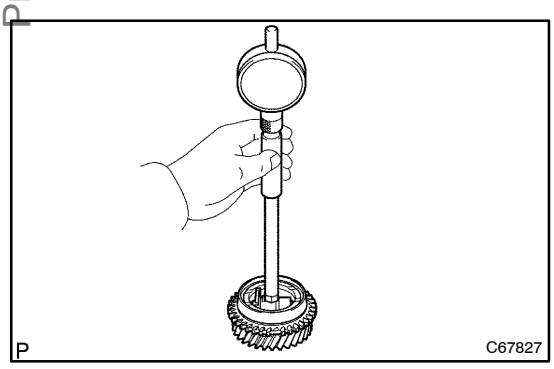


13. INSPECT 4TH GEAR

(a) Using a cylinder gauge, measure the inside diameter of the 4th gear.

Standard inside diameter:
34.015 to 34.031 mm (1.3391 to 1.3398 in.)
Maximum inside diameter:
34.031 mm (1.3398 in.)

If the diameter exceeds the maximum, replace the 4th gear.



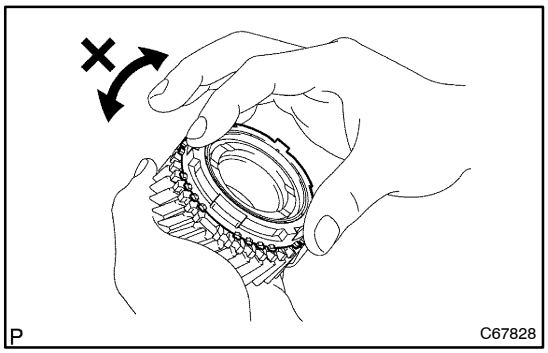
14. INSPECT 3RD GEAR

(a) Using a cylinder gauge, measure the inside diameter of the 3rd gear.

Standard inside diameter:
36.015 to 36.031 mm (1.4179 to 1.4185 in.)
Maximum inside diameter:
36.031 mm (1.4185 in.)

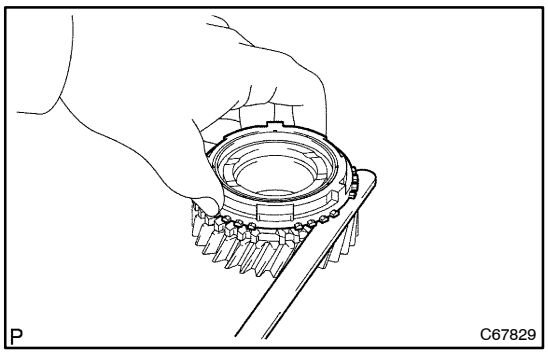
If the diameter exceeds the maximum, replace the 3rd gear.

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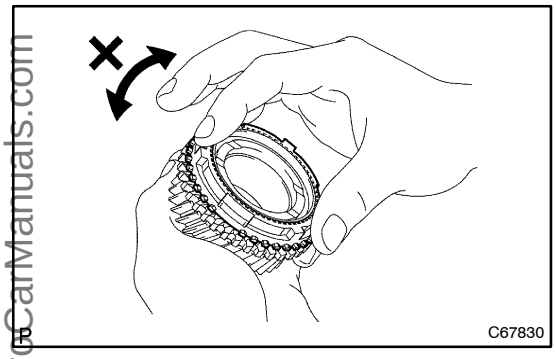
15. INSPECT SYNCHRONIZER RING NO.2 (FOR FOURTH SYNCHRONIZER RING)

(a) Apply gear oil to the cone of the 4th gear and check with the synchronizer ring No.2 contacted by pressure that the gear does not turn in the direction as shown in the illustration.



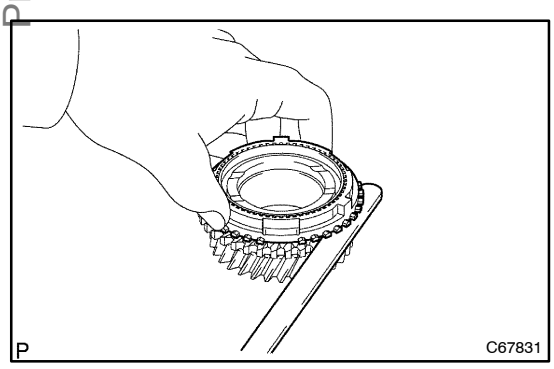
(b) Using a feeler gauge, check the clearance between the synchronizer ring No.2 and 4th gear with the synchronizer ring No.2 contacted to the cone of the 4th gear.
Standard clearance:
0.75 to 1.65 mm (0.0295 to 0.0649 in.)
Minimum clearance:
0.75 mm (0.0295 in)

If the clearance is less than the minimum, replace the synchronizer ring.



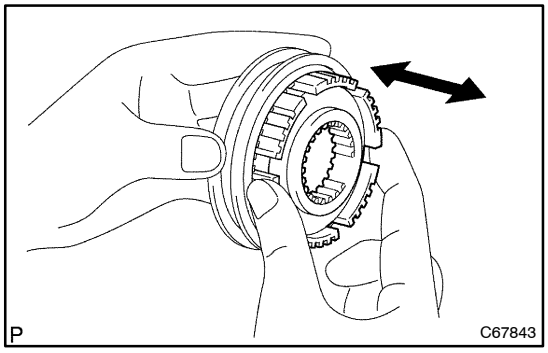
16. INSPECT SYNCHRONIZER RING NO.2 (FOR THIRD SYNCHRONIZER RING)

(a) Apply gear oil to the cone of the 3rd gear and check with the synchronizer ring No.2 contacted by pressure that the gear does not turn in the direction as shown in the illustration.



(b) Using a feeler gauge, check the clearance between the synchronizer No.2 and 3rd gear with the synchronizer ring No.2 contacted to the cone of the 3rd gear.
Standard clearance:
0.75 to 1.65 mm (0.0295 to 0.0649 in.)
Minimum clearance:
0.75 mm (0.0295 in)

If the clearance is less than the minimum, replace the synchronizer ring.

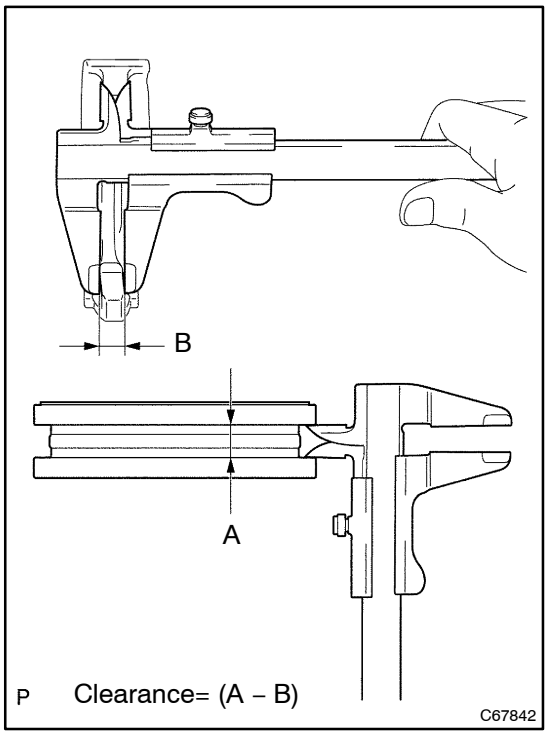


17. INSPECT TRANSMISSION HUB SLEEVE NO.2

(a) Check the sliding condition between the transmission hub sleeve No.2 and transmission clutch hub No.2.

(b) Check that the spline gear's edges of the transmission hub sleeve No.2 are not worn down.

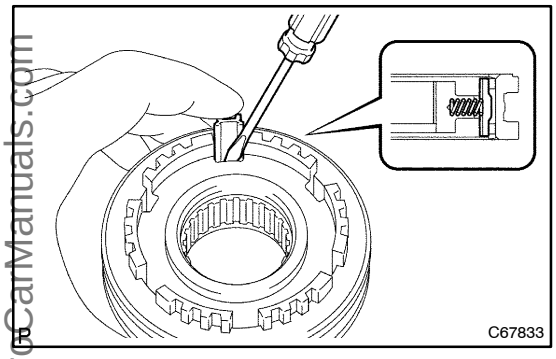
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(c) Using vernier calipers, measure the clearance between the transmission hub sleeve No.2 and gear shift fork No.2.

Standard clearance:
0.15 to 0.35 mm (0.0059 to 0.0137 in.)

If the clearance exceeds the standard clearance, replace the transmission hub sleeve No.2 and gear shift fork No.2.



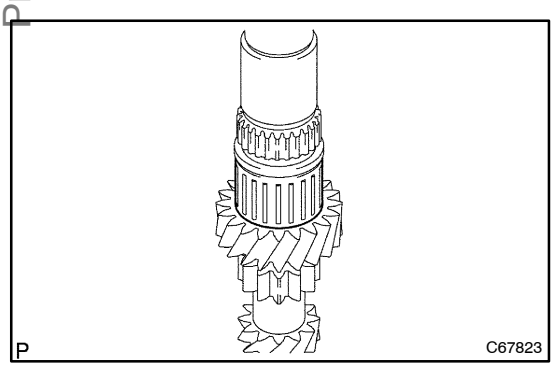
18. INSTALL TRANSMISSION HUB SLEEVE NO.2

(a) Coat the transmission hub sleeve No.2 with gear oil, and install it to the transmission clutch hub No.2.

NOTICE:

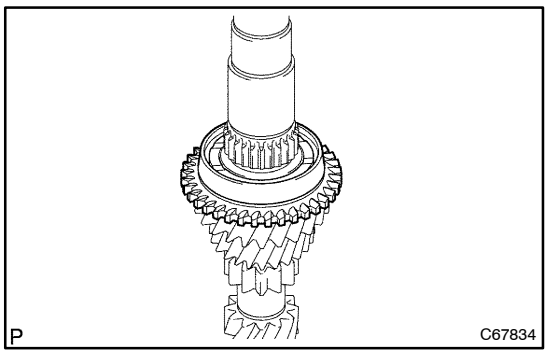
Do not set the transmission hub sleeve No.2 and the transmission clutch hub No.2 in incorrect orientation.

(b) Install the 3 synchromesh key springs with 3 synchromesh shifting keys No.2 to the transmission clutch hub No.2.



19. INSTALL 3RD GEAR NEEDLE ROLLER BEARING

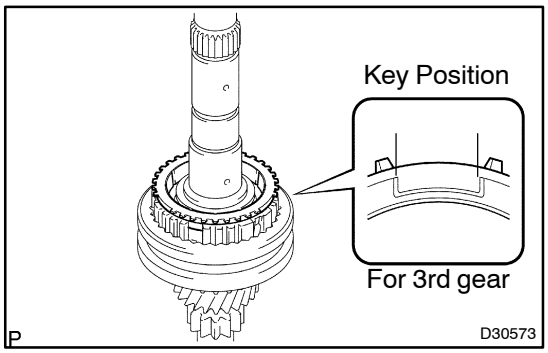
(a) Coat the 3rd gear needle roller bearing with gear oil, and install it to the input shaft.



20. INSTALL 3RD GEAR

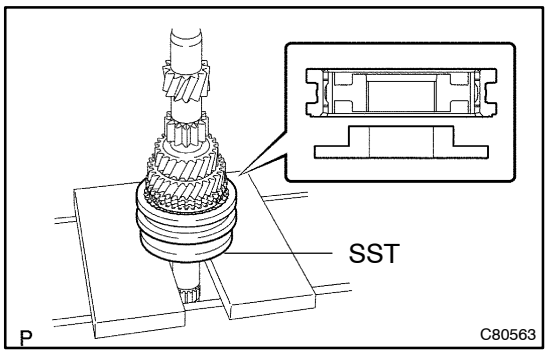
(a) Coat the 3rd gear with gear oil, and install it to the input shaft.

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21. INSTALL SYNCHRONIZER RING NO.2 (FOR THIRD SYNCHRONIZER RING)

(a) Coat the synchronizer ring No.2 with gear oil, and install it to the 3rd gear.



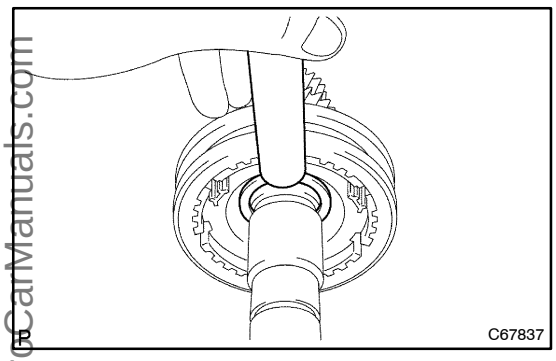
22. INSTALL TRANSMISSION CLUTCH HUB NO.2

(a) Using SST and a press, install the transmission clutch hub No.2 to the input shaft.

SST 09316–60011 (09316–00021)

NOTICE:

Fit the synchromesh shifting key into the groove of the synchronizer ring correctly.



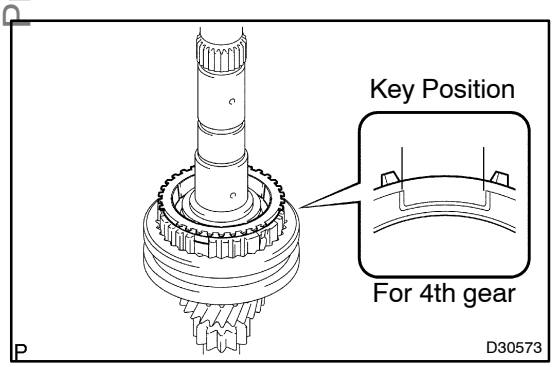
(b) Select a snap ring that will allow the minimum axial play.

Standard clearance:

0.1 mm (0.0039 in.) or less.

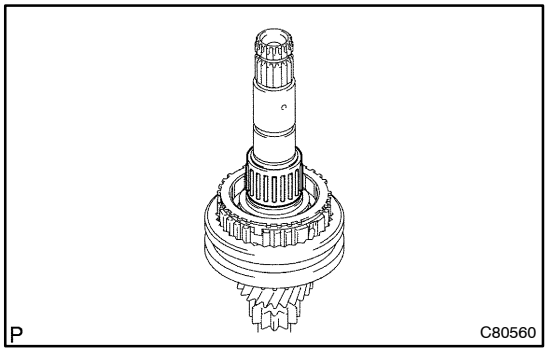
Mark	Thickness mm (in)	Mark	Thickness mm (in)
0	2.30 (0.0906)	3	2.48 (0.0976)
1	2.36 (0.0929)	4	2.54 (0.1000)
2	2.42 (0.0953)	5	2.60 (0.1024)

(c) Using a brass bar and a hammer, install the snap ring to the input shaft.



23. INSTALL SYNCHRONIZER RING NO.2 (FOR FOURTH SYNCHRONIZER RING)

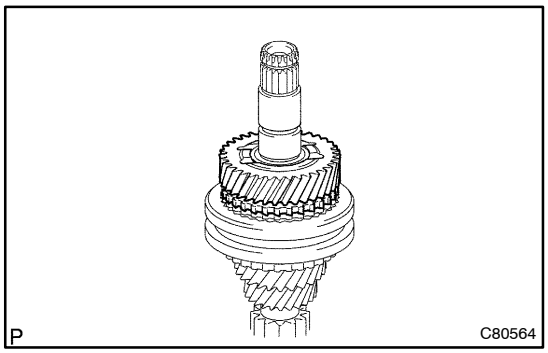
(a) Coat the synchronizer ring No.2 with gear oil, and install it to the transmission clutch hub No.2.



24. INSTALL 4TH GEAR NEEDLE ROLLER BEARING

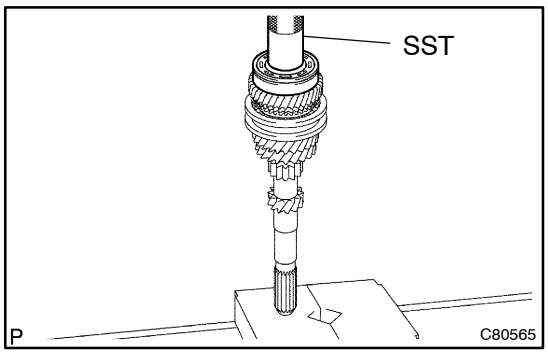
(a) Coat the 4th gear needle roller bearing and 4th gear bearing spacer with gear oil, and install them to the transmission clutch hub No.2.

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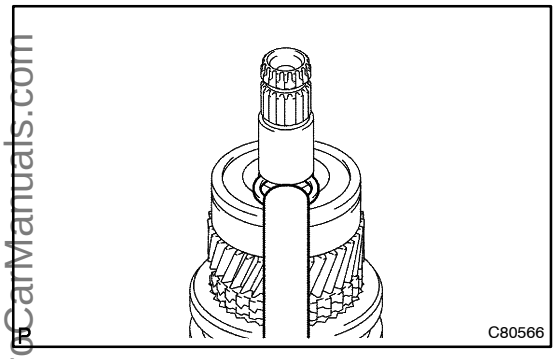
25. INSTALL 4TH GEAR

(a) Coat the 4th gear with gear oil, and install it to the input shaft.



26. INSTALL INPUT SHAFT REAR RADIAL BALL BEARING

(a) Using SST and a press, install the input shaft rear radial ball bearing to the input shaft.
SST 09636–20010



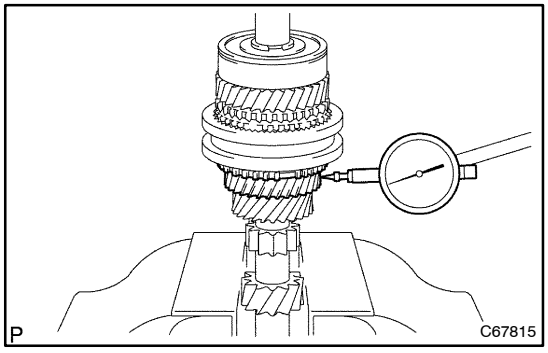
(b) Select a snap ring from the table below that will make the thrust clearance of the input shaft rear radial ball bearing below 0.1 mm (0.0039 in.).

Snap ring thickness:

Mark	Thickness mm (in)	Mark	Thickness mm (in)
A	2.29 (0.0901)	D	2.47 (0.0972)
B	2.35 (0.0925)	E	2.53 (0.0996)
C	2.41 (0.0948)	F	2.59 (0.1019)

(c) Using a brass bar and a hammer, install the snap ring to the input shaft.

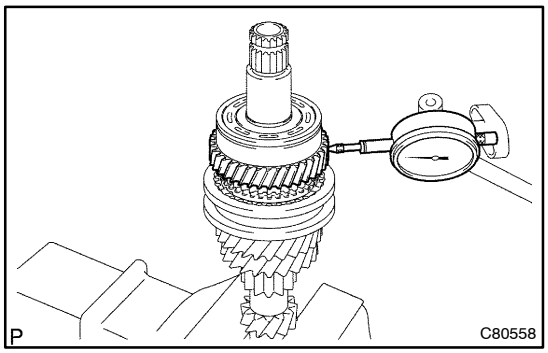
ProCarManuals.com



27. INSPECT 3RD GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the radial clearance between the 3rd gear.
Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum diameter:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 3rd gear, needle roller bearing or input shaft.

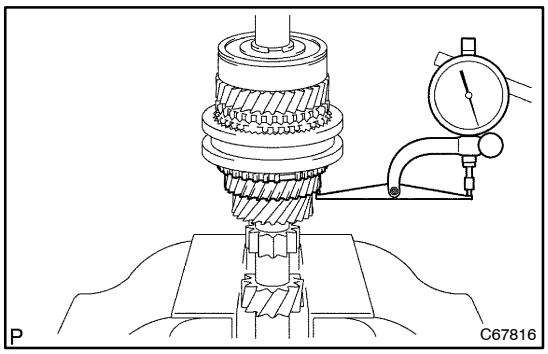


28. INSPECT 4TH GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the radial clearance between the 4th gear.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum diameter:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 4th gear, needle roller bearing or input shaft.

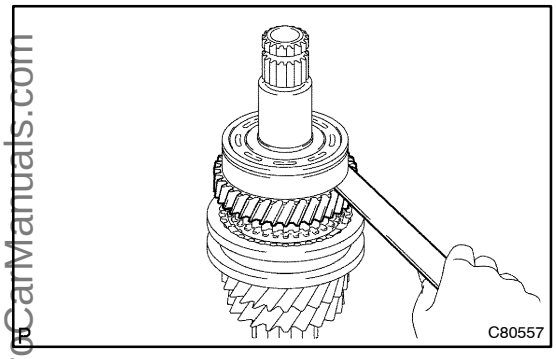


29. INSPECT 3RD GEAR THRUST CLEARANCE

(a) Using a dial gauge, measure the 3rd thrust clearance.

Standard clearance:
0.1 to 0.35 mm (0.0039 to 0.0138 in.)
Maximum diameter:
0.35 mm (0.0138 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.2, 3rd gear or input shaft.



30. INSPECT 4TH GEAR THRUST CLEARANCE

(a) Using a feeler gauge, measure the 4th thrust clearance.

Standard clearance:
0.1 to 0.55 mm (0.0039 to 0.0217 in.)
Maximum diameter:
0.055 mm (0.00217 in.)

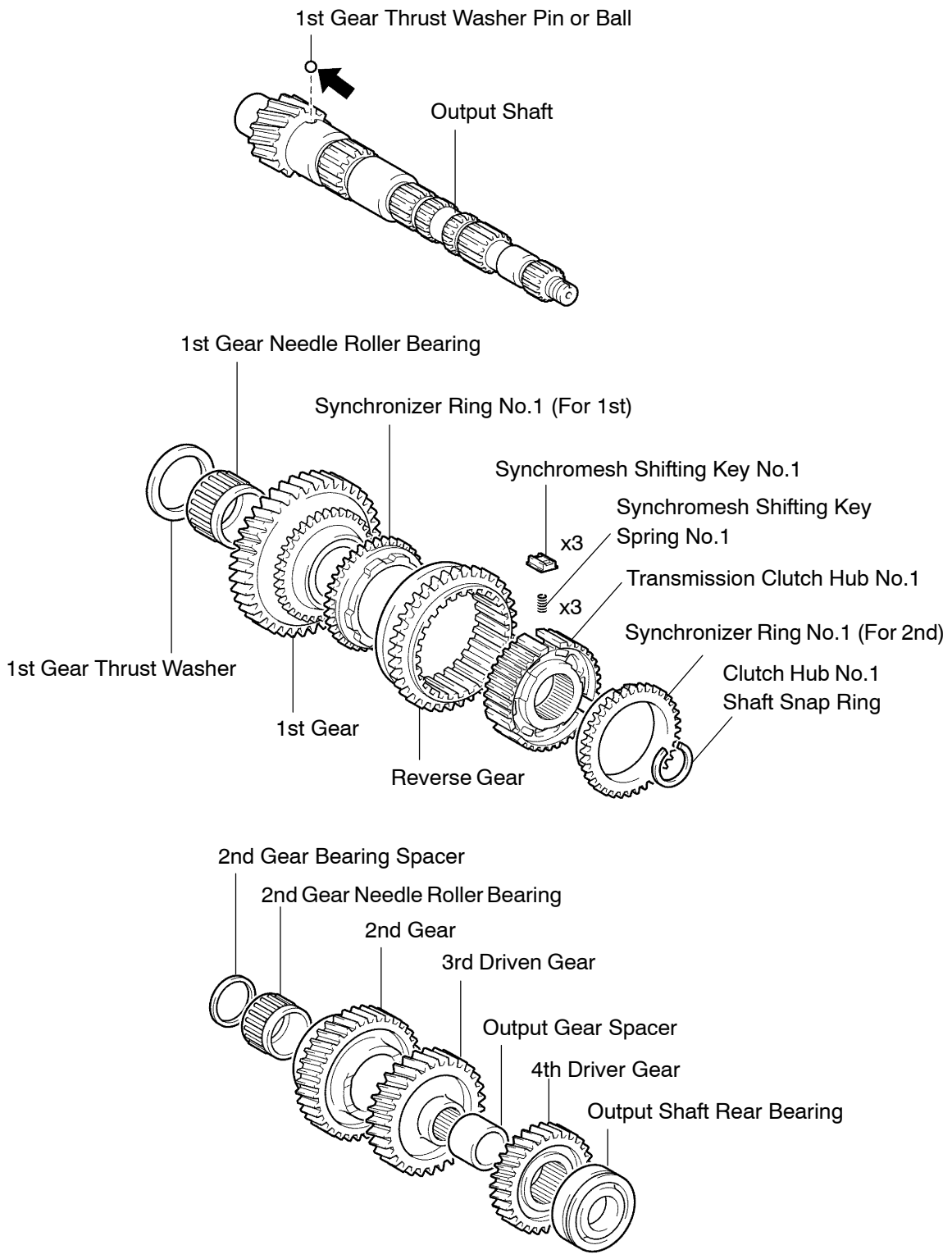
If the clearance exceeds the maximum, replace the transmission clutch hub No.2, 4th gear or input shaft rear radial ball bearing.

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OUTPUT SHAFT ASSY (C50)

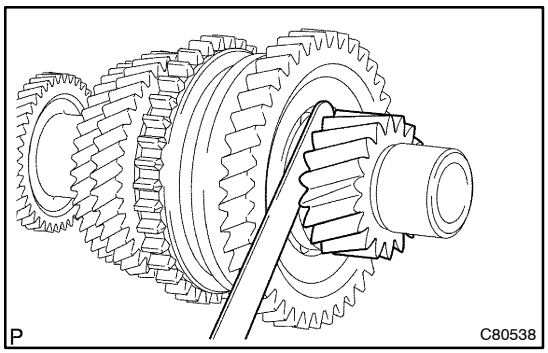
COMPONENTS

410DG–02



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OVERHAUL

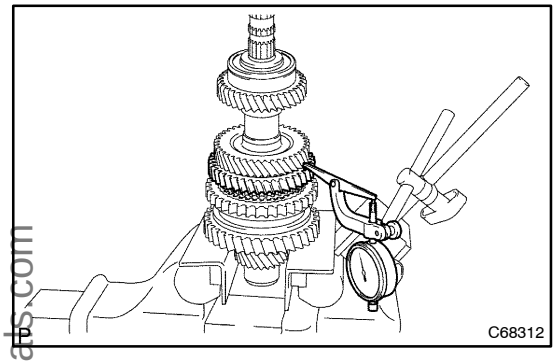


1. INSPECT 1ST GEAR THRUST CLEARANCE

(a) Using a feeler gauge, measure the 1st gear thrust clearance.

Standard clearance:
0.10 to 0.40 mm (0.0039 to 0.0157 in.)
Maximum clearance:
0.40 mm (0.0157 in.)

If the clearance exceeds the maximum, replace the 1st gear thrust washer, 1st gear or transmission clutch hub No.1.

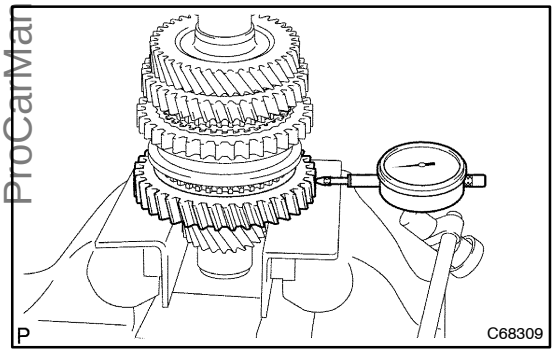


2. INSPECT 2ND GEAR THRUST CLEARANCE

(a) Using a dial gauge, measure the 2nd gear thrust clearance.

Standard clearance:
0.1 to 0.45 mm (0.0039 to 0.0177 in.)
Maximum clearance:
0.45 mm (0.0177 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.1, 2nd gear or 3rd driven gear.

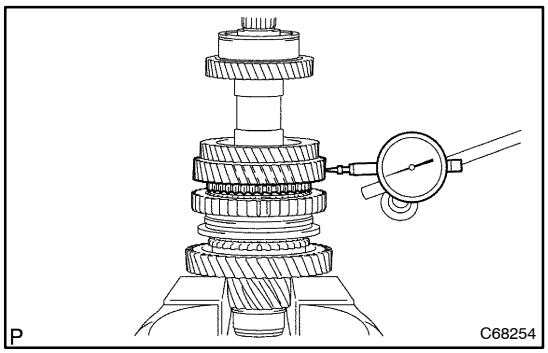


3. INSPECT 1ST GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the 1st gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 1st gear and 1st gear needle roller bearing.



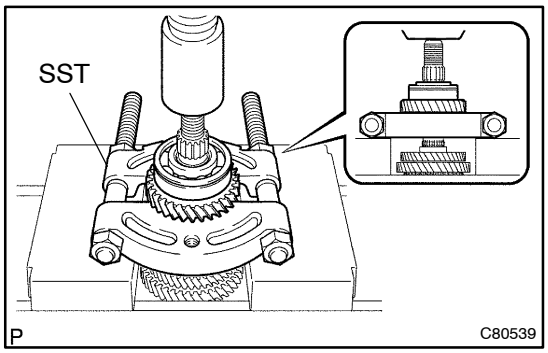
4. INSPECT 2ND GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the 2nd gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

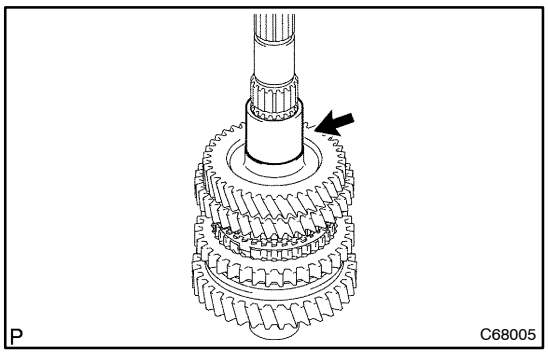
If the clearance exceeds the maximum, replace the 2nd gear and 2nd gear needle roller bearing.

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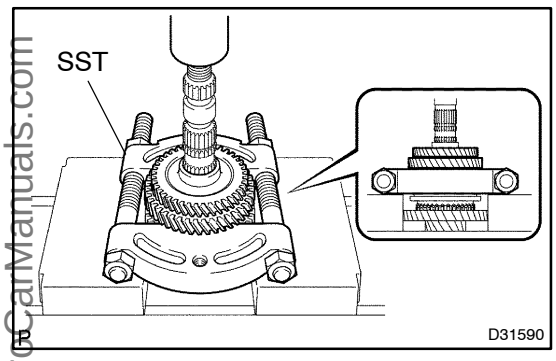
5. REMOVE 4TH DRIVEN GEAR

(a) Using SST and a press, remove the output shaft rear bearing and 4th driven gear from the output shaft.
SST 09950–00020



6. REMOVE OUTPUT GEAR SPACER

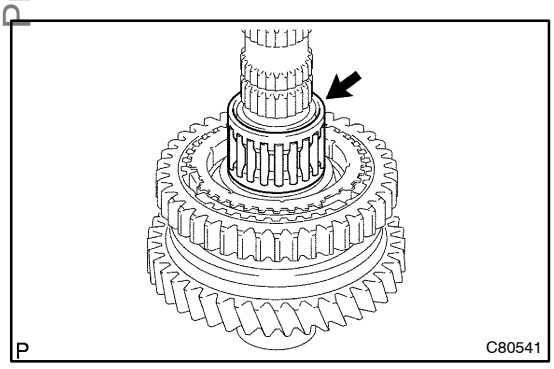
(a) Remove the output gear spacer from the output shaft.



7. REMOVE 2ND GEAR

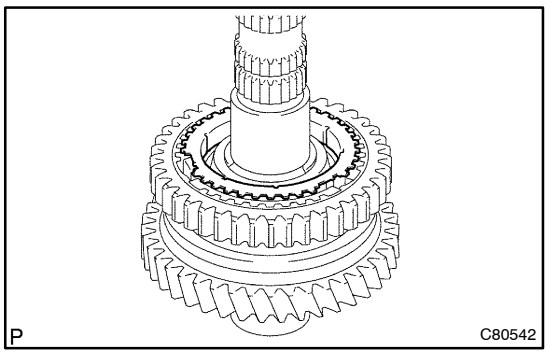
(a) Using SST, remove the 3rd driven gear and 2nd gear from the out put shaft.
SST 09950–00020

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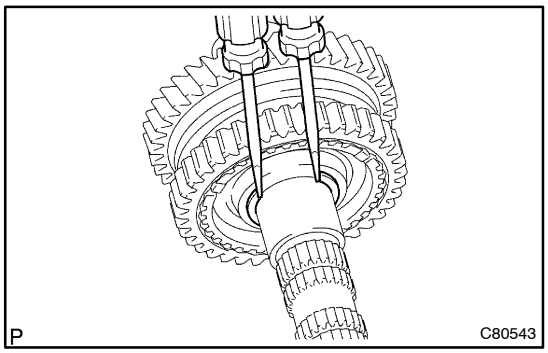
8. REMOVE 2ND GEAR NEEDLE ROLLER BEARING

(a) Remove the 2nd gear needle roller bearing and 2nd gear bearing spacer from the output shaft.



9. REMOVE SYNCHRONIZER RING NO.1 (FOR SECOND SYNCHRONIZER RING)

(a) Remove the synchronizer ring No.1 from the output shaft.

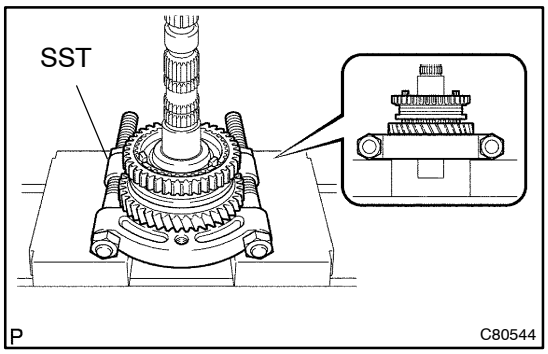


10. REMOVE 1ST GEAR

(a) Using 2 screwdrivers and a hammer, remove the clutch hub No.1 shaft snap ring from the output shaft.

NOTICE:

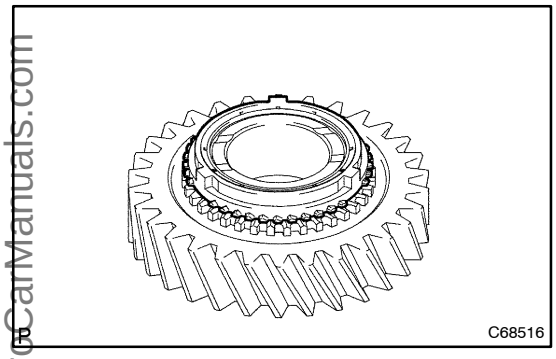
Use a shop rag to keep the shaft snap ring from flying.



(b) Using SST and a press, remove the clutch hub No.1 assy and 1st gear from the output shaft.
SST 09950–00020

NOTICE:

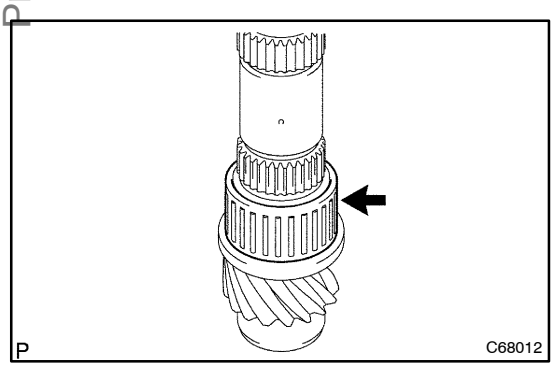
- **Do not tighten SST excessively.**
- **Support the input shaft by hand so that it will not be dropped off.**



11. REMOVE SYNCHRONIZER RING NO.1 (FOR FIRST SYNCHRONIZER RING)

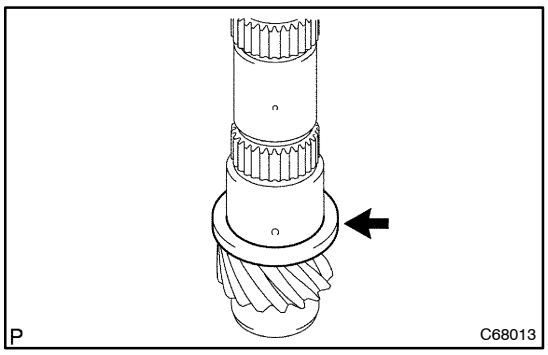
(a) Remove the synchronizer ring No.1 from the 1st gear.

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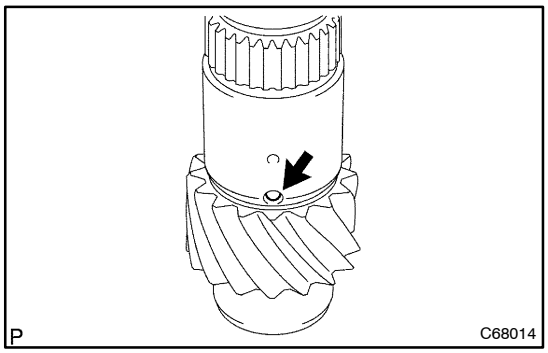
12. REMOVE 1ST GEAR NEEDLE ROLLER BEARING

(a) Remove the 1st gear needle roller bearing from the output shaft.



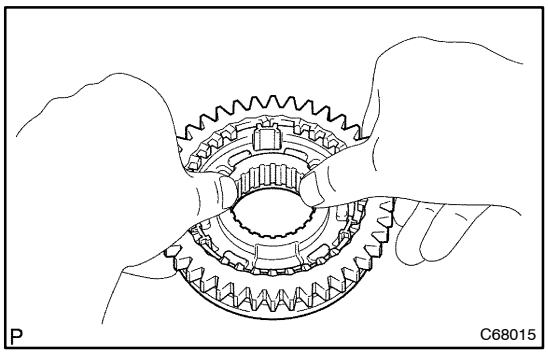
13. REMOVE 1ST GEAR THRUST WASHER

(a) Remove the 1st gear thrust washer from the output shaft.



14. REMOVE 1ST GEAR THRUST WASHER PIN OR BALL

(a) Remove the 1st gear thrust washer pin or ball from the output shaft.

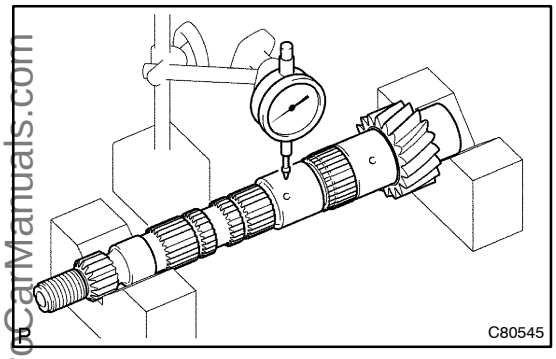


15. REMOVE REVERSE GEAR

(a) Remove the reverse gear, 3 synchromesh shifting keys and 3 synchromesh key springs.

NOTICE:

Use a shop rag to keep the shifting key and shifting key spring from flying.



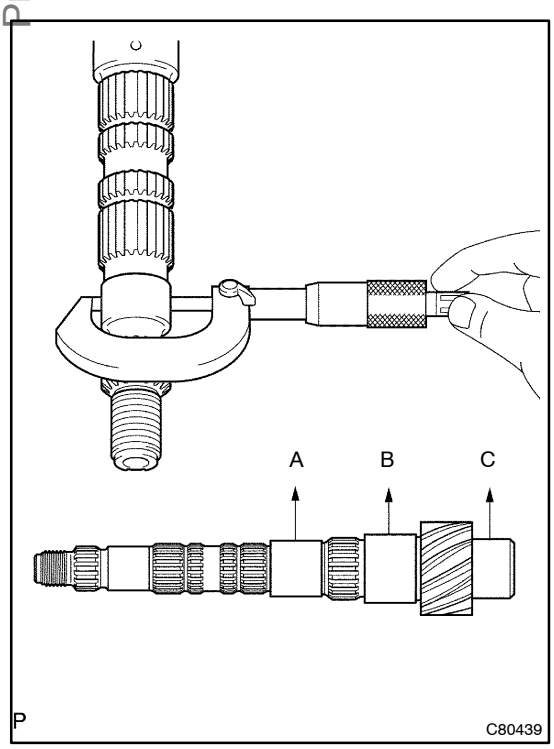
16. INSPECT OUTPUT SHAFT

(a) Using a dial gauge, check the output shaft runout.

Minimum runout: 0.015 mm (0.0006 in.)

If the runout exceeds the maximum, replace the output shaft.

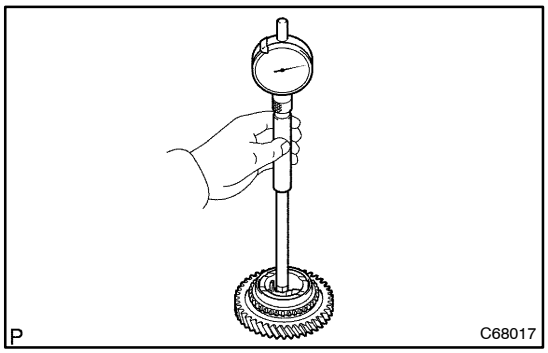
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(b) Using a micrometer, measure the outer diameter of the output shaft journal surface.

Standard outer diameter:
Part A: 31.985 to 32.000 mm (1.2592 to 1.2598 in.)
Part B: 37.985 to 38.000 mm (1.4955 to 1.2598 in.)
Part C: 32.985 to 33.000 mm (1.2986 to 1.2992 in.)
Minimum outer diameter:
Part A: 31.985 mm (1.2592 in.)
Part B: 37.985 mm (1.4955 in.)
Part C: 32.985 mm (1.2986 in.)

If the outer diameter is less than the minimum, replace the output shaft.

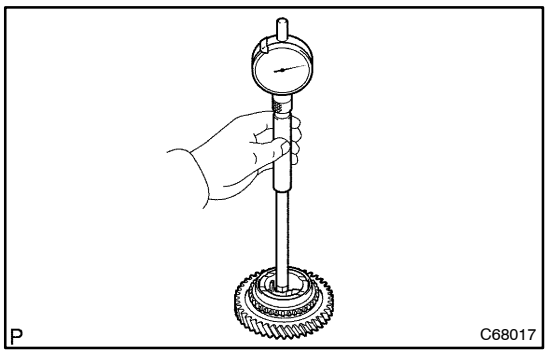


17. INSPECT 2ND GEAR

(a) Using a cylinder gauge, measure the inside diameter of the 2nd gear.

Standard inside diameter:
38.015 to 38.031 mm (1.4966 to 1.4972 in.)
Maximum inside diameter:
38.031 mm (1.4972 in.)

If the inside diameter exceeds the maximum, replace the 2nd gear.

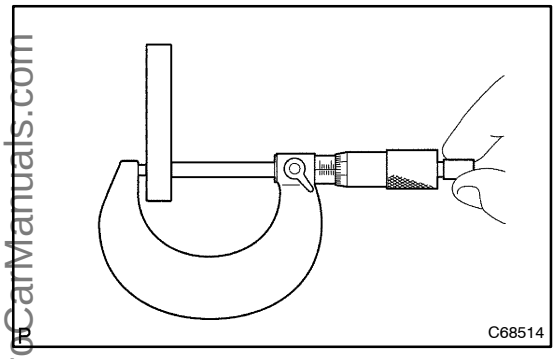


18. INSPECT 1ST GEAR

(a) Using a cylinder gauge, measure the clearance of the 1st gear.

Standard inside diameter:
44.015 to 44.031 mm (1.7328 to 1.7335 in.)
Maximum inside diameter:
44.031 mm (1.7335 in.)

If the inside diameter exceeds the maximum, replace the 1st gear.



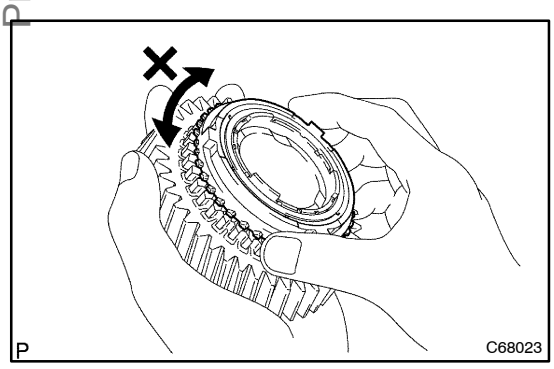
19. INSPECT 1ST GEAR THRUST WASHER

(a) Using a micrometer, measure the 1st gear thrust washer.

Standard thickness:
5.975 to 6.025 mm (0.2352 to 0.2372 in.)
Minimum thickness:
5.975 mm (0.2352 in.)

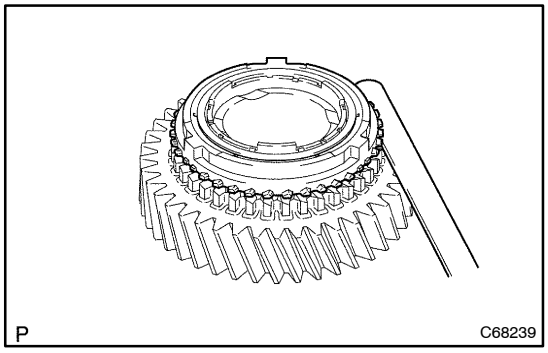
If the clearance is less than the minimum, replace the 1st gear.

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20. INSPECT SYNCHRONIZER RING NO.1 (FOR SECOND SYNCHRONIZER RING)

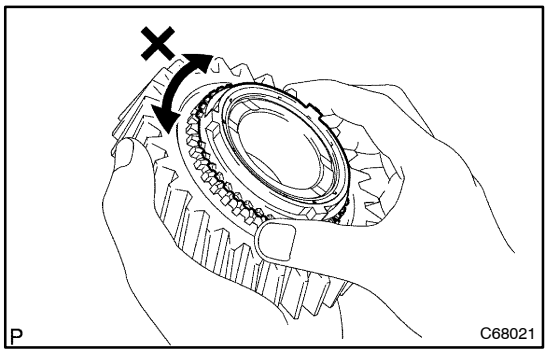
(a) Coat the 2nd gear cone with gear oil. Check the braking effect of the synchronizer ring. Turn the synchronizer ring in one direction while pushing it to the gear cone. Check that the ring locks.



(b) Using a feeler gauge, measure the clearance between the back of the synchronizer ring back and gear spline end.

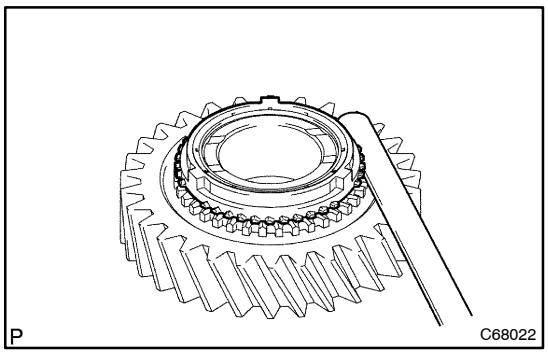
Standard clearance:
0.70 to 1.30 mm (0.0276 to 0.0512 in.)
Minimum clearance:
0.70 mm (0.0276in.)

If the clearance is less than the minimum, replace the synchronizer ring.



21. INSPECT SYNCHRONIZER RING NO.1 (FOR FIRST SYNCHRONIZER RING)

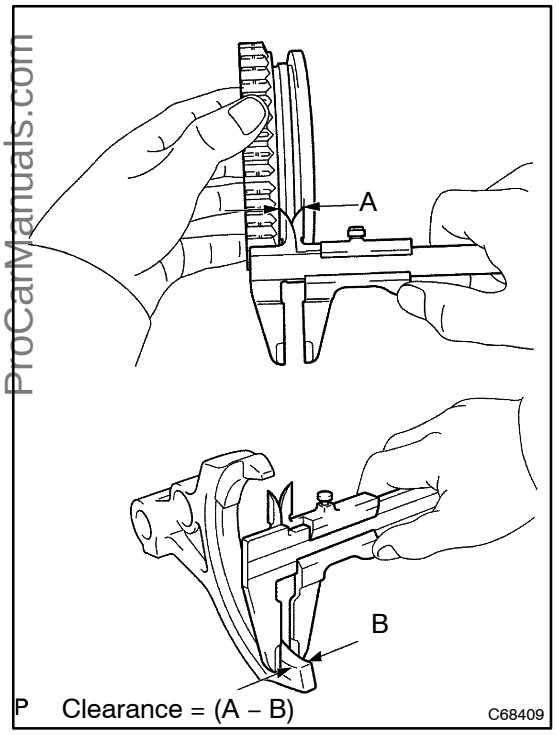
(a) Coat the 1st gear cone with gear oil. Check the braking effect of the synchronizer ring. Turn the synchronizer ring in one direction while pushing it to the gear cone. Check that the ring locks.



(b) Using a feeler gauge, measure the clearance between the synchronizer ring back and gear spline end.

Standard clearance:
0.75 to 1.65 mm (0.0295 to 0.0650 in.)
Minimum clearance:
0.75 mm (0.0295 in.)

If the clearance is less than the minimum, replace the synchronizer ring.

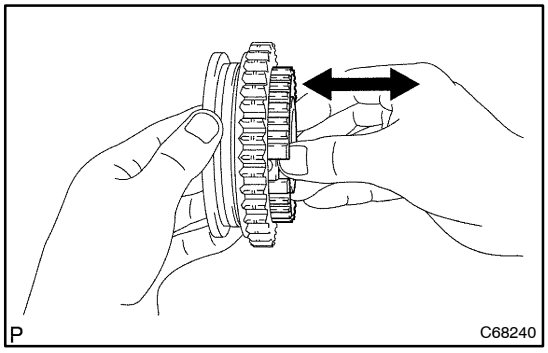


22. INSPECT REVERSE GEAR

(a) Using vernier calipers, measure the clearance between the reverse gear and shift fork.

Maximum clearance: 0.35 mm (0.014 in.)

If the clearance exceeds the maximum, replace the shift fork and reverse gear.

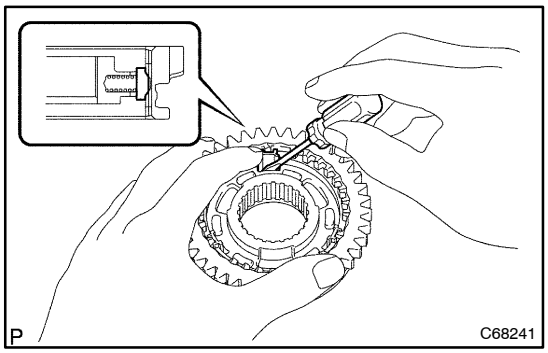


23. INSPECT TRANSMISSION CLUTCH HUB NO.1

(a) Check that the transmission clutch hub No.1 and reverse gear slide smoothly.

(b) Check that the spline gear's edges of the reverse gear are not worn down.

ProCarManuals.com



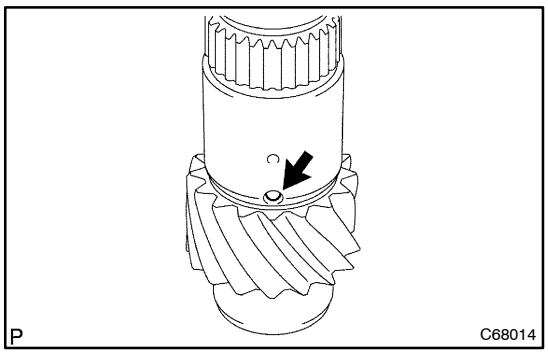
24. INSTALL REVERSE GEAR

(a) Coat the reverse gear with gear oil, and install it to the transmission clutch hub No.1.

NOTICE:

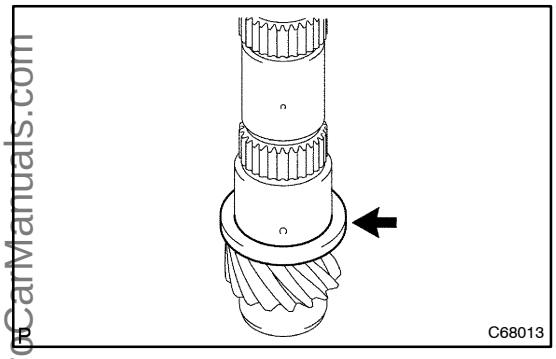
Do not set the reverse gear and transmission clutch hub No.1 in incorrect orientation.

(b) Install the 3 synchromesh shifting key springs and 3 synchromesh shifting keys.



25. INSTALL 1ST GEAR THRUST WASHER PIN OR BALL

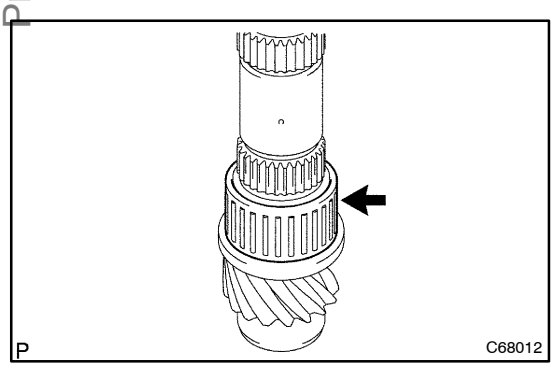
(a) Coat the 1st gear thrust washer pin or ball with gear oil, and install it to the output shaft.



26. INSTALL 1ST GEAR THRUST WASHER

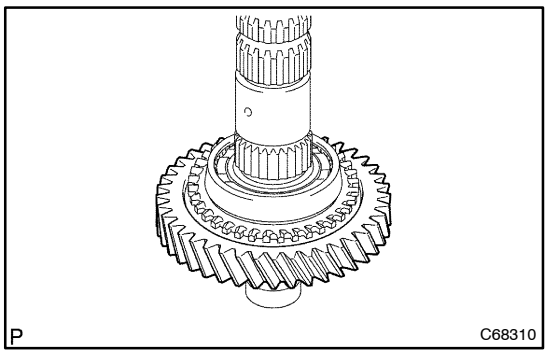
(a) Coat the 1st gear thrust washer with gear oil, and install it to the output shaft.

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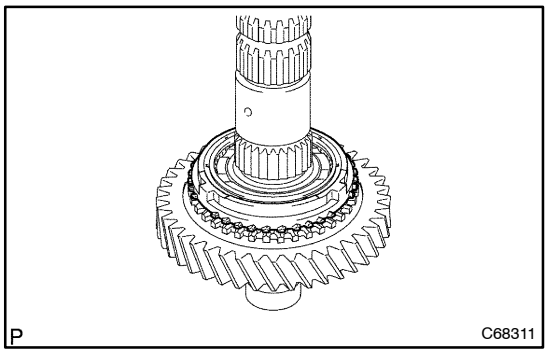
27. INSTALL 1ST GEAR NEEDLE ROLLER BEARING

(a) Coat the 1st gear needle roller bearing with gear oil, and install it to the output shaft.



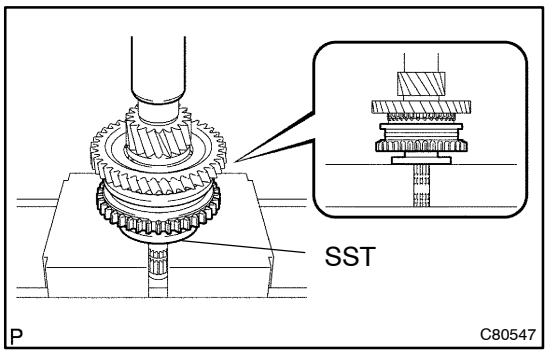
28. INSTALL 1ST GEAR

(a) Coat the 1st gear with gear oil, and install it to the output shaft.



29. INSTALL SYNCHRONIZER RING NO.1 (FOR FIRST SYNCHRONIZER RING)

(a) Coat the synchronizer ring No.1 with gear oil, and install it to the 1st gear.



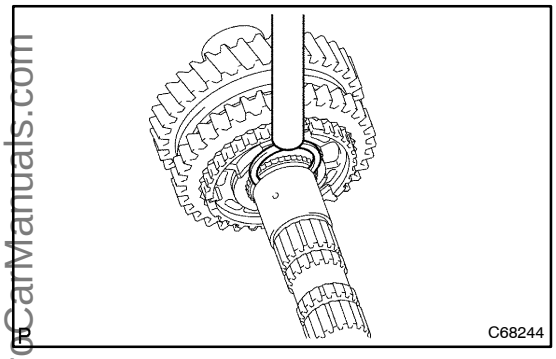
30. INSTALL TRANSMISSION CLUTCH HUB NO.1

(a) Using SST and a press, install the transmission clutch hub No.1 to the output shaft

SST 09316–60011 (09316–00031)

HINT:

- The 1st gear can be turned.
- While checking that the 1st gear thrust washer pin or ball is inserted into the groove of the 1st gear thrust washer, press and fit the clutch hub No.1.



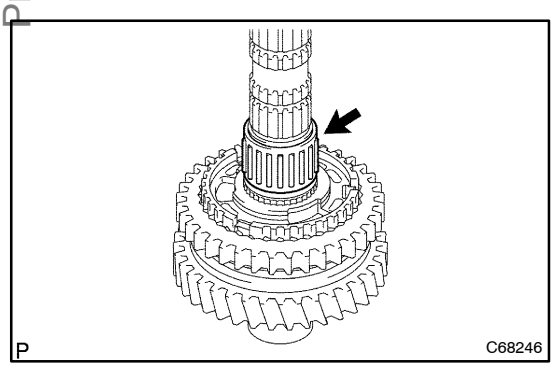
(b) Select a snap ring that will allow the minimum axial play.

Standard clearance:
0.1 mm (0.039 in.) or less

Mark	Thickness mm (in.)	Mark	Thickness mm (in.)
A	2.50 (0.0984)	D	2.68 (0.1055)
B	2.56 (0.1008)	E	2.74 (0.1079)
C	2.62 (0.1031)	F	2.80 (0.1102)

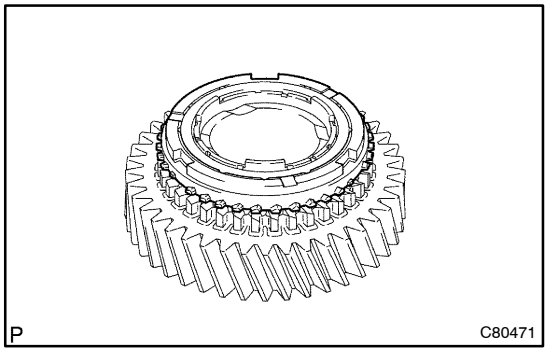
(c) Using a brass bar and a hammer, install the snap ring to the output shaft.

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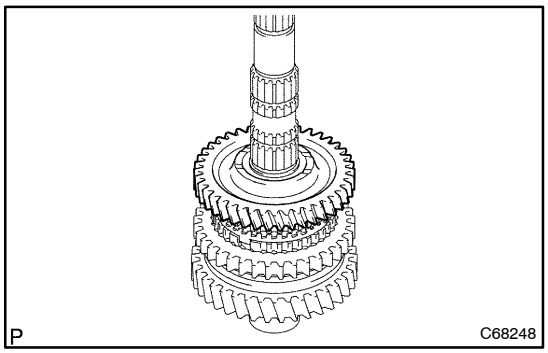
31. INSTALL 2ND GEAR NEEDLE ROLLER BEARING

(a) Coat the 2nd gear needle roller bearing and 2nd gear bearing spacer with gear oil, and install them to the output shaft.



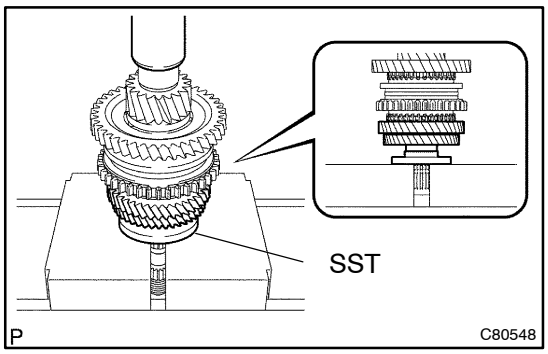
32. INSTALL SYNCHRONIZER RING NO.1 (FOR SECOND SYNCHRONIZER RING)

(a) Coat the synchronizer ring No.1 with gear oil, and install it to the 2nd gear.



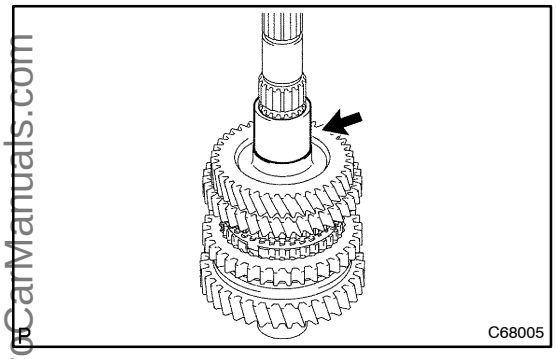
33. INSTALL 2ND GEAR

(a) Coat the 2nd gear with gear oil, and install it to the output shaft.



34. INSTALL 3RD DRIVEN GEAR

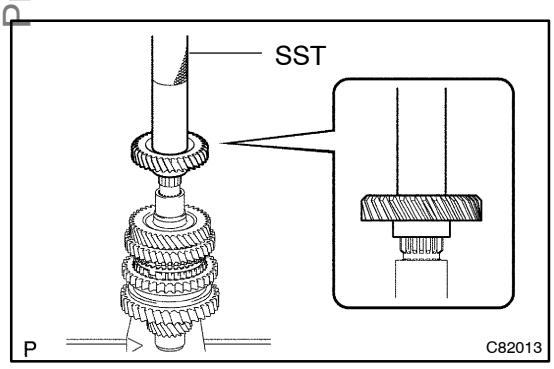
(a) Using SST and a press, install the 3rd driven gear to the output shaft.
SST 09950–00020



35. INSTALL OUTPUT GEAR SPACER

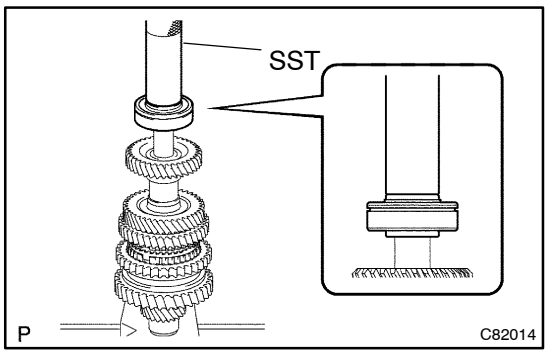
(a) Install the output gear spacer to the output shaft.

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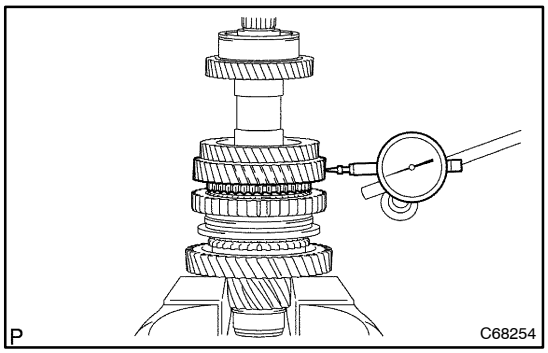
36. INSTALL 4TH DRIVEN GEAR

(a) Using SST and a press, install the 4th driven gear to the output shaft.
SST 09612–22011



37. INSTALL OUTPUT SHAFT REAR BEARING

(a) Using SST and a press, install the output shaft rear bearing to the output shaft.
SST 09612–22011

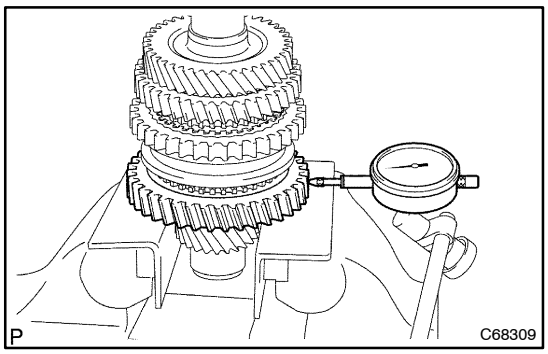


38. INSPECT 2ND GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the 2nd gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 2nd gear and 2nd gear needle roller bearing.

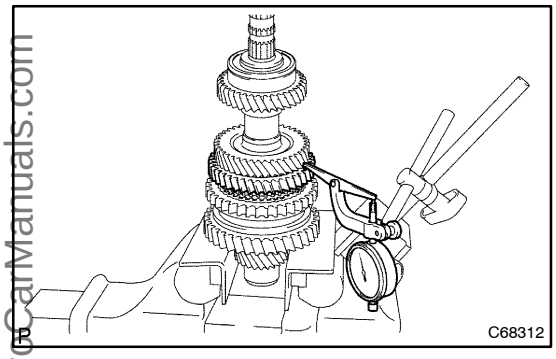


39. INSPECT 1ST GEAR RADIAL CLEARANCE

(a) Using a dial gauge, measure the 1st gear radial clearance between the gear and shaft.

Standard clearance:
0.015 to 0.058 mm (0.0006 to 0.0023 in.)
Maximum clearance:
0.058 mm (0.0023 in.)

If the clearance exceeds the maximum, replace the 1st gear and 1st gear needle roller bearing.

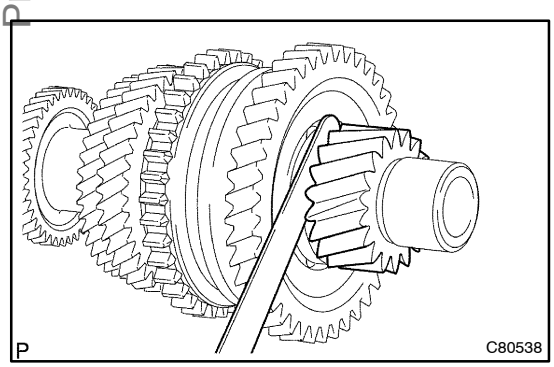


40. INSPECT 2ND GEAR THRUST CLEARANCE

(a) Using a dial gauge, measure the 2nd gear thrust clearance.

Standard clearance:
0.10 to 0.45 mm (0.0039 to 0.0177 in.)
Maximum clearance:
0.45 mm (0.0177 in.)

If the clearance exceeds the maximum, replace the transmission clutch hub No.1, 2nd gear or 3rd driven gear.



41. INSPECT 1ST GEAR THRUST CLEARANCE

(a) Using a feeler gauge, measure the 1st gear thrust clearance.

Standard clearance:
0.10 to 0.40 mm (0.0039 to 0.0157 in.)
Maximum clearance:
0.40 mm (0.0157 in.)

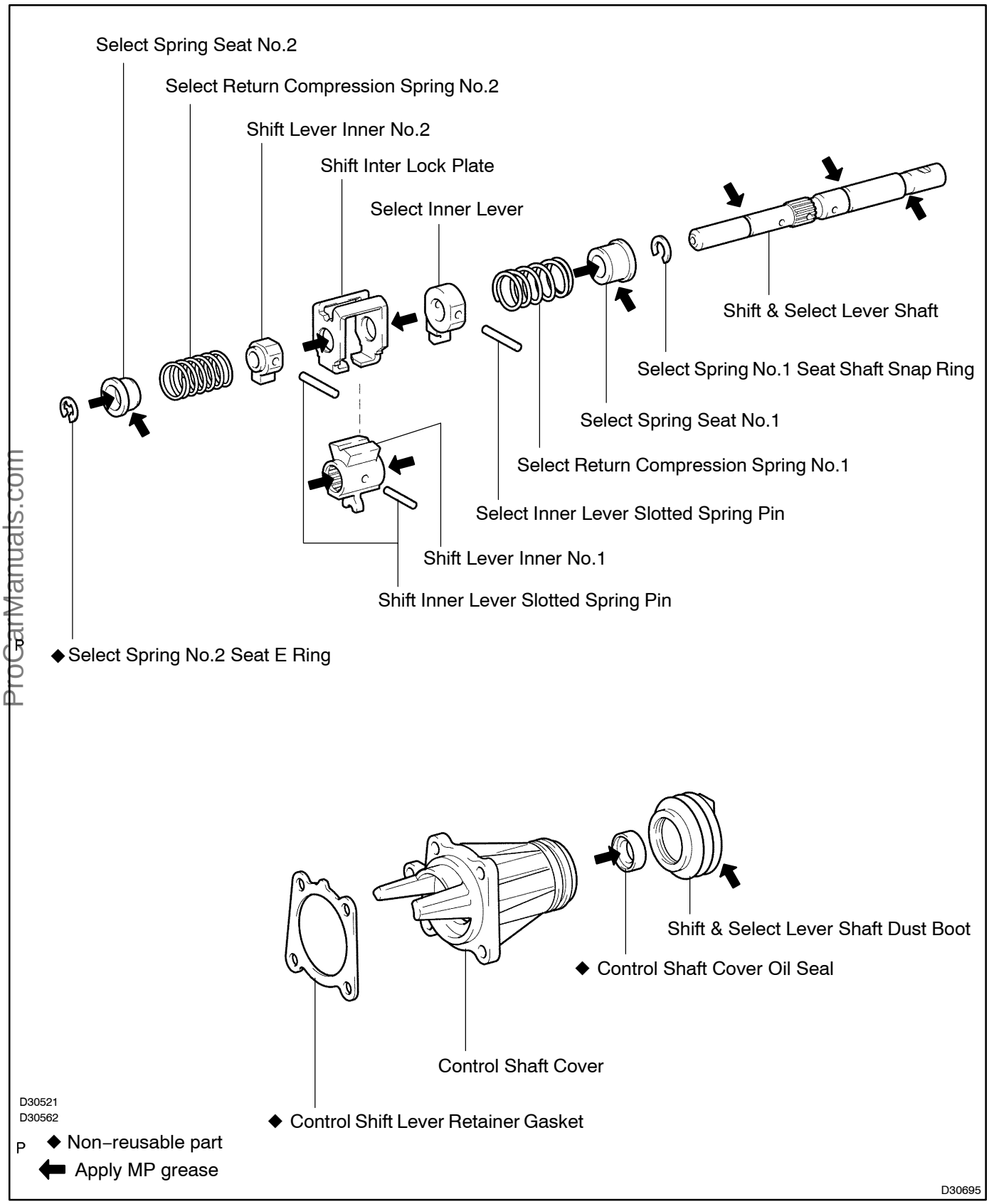
If the clearance exceeds the maximum, replace the 1st gear thrust washer, 1st gear or transmission clutch hub No.1.

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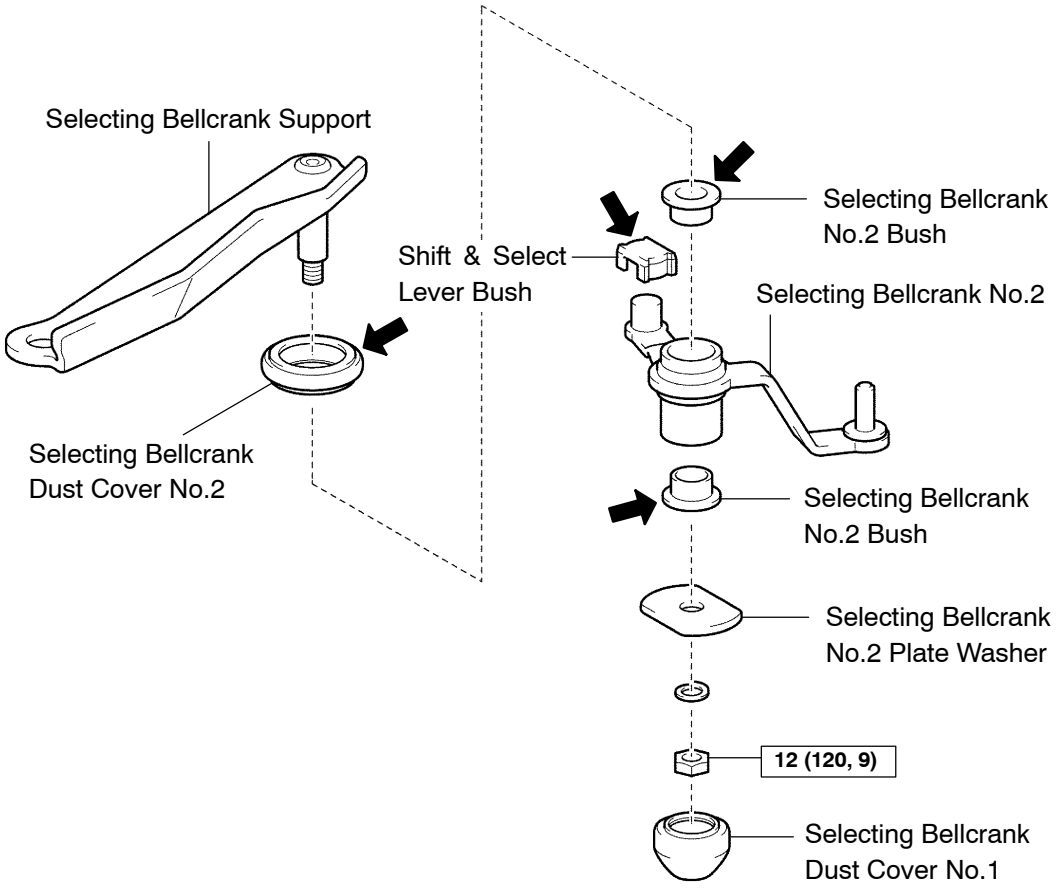
SHIFT & SELECT LEVER SHAFT ASSY (C50)

COMPONENTS

410DI–02



ProCarManuals.com



N·m (kgf·cm, ft·lbf) : Specified torque

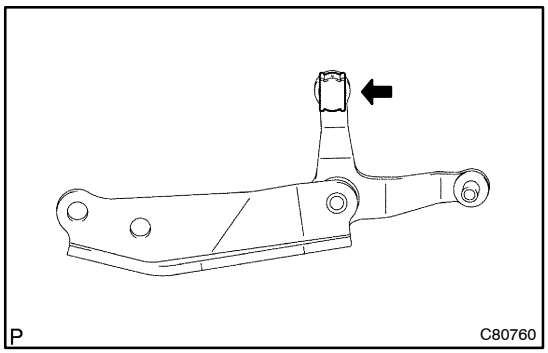
Apply MP grease

C94578

ProCarManuals.com

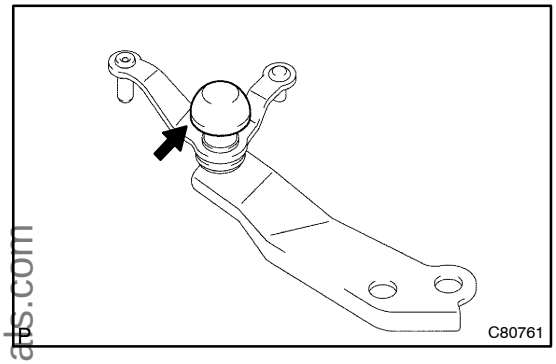
410DJ–02

OVERHAUL



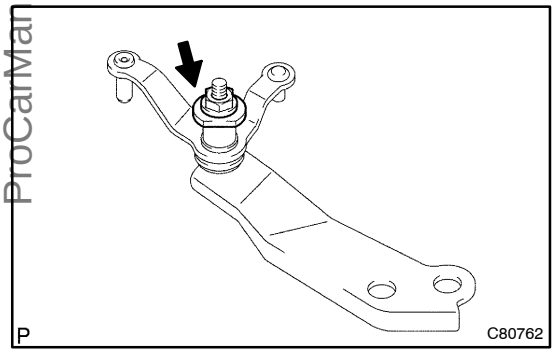
1. REMOVE SHIFT & SELECT LEVER BUSH

(a) Remove the shift & select lever bush from the selecting bell crank assy.



2. REMOVE SELECTING BELLCRANK DUST COVER NO.1

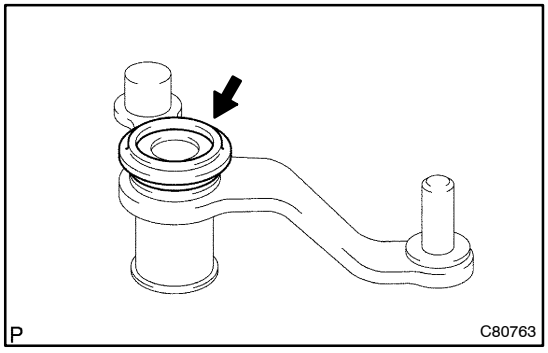
(a) Remove the selecting bellcrank dust cover No.1 from the selecting bellcrank assy.



3. REMOVE SELECTING BELL CRANK NO.2

(a) Remove the nut, spring washer and selecting bellcrank No.2 plate washer.

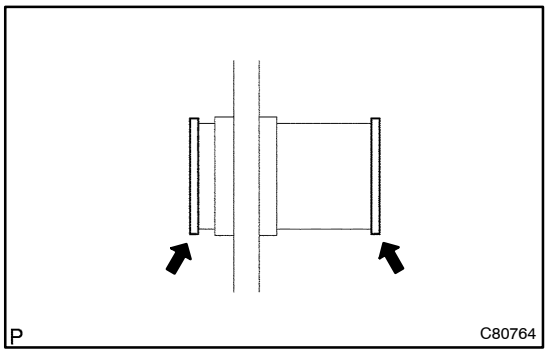
(b) Remove selecting bellcrank No.2 from the selecting bellcrank support.



4. REMOVE SELECTING BELLCRANK DUST COVER NO.2

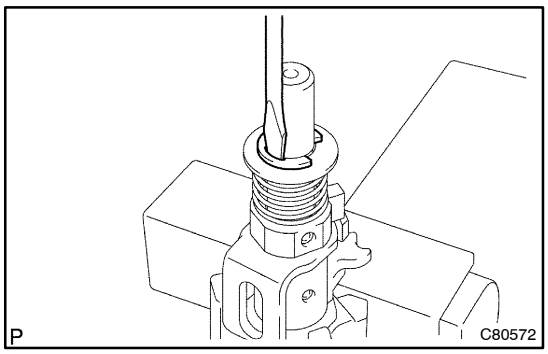
(a) Remove the selecting bellcrank dust cover No.2 from the selecting bellcrank No.2.

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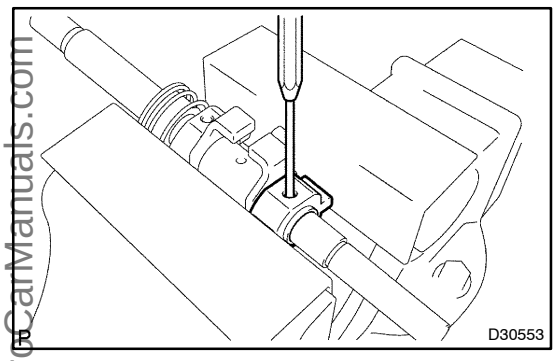
5. REMOVE SELECTING BELLCRANK NO.2 BUSH

(a) Remove the 2 selecting bellcrank No.2 bushes from the selecting bellcrank No.2.



6. REMOVE SELECT SPRING SEAT NO.2

(a) Using a screwdriver, remove the select spring seat No.2 E–ring, select spring seat No.2 and select return spring No.2 from the shift & select lever shaft.



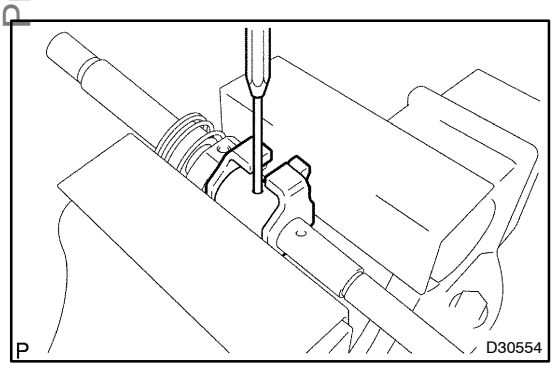
7. REMOVE SHIFT LEVER INNER NO.2

(a) Using a pin punch (ϕ 5 mm) and a hammer, remove the shift inner lever slotted spring pin and shift lever inner No.2 from the shift & select lever shaft.

HINT:

Make sure of the orientation of the shift lever inner No.2.

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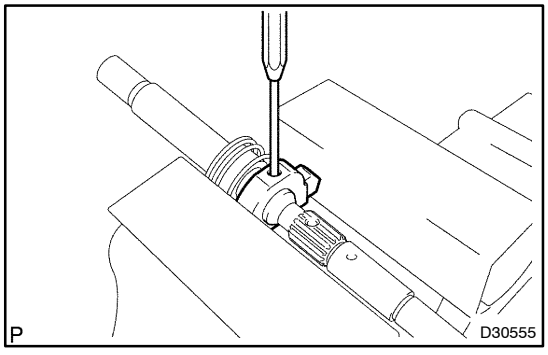


8. REMOVE SHIFT LEVER INNER NO.1

(a) Using a pin punch (ϕ 5 mm) and a hammer, remove the shift inner lever slotted spring pin, shift lever inner No.1 and shift inter lock plate from the shift & select lever shaft.

HINT:

Make sure of the orientation of the shift lever inner No.1.

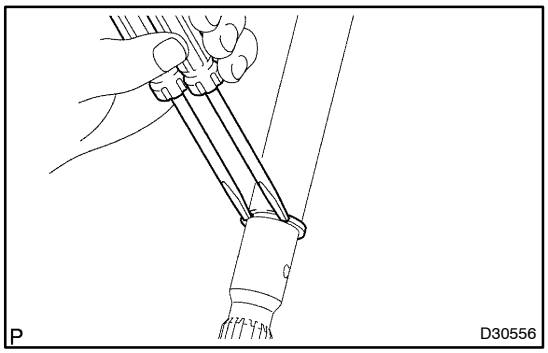


9. REMOVE SELECT INNER LEVER

(a) Using a pin punch (ϕ 5 mm) and a hammer, remove the select inner lever slotted spring pin, select inner lever select return compression spring No.1 and select return spring No.1 seat from shift & select lever shaft.

HINT:

Make sure of the orientation of the select inner lever.



10. REMOVE SELECT SPRING NO.1 SEAT SHAFT SNAP RING

(a) Using 2 screwdrivers and a hammer, remove the select spring No.1 seat shaft snap ring.

NOTICE:

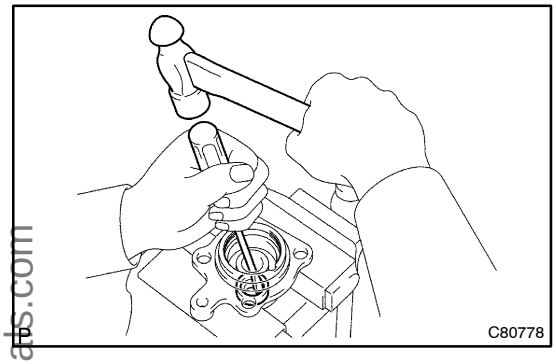
Do not damage the shaft.

HINT:

Use a shop rag to keep the shaft snap ring from flying.

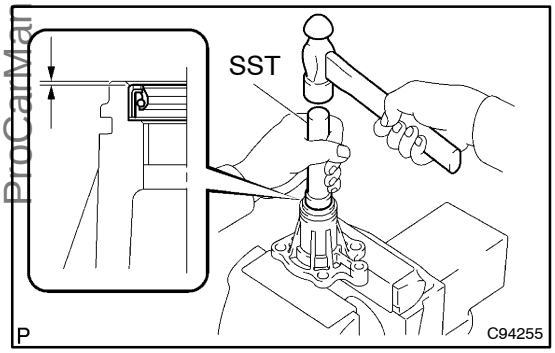
11. REMOVE SHIFT & SELECT LEVER SHAFT DUST BOOT

(a) Remove the shift & select lever shaft dust boot from the shift & select lever shaft.



12. REMOVE CONTROL SHAFT COVER OIL SEAL

(a) Using a screwdriver, remove the control shaft cover oil seal from the control shaft cover.



13. INSTALL CONTROL SHAFT COVER OIL SEAL

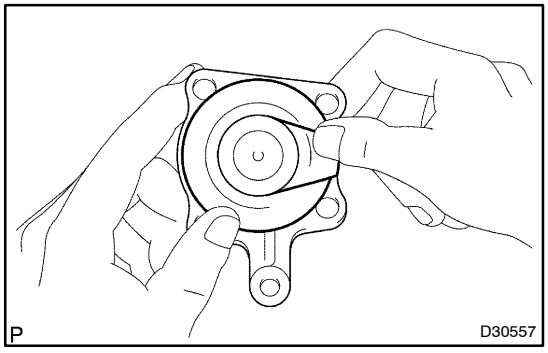
(a) Using SST and a hammer, install a new control shaft cover oil seal to the control shaft cover.

SST 09950–60010 (09951–00220), 09950–70010 (09951–07100)

Drive in depth:

0.2 to 1.2 mm (0.079 to 0.0472 in.)

(b) Coat the control shaft cover oil seal with MP grease.



14. INSTALL SHIFT & SELECT LEVER SHAFT DUST BOOT

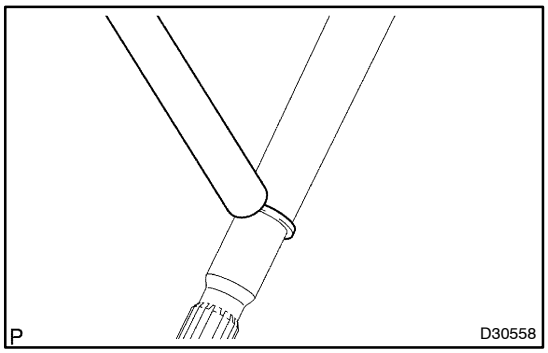
(a) Coat the shift & select lever shaft dust boot with MP grease.

(b) Install the shift & select lever shaft dust boot to the shift & select lever shaft.

HINT:

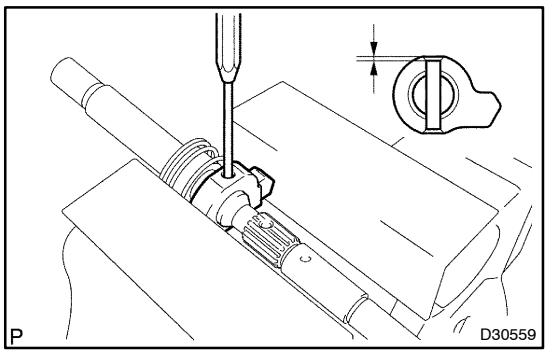
Install the boot with the projection facing up and the hole side down.

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15. INSTALL SELECT SPRING NO.1 SEAT SHAFT SNAP RING

(a) Coat the shift & select lever shaft with MP grease, and install it and snap ring with a brass bar and a hammer.

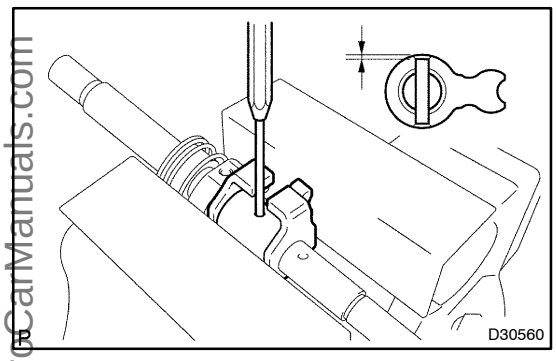


16. INSTALL SELECT INNER LEVER

(a) Coat the select spring No.1 seat with MP grease.

(b) Using a pin punch (ϕ 5 mm) and a hammer, install the select spring No.1 seat, select return compression spring No.1, select inner lever and select inner lever slotted spring pin to the shift & select lever.

Drive in depth: 3.0 to 4.0 mm (0.1181 to 0.1575 in)



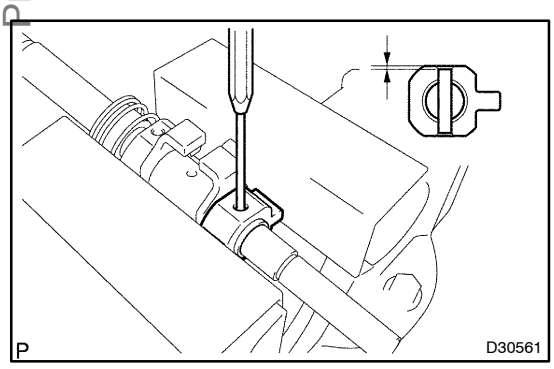
17. INSTALL SHIFT LEVER INNER NO.1

(a) Coat the shift inter lock plate and shift lever inner No.1 with MP grease.

(b) Using a pin punch (ϕ 5 mm) and a hammer, install the shift inter lock plate, shift lever inner No.1 and shift inner lever slotted spring pin to the shift & select lever shaft.

Drive in depth: –0.5 to 0.5 mm (–0.0197 to 0.0197 in)

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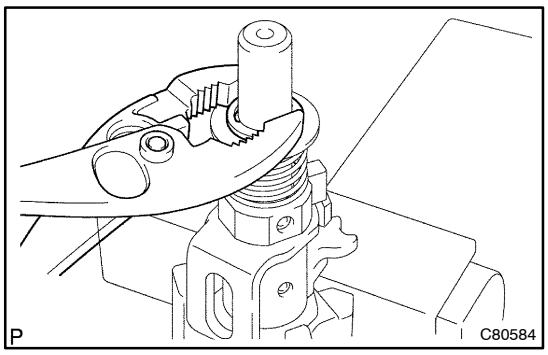


18. INSTALL SHIFT LEVER INNER NO.2

(a) Coat the shift lever inner No.2 with Mp grease.

(b) Using a pin punch (ϕ 5 mm) and a hammer, install the shift lever inner No.2 and shift inner lever slotted spring pin to the shift & select lever shaft.

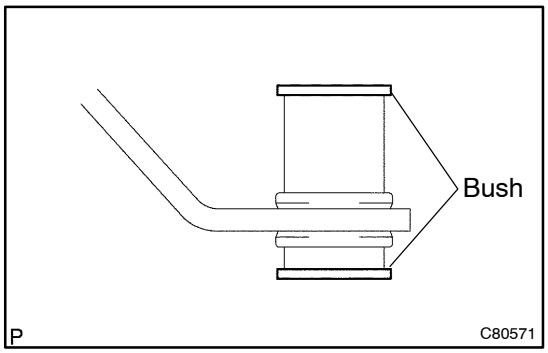
Drive in depth: –0.5 to 0.5 mm (–0.0197 to 0.0197 in)



19. INSTALL SELECT SPRING SEAT NO.2

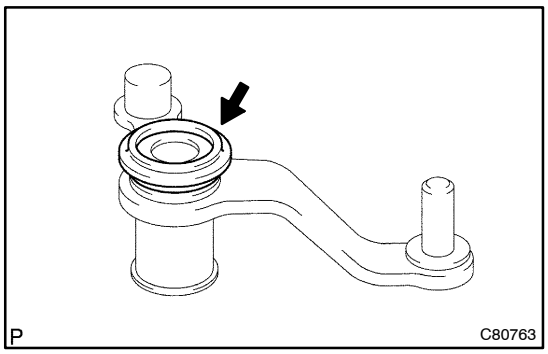
(a) Coat the select spring seat No.2 with MP grease.

(b) Install the select spring seat No.2 and select return compression spring No.2. Using a plier, install a new select spring No.2 seat E–ring to the shift & select lever shaft.



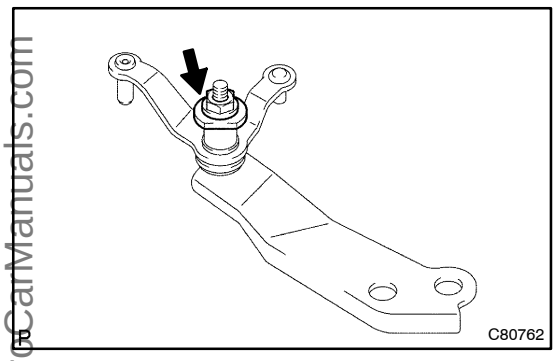
20. INSTALL SELECTING BELLCRANK NO.2 BUSH

(a) Coat the 2 selecting bellcrank No.2 bushes with MP grease, and install it to the selecting bellcrank No.2.



21. INSTALL SELECTING BELLCRANK DUST COVER NO.2

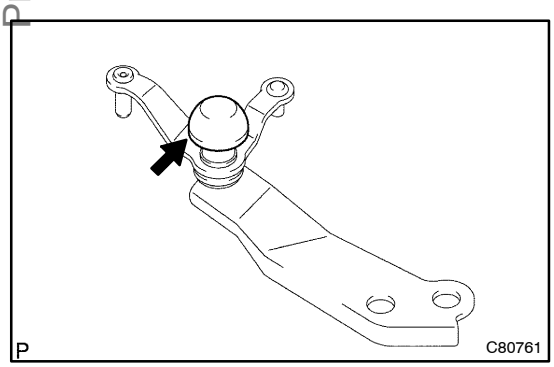
(a) Coat the selecting bellcrank dust cover No.2 with MP grease, and install it to the selecting bellcrank support.



22. INSTALL SELECTING BELL CRANK NO.2

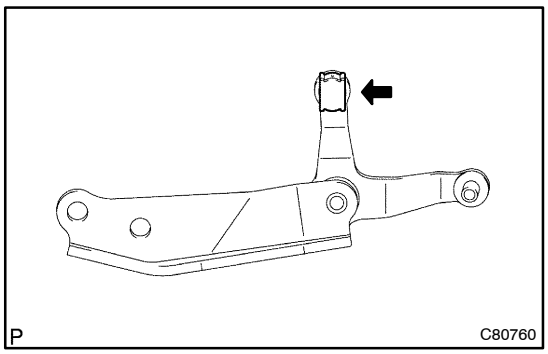
(a) Install the selecting bellcrank support, selecting bellcrank No.2 plate washer, spring washer and nut.
Torque: 12 N·m (120 kgf·cm, 9 ft·lbf)

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23. INSTALL SELECTING BELLCRANK DUST COVER NO.1

(a) Install the selecting bellcrank dust cover No.1 to the selecting bell crank assy.



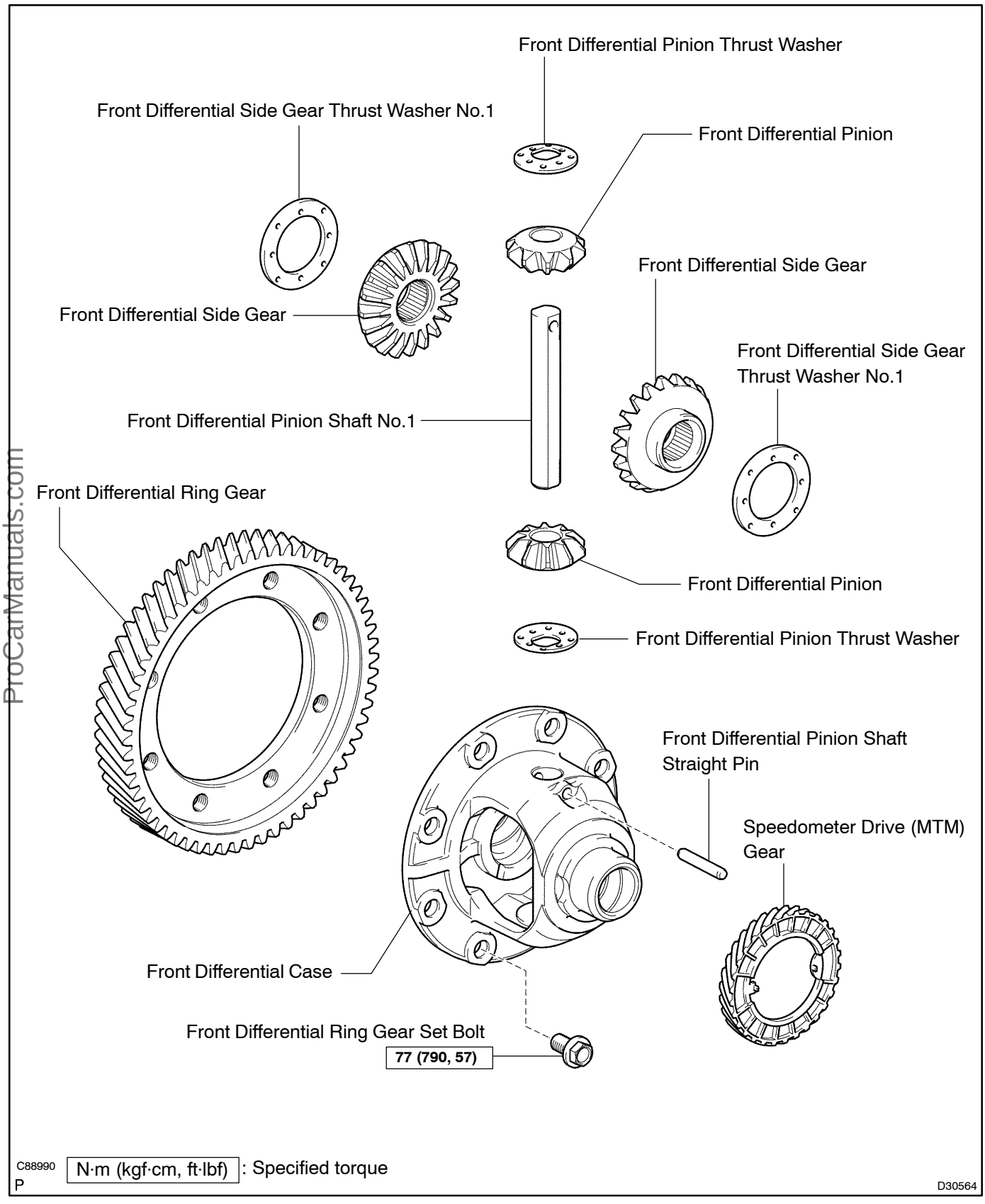
24. INSTALL SHIFT & SELECT LEVER BUSH

(a) Install shift & select lever bush to the selecting bellcrank assy.

DIFFERENTIAL CASE ASSY (C50)

COMPONENTS

410DK–02



C88990
P

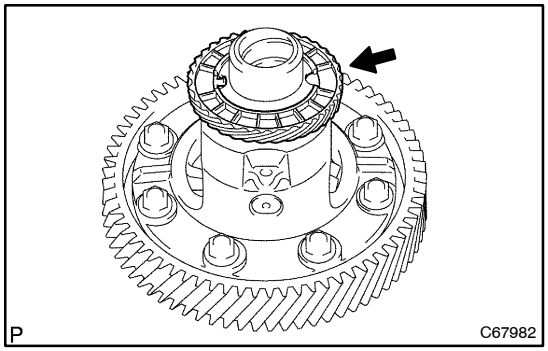
N·m (kgf·cm, ft·lbf) : Specified torque

D30564

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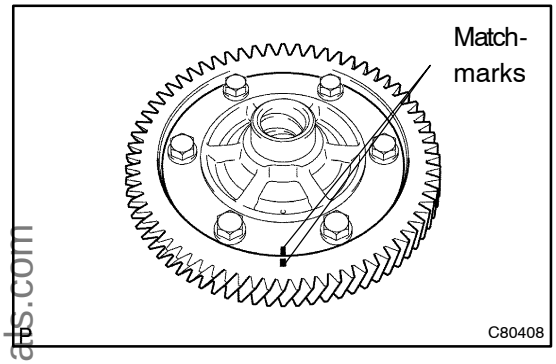
4101E–03

OVERHAUL



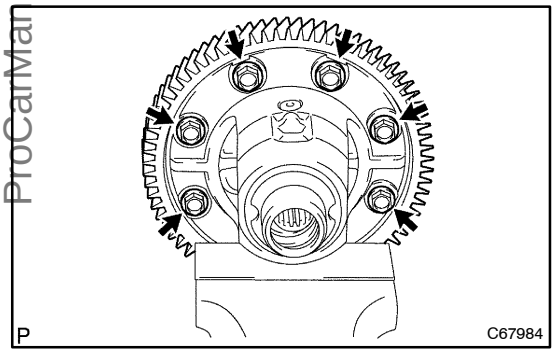
1. REMOVE SPEEDOMETER DRIVE (MTM) GEAR

(a) Remove the speedometer drive gear from the front differential case.

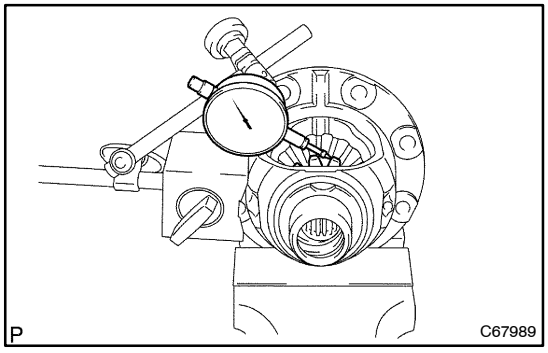


2. REMOVE FRONT DIFFERENTIAL RING GEAR

(a) Put matchmarks on the front differential ring gear and the front differential case.



(b) Remove the 8 bolts. Using a hammer, remove the front differential ring gear from the front differential case.



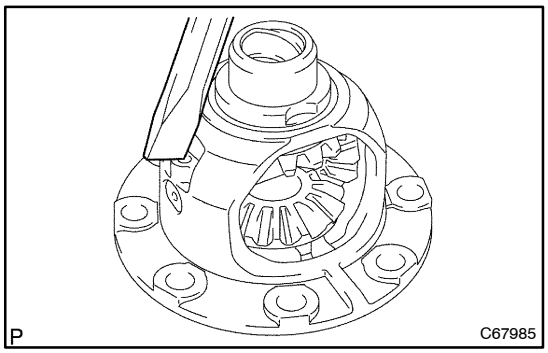
3. INSPECT FRONT DIFFERENTIAL SIDE GEAR BACKLASH

(a) Fixing the front differential pinion to the front differential case side. Using a dial indicator, measure the front differential side gear backlash.

Standard backlash:
0.05 to 0.20 mm (0.0020 to 0.0079 in.)

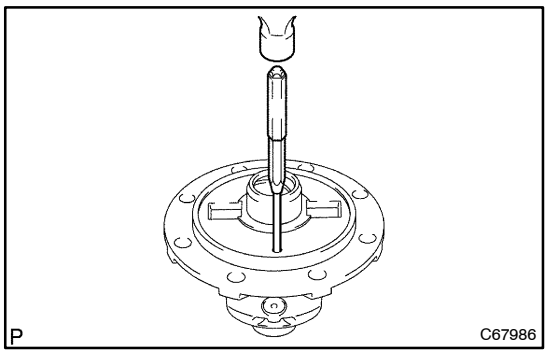
If the backlash is not within the specification, install the connect side gear thrust washers.

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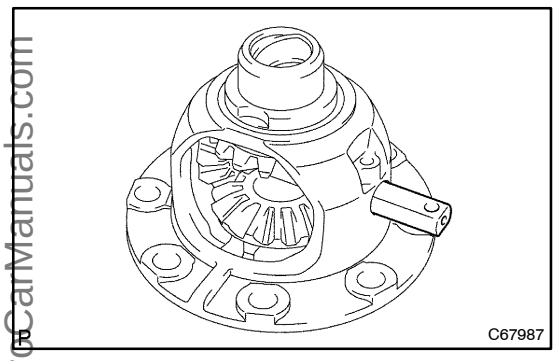


4. REMOVE FRONT DIFFERENTIAL PINION SHAFT STRAIGHT PIN

(a) Using a chisel and a hammer, loosen the staked part of the front differential case.



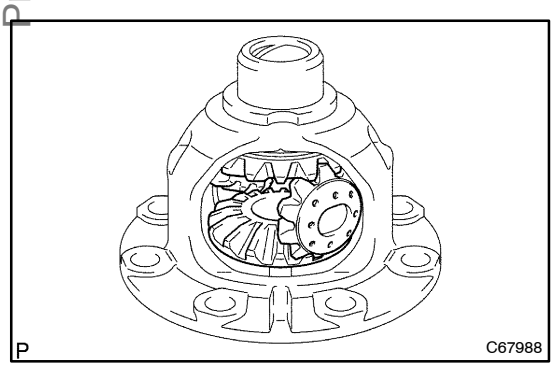
(b) Using a pin punch (ϕ 3 mm) and a hammer, remove the front differential pinion shaft straight pin from the front differential case.



5. REMOVE FRONT DIFFERENTIAL PINION SHAFT NO.1

(a) Remove the front differential pinion shaft No.1 from the front differential case.

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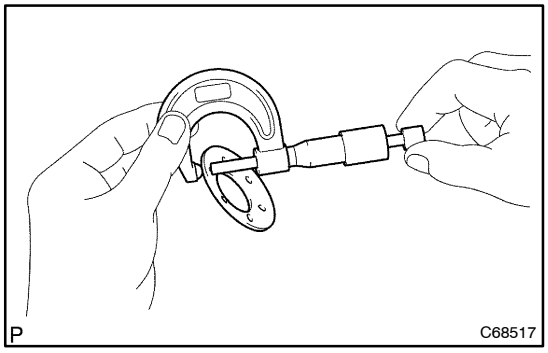


6. REMOVE FRONT DIFFERENTIAL SIDE GEAR

(a) Remove the 2 front differential pinions, front differential pinion thrust washers and front differential side gear thrust washers No.1 from the front differential case.

HINT:

Revolving the front differential pinion, remove the 2 pinions, pinion thrust washers and side gear thrust washers.

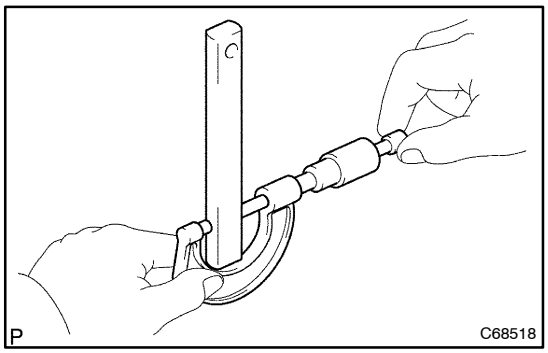


7. INSPECT FRONT DIFFERENTIAL PINION THRUST WASHER

(a) Using a micrometer, measure the thickness of the front differential pinion thrust washer.

Minimum thickness: 0.94 mm (0.03701 in.)

If the thickness is less than the minimum, replace the front differential pinion thrust washer.

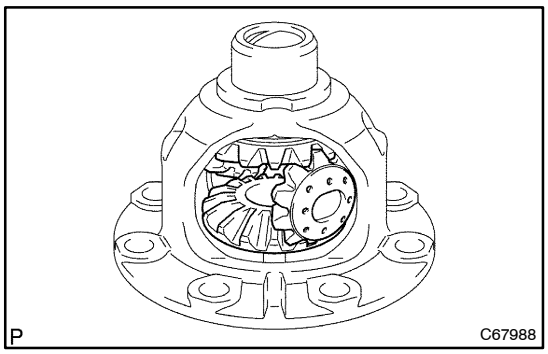


8. INSPECT FRONT DIFFERENTIAL PINION SHAFT NO.1

(a) Using a micrometer, measure the outer diameter of the front differential pinion shaft No.1.

Minimum outer diameter: 16.982 mm (0.6685 in.)

If the outer diameter is less than the minimum, replace the front differential pinion shaft No.1.



9. INSTALL FRONT DIFFERENTIAL SIDE GEAR

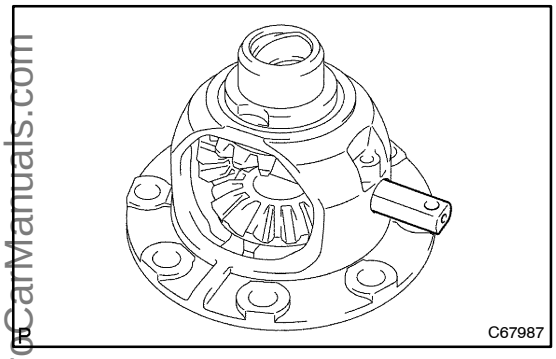
(a) Coat the sliding and rotating surface of the front differential side gear with MP grease.

(b) Install the 2 front differential side gear thrust washers to the 2 front differential side gear.

(c) Install the 2 front differential pinions with front differential pinion thrust washers to the front differential case.

HINT:

Revolving the front differential pinion, install the 2 front differential pinions with front differential pinion thrust washers.

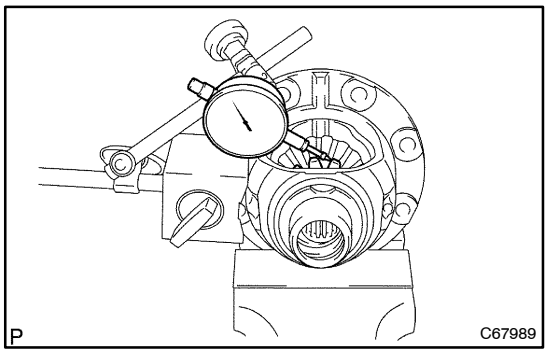


10. INSTALL FRONT DIFFERENTIAL PINION SHAFT NO.1

(a) Coat the front differential pinion shaft No.1 with MP grease.

(b) Install the front differential pinion shaft No.1 to the front differential case so that the hole for the front differential pinion shaft straight pin is aligned with the hole in the front differential case.

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11. ADJUST FRONT DIFFERENTIAL SIDE GEAR BACKLASH

(a) Fix the front differential pinion to the front differential case side. Using a dial gauge, measure the front differential side gear backlash.

Standard backlash:
0.05 to 0.20 mm (0.0020 to 0.0079 in.)

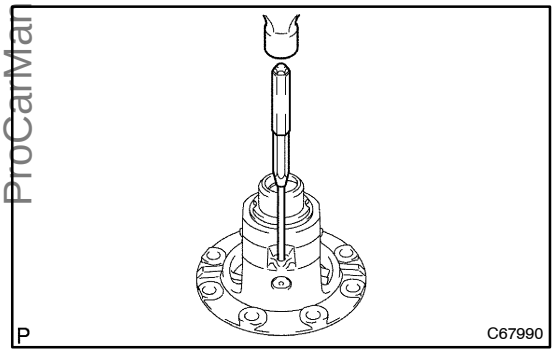
HINT:

- Because the thrust washer does not have any marks for distinction, measure the thickness with a micrometer to select a proper thrust washer.
- Select washers of the same thickness for both the right and left.

Thrust washer thickness

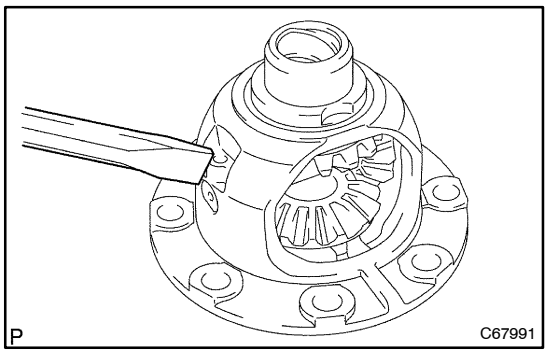
Parts No.	Thickness mm (in.)
41361 – 22140	0.95 (0.0374 in.)
41361 – 22020	1.00 (0.0394 in.)
41361 – 22150	1.05 (0.0413 in.)
41361 – 22030	1.10 (0.0433 in.)
41361 – 22160	1.15 (0.0452 in.)
41361 – 22040	1.20 (0.0472 in.)

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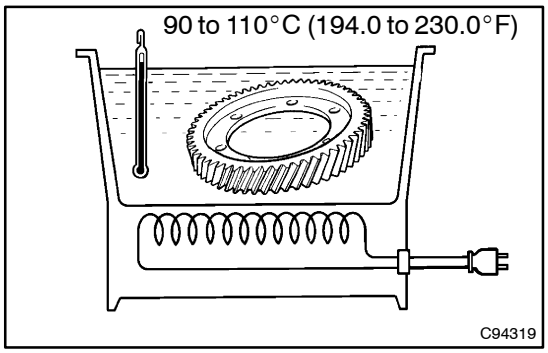


12. INSTALL FRONT DIFFERENTIAL PINION SHAFT STRAIGHT PIN

(a) Using a pin punch (ϕ 3 mm) and a hammer, install the front differential pinion shaft straight pin to the front differential case.



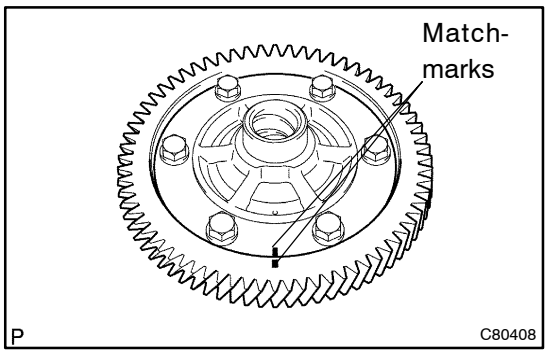
(b) Using a chisel and a hammer, stake the front differential case hole.



13. INSTALL FRONT DIFFERENTIAL RING GEAR

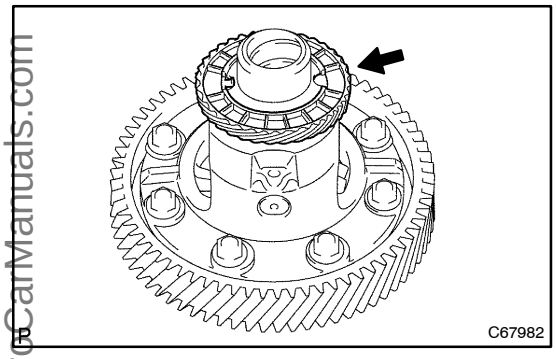
(a) Using a heater, heat the front differential ring gear to 90 to 110 °C (194.0 to 230.0 °F.)

(b) Clean the contact surface of the front differential case.



(c) Aligning the both matchmarks, quickly install the front differential ring gear to the front differential case.

(d) Install the 8 bolts.
Torque: 77 N·m (790 kgf·cm, 57 ft·lbf)



14. INSTALL SPEEDOMETER DRIVE (MTM) GEAR

(a) Install the speedometer drive gear to the front differential case.

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